CMS Draft Analysis Note

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Measurement of dilepton trigger efficiencies and scale factors with the Run 3 dataset

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Abstract

Dilepton trigger efficiencies and data-to-simulation scale factors are studied in the ee, $e\mu$, and $\mu\mu$ channels using Run 3 proton–proton collision data at $\sqrt{s} = 13.6$ TeV. The present draft includes the available 2024 and 2025 results; results for other Run 3 years will be added as they become available. The 2024 and 2025 results are labelled with integrated luminosities of 109.82 and 110.37 fb^{-1}, respectively. An independent MET-reference trigger selection defines the efficiency denominator. This note presents one-dimensional efficiency comparisons, two-dimensional scale-factor maps, and studies of statistical precision, trigger correlation, event topology, pileup, and data-taking periods. The correction axes are the leading and subleading lepton transverse momenta for same-flavour channels, and electron and muon transverse momenta for the mixed-flavour channel. This is a working draft: the exact production configuration and uncertainty prescription require verification before release.

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1 Introduction

Accurate modelling of the trigger response is required for dileptonic top-quark measurements. The efficiency of the analysis trigger OR is measured in data and simulated events selected with MET reference triggers. Their ratio defines the event-level trigger correction. The present note documents the Run 3 measurement. This draft currently presents the supplied 2024 and 2025 results for ee, $e\mu$, and $\mu\mu$, motivated by the top-quark CP-violation analysis. The MET-reference strategy follows the approach described in the earlier studies [1, 2]. The existing Run 3 draft [3]

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supplies the provisional trigger and offline-selection descriptions. Numerical results shown here come from the supplied 2024 and 2025 plot sets; their inclusion does not independently establish that the provisional selection matches the production configuration.

The 2024 and 2025 results are reported separately; results for 2022 and 2023 will be added in subsequent updates.

2 Data and simulation

The measurement uses separate 2024 and 2025 datasets at $\sqrt{s} = 13.6$ TeV. The supplied one-dimensional plots are labelled with integrated luminosities of 109.82 fb^{-1} for 2024 and 110.37 fb^{-1} for 2025. These values are retained as the measurement figure metadata; they are not replaced by luminosities quoted for different analysis selections or input versions. The reference data are collected with MET/JetMET triggers, and dileptonic $t\bar{t}$ simulation provides the simulated reference. The two years are kept separate in the result figures and correction lookup. The supplied files `TriggerSF_2024.root` and `TriggerSF_2025.root` are distributed unchanged. **[TO VERIFY: Confirm the actual data and MC input lists, campaign, certified run intervals, luminosity calculation, and dataset-overlap removal for each year.]**

3 Trigger paths

For each final state, the signal-trigger decision is the logical OR of the applicable single-lepton and dilepton paths. Table 1 lists the Run 3 signal paths; every path must be checked against the HLT menu and NanoAOD branches in each period. A recorded event may pass several paths but enters each numerator once. Any run-dependent changes, prescales, dataset-overlap removal, or trigger-object matching requirements must be documented before finalization.

Table 1: Signal-trigger OR used for the dilepton trigger measurement. Path suffixes `_v*` are suppressed. Confirm per-era availability and signal dataset overlap removal.

Channel	Signal HLT path
ee	`HLT_Ele23_Ele12_CaloIdL_TrackIdL_IsoVL`
	`HLT_Ele32_WPTight_Gsf`
	`HLT_DoubleEle25_CaloIdL_MW`
$e\mu$	`HLT_Mu23_TrkIsoVVL_Ele12_CaloIdL_TrackIdL_IsoVL`
	`HLT_Mu8_TrkIsoVVL_Ele23_CaloIdL_TrackIdL_IsoVL_DZ`
	`HLT_Mu12_TrkIsoVVL_Ele23_CaloIdL_TrackIdL_IsoVL_DZ`
	`HLT_Ele32_WPTight_Gsf`
	`HLT_IsoMu24`
$\mu\mu$	`HLT_Mu17_TrkIsoVVL_Mu8_TrkIsoVVL_DZ_Mass3p8`
	`HLT_IsoMu24`

3.1 MET reference triggers

Events for the efficiency denominator are recorded with a logical OR of MET or MET+HT reference paths. These triggers select events without directly requiring an electron or muon. The candidate reference set is given in Table 2. The unprescaled status and validity interval of each path, and its actual NanoAOD implementation, must be verified. The MET requirement

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of the reference trigger can bias the measured efficiency if it is correlated with the dilepton trigger; this assumption is examined in Section 6.3.

Table 2: Candidate MET-reference paths. Their availability must be verified separately for each 2025 period. Preserve only paths verified in each run and in the input files.

Reference category	HLT path
MET	`HLT_PFMET120_PFMHT120_IDTight`
	`HLT_PFMET200_BeamHaloCleaned`
Type-1 MET	`HLT_PFMETTypeOne200_BeamHaloCleaned`
MET (no muons)	`HLT_PFMETNoMu120_PFMHTNoMu120_IDTight`
MET + HT	`HLT_PFMET120_PFMHT120_IDTight_PFHT60`
MET (no muons) + HT	`HLT_PFMETNoMu120_PFMHTNoMu120_IDTight_PFHT60`
HT + MET	`HLT_PFHT500_PFMET100_PFMHT100_IDTight`
	`HLT_PFHT700_PFMET85_PFMHT85_IDTight`
	`HLT_PFHT800_PFMET75_PFMHT75_IDTight`
	`HLT_PFHT500_PFMET110_PFMHT110_IDTight`
MET (no muons), HF	`HLT_PFMETNoMu120_PFMHTNoMu120_IDTight_FilterHF`

4 Object and event selection

The offline selection used for the trigger-efficiency measurement uses the Run 3 configuration described below. It is distinct from the final CPV analysis selection used for the final event sample. In particular, the trigger study uses a subleading-lepton threshold of 15 GeV and a missing-transverse-momentum threshold of 100 GeV. The final CPV analysis requirements of at least two jets, a same-flavour Z veto, and a same-flavour missing-momentum threshold of 40 GeV are not substituted for the trigger-study selection. The 2025 extension uses the same object and event selection as the 2024 measurement. Year-dependent calibrations and the implemented trigger configuration remain subject to verification against the production code.

4.1 Electron selection

Electrons satisfy $p_{\mathrm{T}} > 15\,\mathrm{GeV}$ and $|\eta_{\mathrm{SC}}| < 2.4$, where η_{SC} is the pseudorapidity of the associated ECAL supercluster. Electrons in the barrel–endcap transition region, $1.4442 < |\eta_{\mathrm{SC}}| < 1.566$, are excluded. The tight cut-based electron identification working point is required. Electron identification scale factors and the corresponding momentum corrections are applied according to the production configuration. **[TO VERIFY: Document the exact identification branch, impact-parameter requirements, and calibration identifiers used in each year.]**

4.2 Muon selection

Muons satisfy $p_{\mathrm{T}} > 15\,\mathrm{GeV}$ and $|\eta| < 2.4$. Global muons passing the recommended tight muon identification are selected, with PF-based relative isolation below 0.15. Muon identification and isolation scale factors and the corresponding momentum corrections are applied according to the production configuration. **[TO VERIFY: Document the isolation cone and branch, impact-parameter requirements, and calibration identifiers used in each year.]**

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4.3 Jet selection and veto map

Jets are clustered with the anti-k_T algorithm with radius $R = 0.4$ from PUPPI-weighted particle-flow candidates. Selected analysis jets satisfy $p_T > 30$ GeV, $|\eta| < 2.4$, and the tight lepton-veto jet identification. Jets within $\Delta R < 0.4$ of a selected lepton are removed. Jet energy corrections and jet energy resolution corrections follow the period-dependent production configuration.

The Jet Veto Map is applied as an event veto. Its jet eligibility threshold is $p_T > 15$ GeV, with tight lepton-veto jet identification and the recommended charged and neutral electromagnetic-fraction requirements. This threshold is separate from the 30 GeV threshold used to select analysis jets. An event is rejected if an eligible jet falls in a vetoed region. **[TO VERIFY: Document the map identifiers, exact electromagnetic-fraction requirements, and period-dependent application from code.]**

4.4 b jet identification

At least one selected jet must pass the medium Unified Particle Transformer (UParT) b-tagging working point. **[TO VERIFY: Insert the exact discriminator branch, working-point thresholds, efficiency inputs, and scale-factor calibration used in each period.]**

4.5 Missing transverse momentum

PUPPI missing transverse momentum is used, with $p_T^{\text{miss}} > 100$ GeV required in all three channels. The same offline requirement is imposed on the efficiency numerator and denominator. Jet energy scale and resolution corrections are propagated to the missing transverse momentum.

4.6 Event selection and corrections

Events contain at least two oppositely charged selected leptons in the ee, $e\mu$, or $\mu\mu$ channel. The leading and subleading leptons satisfy $p_T > 25$ and 15 GeV, respectively, and the dilepton invariant mass satisfies $m_{\ell\ell} > 20$ GeV. At least one selected jet passes the medium UParT working point, and $p_T^{\text{miss}} > 100$ GeV is required. The documented trigger-study selection does not specify an additional minimum of two jets or a same-flavour Z-mass veto. An exact-two-lepton requirement is likewise not inferred from the stated requirement of at least two leptons.

The event-quality flags documented for the study are listed in Table 4. For the 2024 simulation, the pileup reweighting was updated to match the C–I data-taking periods rather than B–I. Lepton identification and isolation scale factors and the object corrections are applied consistently with the measurement configuration. The measured trigger scale factor is not applied when deriving the simulated trigger efficiency itself. The same non-trigger MC weights are used in the efficiency numerator and denominator.

For the efficiency measurement, the MET-reference trigger OR is imposed on both numerator and denominator, whereas the signal-trigger OR is imposed only on the numerator.

5 Efficiency and scale-factor measurement

Let O denote the offline event selection, R the reference-trigger OR, and S_c the signal-trigger OR for channel c. The measured efficiency in bin i is

$$\widehat{\epsilon}_{d/m,c,i} = \frac{N_{d/m,c,i}(O \cap R \cap S_c)}{N_{d/m,c,i}(O \cap R)}, \qquad \mathrm{SF}_{c,i} = \frac{\widehat{\epsilon}_{d,c,i}}{\widehat{\epsilon}_{m,c,i}}. \tag{1}$$

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Table 3: Offline selection for the trigger-efficiency measurement, with the medium UParT working point.

Object or quantity	Requirement
Electrons	$p_T > 15$ GeV, $\lvert\eta_{SC}\rvert < 2.4$; exclude $1.4442 < \lvert\eta_{SC}\rvert < 1.566$; tight cut-based ID.
Muons	$p_T > 15$ GeV, $\lvert\eta\rvert < 2.4$; global muon, tight ID; PF relative isolation < 0.15.
Analysis jets	AK4 PUPPI; $p_T > 30$ GeV, $\lvert\eta\rvert < 2.4$; tight lepton-veto ID; $\Delta R(j, \ell) \geq 0.4$.
Jet Veto Map	Event veto using eligible jets with $p_T > 15$ GeV and the map-specific quality requirements.
Dilepton topology	$\geq$ 2 opposite-charge leptons; leading/subleading $p_T > 25/15$ GeV; $m_{\ell\ell} > 20$ GeV.
b-tagged jets	≥ 1 selected jet at the medium UParT working point.
PUPPI missing momentum	$p_T^{miss} > 100$ GeV in all channels.
Event quality	Certified data lumisections and the event-quality filters in Table 4.

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Table 4: Event-quality filters for the 2024 measurement. The ECAL bad-calibration flag is included for 2024; the 2025 filter configuration must be checked separately.

Filter	NanoAOD flag
Primary vertex	`Flag_goodVertices`
Beam halo	`Flag_globalSuperTightHalo2016Filter`
ECAL trigger primitive	`Flag_EcalDeadCellTriggerPrimitiveFilter`
Bad PF muon	`Flag_BadPFMuonFilter`
Bad PF muon longitudinal displacement	`Flag_BadPFMuonDzFilter`
HF noisy hits	`Flag_hfNoisyHitsFilter`
Bad ECAL supercluster	`Flag_eeBadScFilter`
ECAL bad calibration (2024)	`Flag_ecalBadCalibFilter`

The numerator is a subset of the denominator. Each event enters once even when multiple signal paths fire. Simulation counts represent the configured sum of non-trigger weights; the same weights are applied in the numerator and denominator. The conditional efficiency $P(S_c \mid O, R)$ represents $P(S_c \mid O)$ only when the residual reference-trigger bias is controlled. The supplied one-dimensional figures already combine data and MC efficiencies with their SF ratio. These are used directly in the main results; selected numerator, denominator, and efficiency diagnostics are provided in Appendix C and Appendix F. The leading/subleading convention is used in one-dimensional plots. For the final two-dimensional correction, same-flavour leptons are ordered by transverse momentum, whereas the mixed-flavour map uses $x = p_{\mathrm{T}}^{e}$ and $y = p_{\mathrm{T}}^{\mu}$. The lowest occupied bins probe the turn-on region and should not be described as plateau bins without checking the offline thresholds and trigger response. Invalid, unpopulated, and inaccessible bins must not be assigned unity by default.**[TO VERIFY: Verify interval construction, MC weight treatment, exact bin edges, and underflow/overflow rules from code.]**

6 Uncertainties

6.1 Statistical precision

The numerator–denominator covariance must be retained when estimating efficiency errors. For data this requires a binomial construction where applicable; weighted simulation requires a treatment appropriate to its weights, including any negative weights. The supplied statistical and breakdown plots are retained without attributing an unverified interval algorithm to them.

6.2 Topology, pileup and era dependence

The supplied legends define jet-multiplicity regions $N_{\mathrm{jet}} < 3$ and $N_{\mathrm{jet}} \geq 3$, and vertex regions $N_{\mathrm{PV}} < 40$ and $N_{\mathrm{PV}} \geq 40$. These boundaries supersede the different provisional jet split in the original draft. The breakdown legends describe an envelope relative to the nominal SF. The intended interpretation is

$$\delta_k(i) = \max_{a \in \{\mathrm{low,high}\}} |\mathrm{SF}_{k,a}(i) - \mathrm{SF}_{\mathrm{nom}}(i)|, \tag{2}$$

with relative uncertainty obtained by division by $|\mathrm{SF}_{\mathrm{nom}}(i)|$. This interpretation, including any statistical subtraction, must be checked against the production code. An era-dependence component is also shown; the plot alone does not specify its luminosity weighting or combination procedure.

6.3 Reference–signal correlation

A useful diagnostic is

$$\alpha_c = \frac{P(R \cap S_c \mid O)}{P(R \mid O) P(S_c \mid O)}, \qquad P(S_c \mid O, R) = \alpha_c P(S_c \mid O). \tag{3}$$

Thus the reference-conditioned SF differs from the inclusive SF by α_d / α_m. An MC-only correlation study cannot determine this ratio in data without additional assumptions. The supplied orange curves are labelled as a correlation uncertainty, not as measurements of α; they must not be relabelled as α distributions.

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6.4 Uncertainty combination and current limitation

The supplied relative-breakdown legend defines total systematic uncertainty using era, jet multiplicity, and vertex multiplicity, and total uncertainty from statistics and that systematic term. Although a correlation curve is shown, it is absent from the stated systematic sum. This discrepancy requires resolution before adopting the total error or the accompanying 2D uncertainty maps. The draft reproduces the figures without changing their numerical content.

6.4.1 2024 uncertainty breakdowns

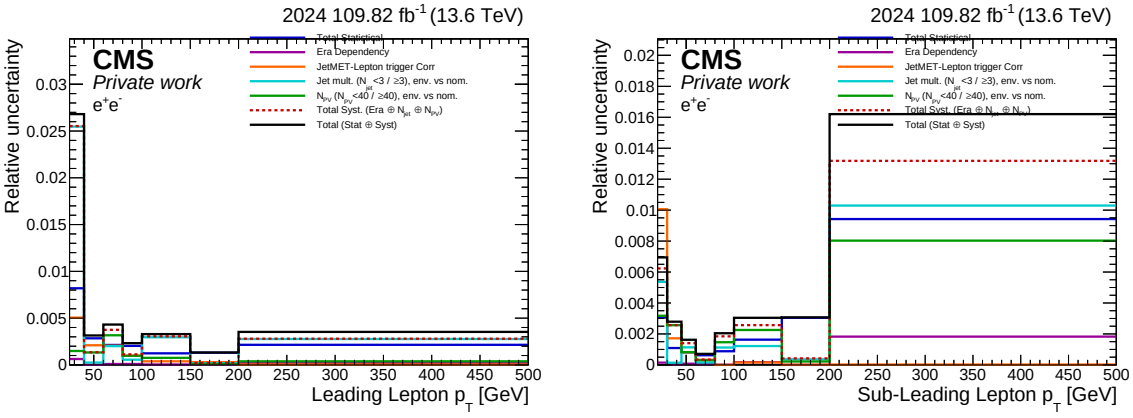


Figure 1: 2024: Relative uncertainty breakdown in the *ee* channel versus leading and subleading lepton transverse momentum. The displayed total-systematic legend lists era, jet, and vertex components; the separately drawn correlation component is not listed in that sum.

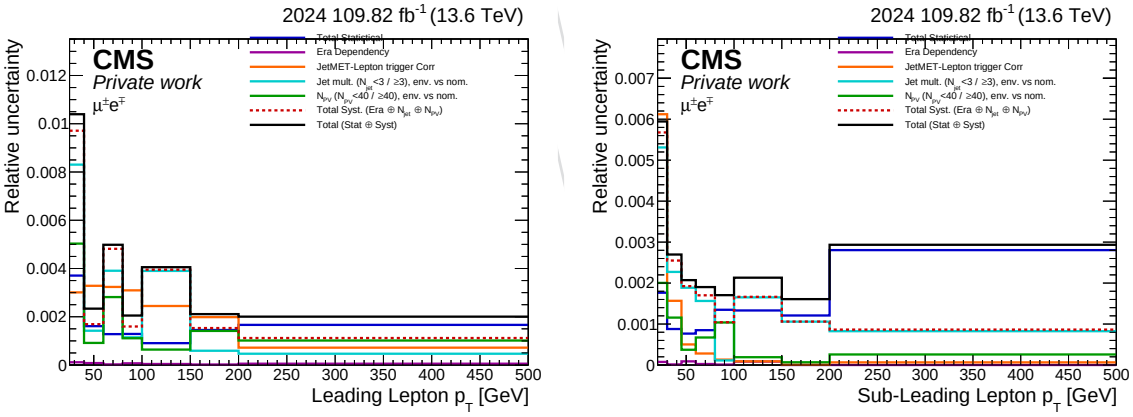


Figure 2: 2024: Relative uncertainty breakdown in the $e\mu$ channel versus leading and subleading lepton transverse momentum. The displayed total-systematic legend lists era, jet, and vertex components; the separately drawn correlation component is not listed in that sum.

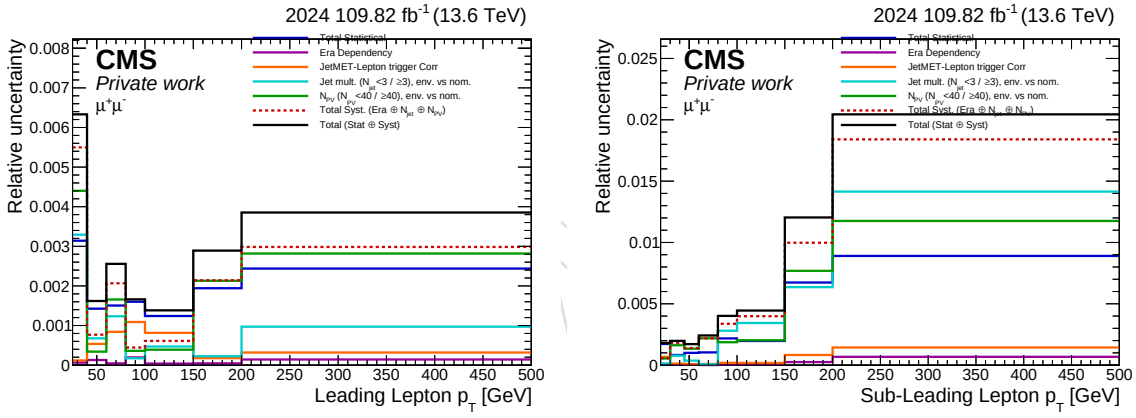


Figure 3: 2024: Relative uncertainty breakdown in the $\mu\mu$ channel versus leading and sub-leading lepton transverse momentum. The displayed total-systematic legend lists era, jet, and vertex components; the separately drawn correlation component is not listed in that sum.

6.4.2 2025 uncertainty breakdowns

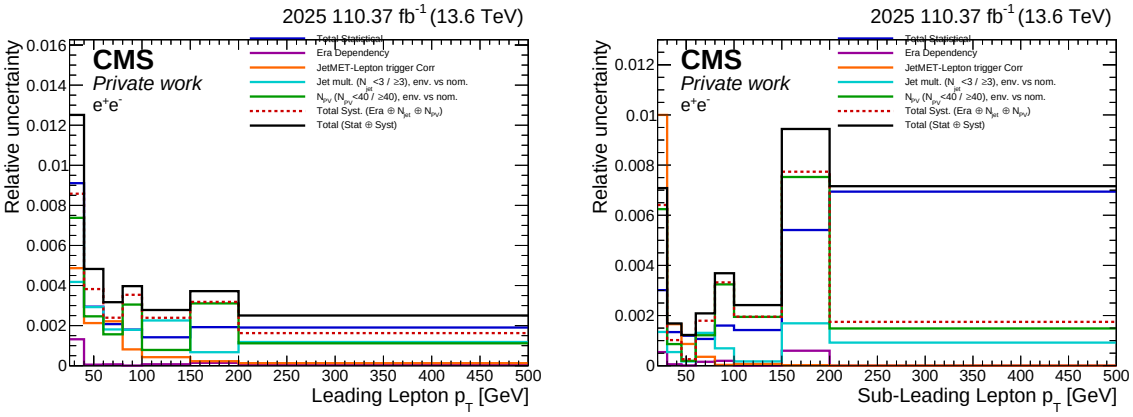


Figure 4: 2025: Relative uncertainty breakdown in the *ee* channel versus leading and subleading lepton transverse momentum. The displayed total-systematic legend lists era, jet, and vertex components; the separately drawn correlation component is not listed in that sum.

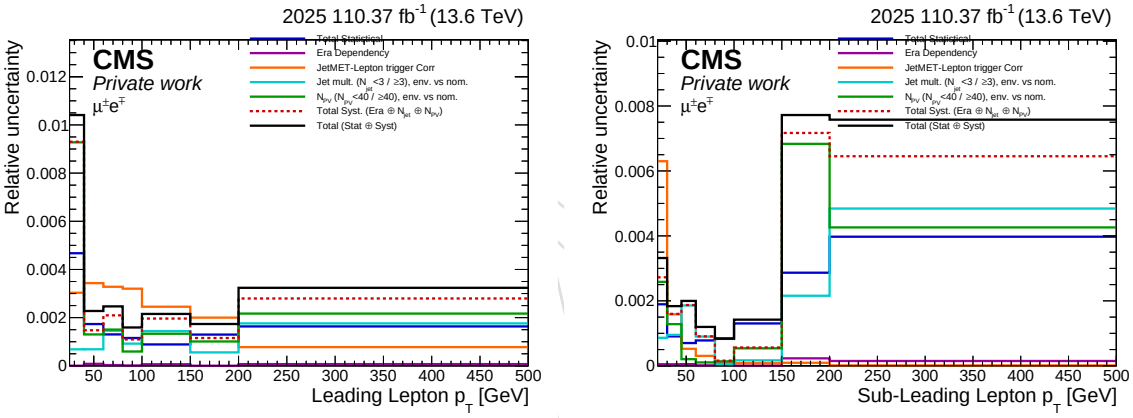


Figure 5: 2025: Relative uncertainty breakdown in the $e\mu$ channel versus leading and subleading lepton transverse momentum. The displayed total-systematic legend lists era, jet, and vertex components; the separately drawn correlation component is not listed in that sum.

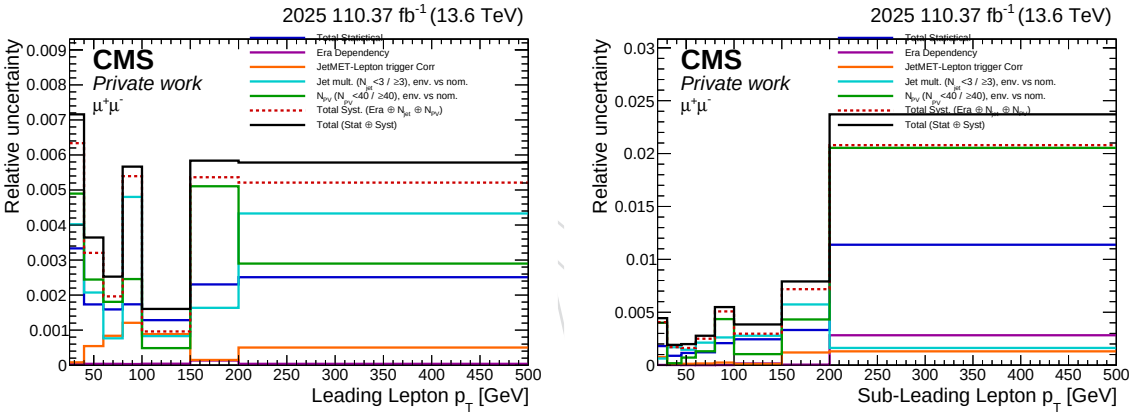


Figure 6: 2025: Relative uncertainty breakdown in the $\mu\mu$ channel versus leading and sub-leading lepton transverse momentum. The displayed total-systematic legend lists era, jet, and vertex components; the separately drawn correlation component is not listed in that sum.

7 Results

The 2024 and 2025 corrections are measured separately. Figures retain the original production binning and annotations.

7.1 2024 results

7.1.1 One-dimensional efficiencies and scale factors

Figures below compare the data and simulation efficiencies as functions of leading and subleading lepton transverse momentum. The lower panels show their ratios. Error bars and constant-fit overlays are retained from the supplied production plots; their confidence levels and fit configuration have not been independently established. A fitted constant is a diagnostic and does not replace the two-dimensional correction.

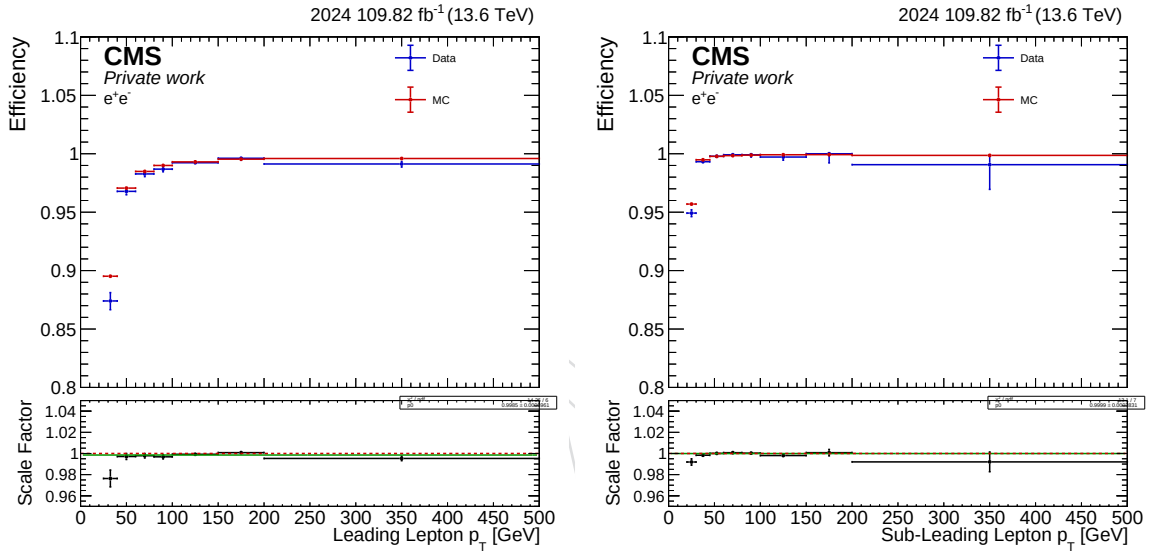


Figure 7: 2024: Trigger efficiencies and data-to-simulation scale factors in the *ee* channel versus leading (left) and subleading (right) lepton transverse momentum. Upper panels show data and MC; lower panels show their ratio. The plotted errors and fit overlays are those of the supplied result set.

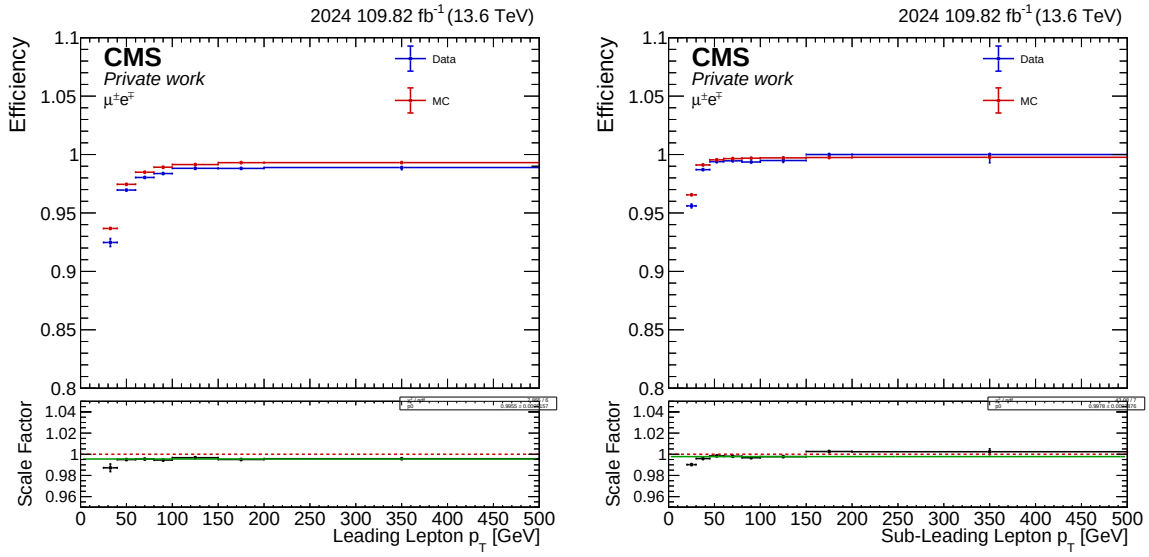


Figure 8: 2024: Trigger efficiencies and data-to-simulation scale factors in the $e\mu$ channel versus leading (left) and subleading (right) lepton transverse momentum. Upper panels show data and MC; lower panels show their ratio. The plotted errors and fit overlays are those of the supplied result set.

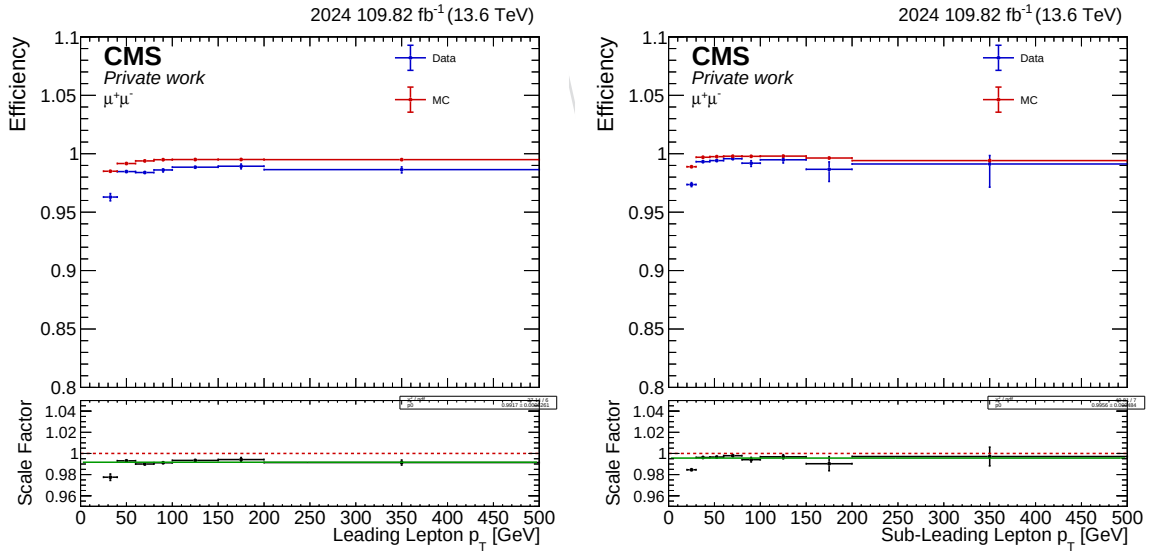


Figure 9: 2024: Trigger efficiencies and data-to-simulation scale factors in the $\mu\mu$ channel versus leading (left) and subleading (right) lepton transverse momentum. Upper panels show data and MC; lower panels show their ratio. The plotted errors and fit overlays are those of the supplied result set.

7.1.2 Two-dimensional correction maps

The central and accompanying uncertainty maps are presented for all three channels. White regions in same-flavour maps lie above the ordered-lepton diagonal and are not physical correction bins. The mixed-flavour map has no such ordering restriction. The associated uncertainty image is labelled here as a supplied uncertainty map: whether its entries are absolute or relative and which components it includes must be established from the producing code. The displayed values are generally close to unity, with more pronounced departures in some low-transverse-momentum bins. Closely spaced annotations at low momentum overlap, so the machine-readable bin content is required for quantitative use.

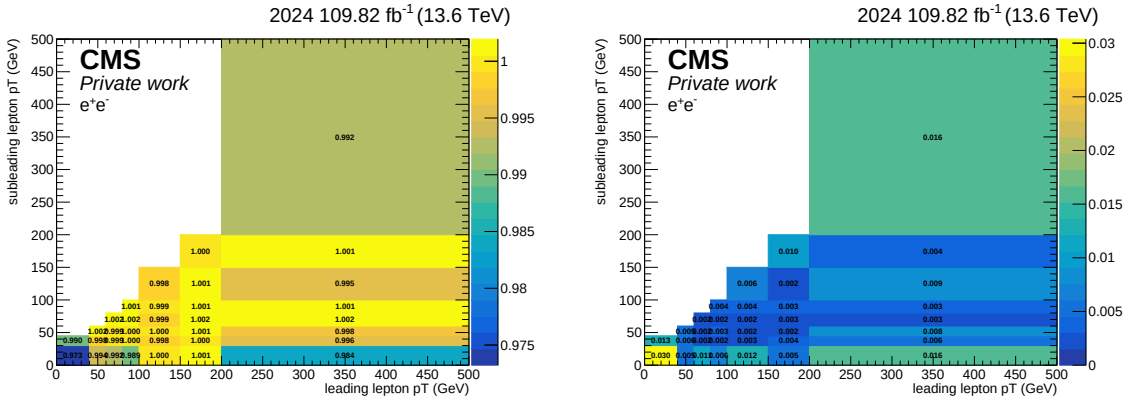


Figure 10: 2024: Two-dimensional central SF (left) and supplied uncertainty map (right) in the *ee* channel. The axes are leading and subleading lepton transverse momentum, respectively; only the ordered phase space is populated. Uncertainty content and lookup boundaries require code verification.

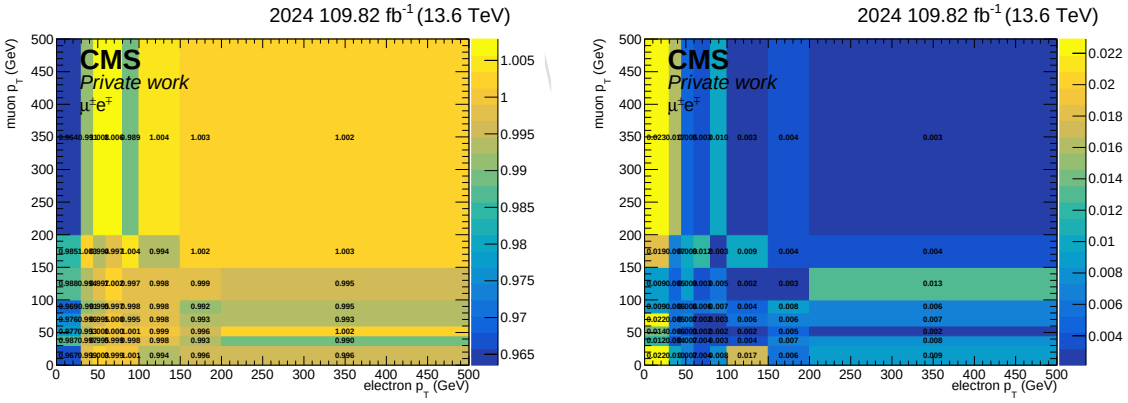


Figure 11: 2024: Two-dimensional central SF (left) and supplied uncertainty map (right) in the *eμ* channel. The horizontal and vertical axes are electron and muon transverse momentum, respectively. Uncertainty content and lookup boundaries require code verification.

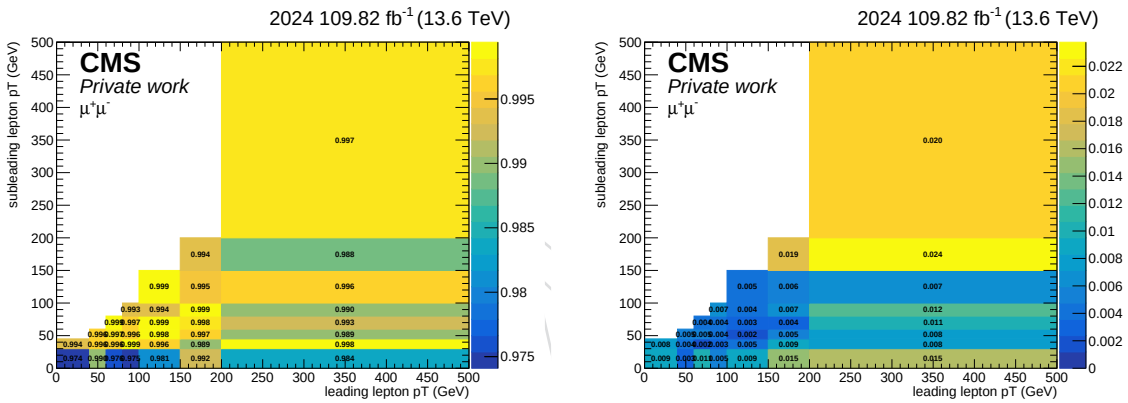


Figure 12: 2024: Two-dimensional central SF (left) and supplied uncertainty map (right) in the $\mu\mu$ channel. The axes are leading and subleading lepton transverse momentum, respectively; only the ordered phase space is populated. Uncertainty content and lookup boundaries require code verification.

7.2 2025 results

7.2.1 One-dimensional efficiencies and scale factors

Figures below compare the data and simulation efficiencies as functions of leading and subleading lepton transverse momentum. The lower panels show their ratios. Error bars and constant-fit overlays are retained from the supplied production plots; their confidence levels and fit configuration have not been independently established. A fitted constant is a diagnostic and does not replace the two-dimensional correction.

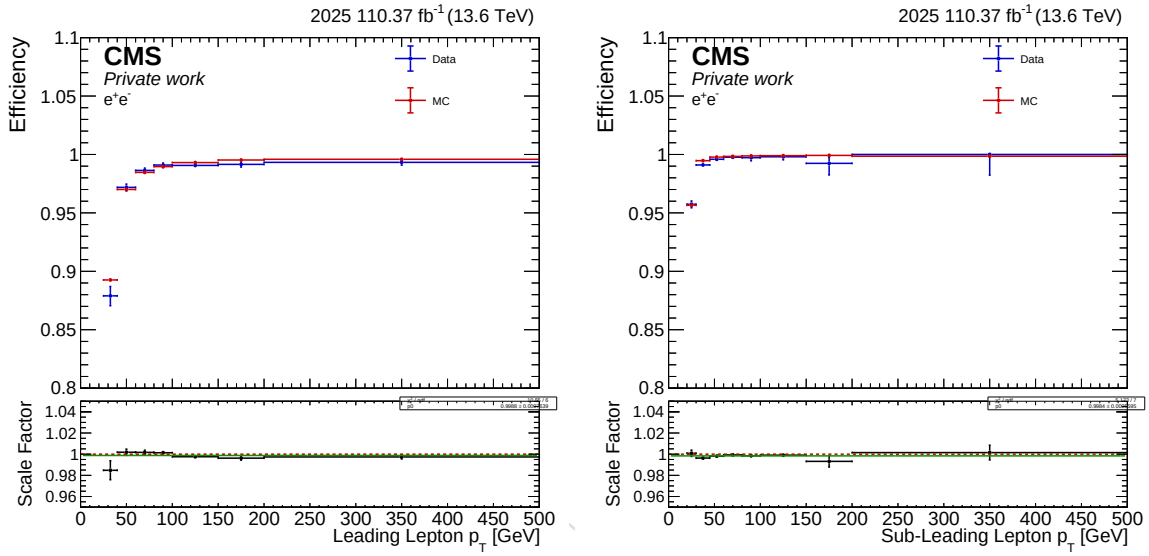


Figure 13: 2025: Trigger efficiencies and data-to-simulation scale factors in the *ee* channel versus leading (left) and subleading (right) lepton transverse momentum. Upper panels show data and MC; lower panels show their ratio. The plotted errors and fit overlays are those of the supplied result set.

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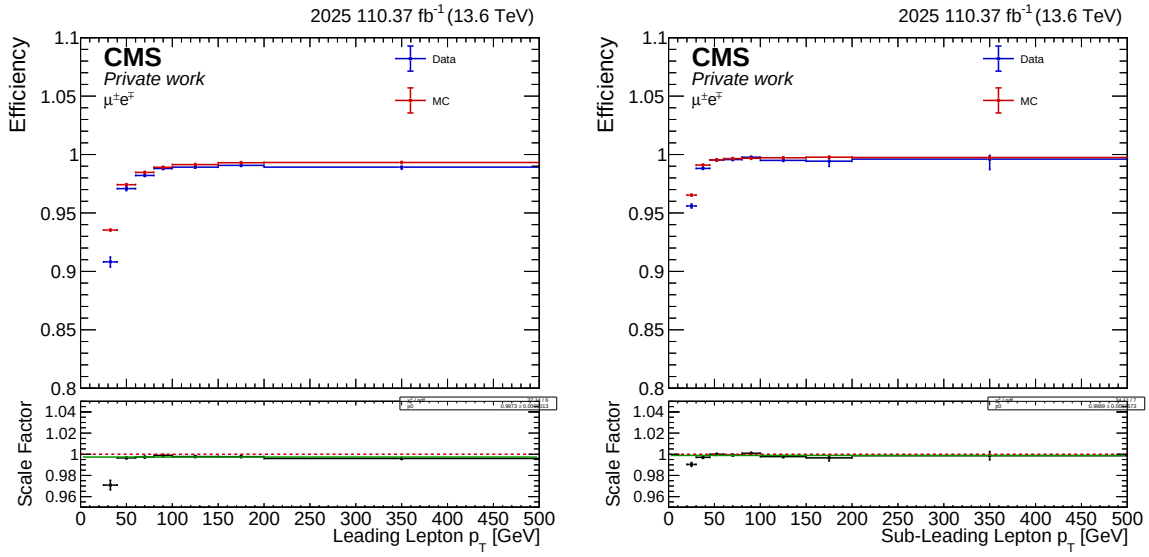


Figure 14: 2025: Trigger efficiencies and data-to-simulation scale factors in the $e\mu$ channel versus leading (left) and subleading (right) lepton transverse momentum. Upper panels show data and MC; lower panels show their ratio. The plotted errors and fit overlays are those of the supplied result set.

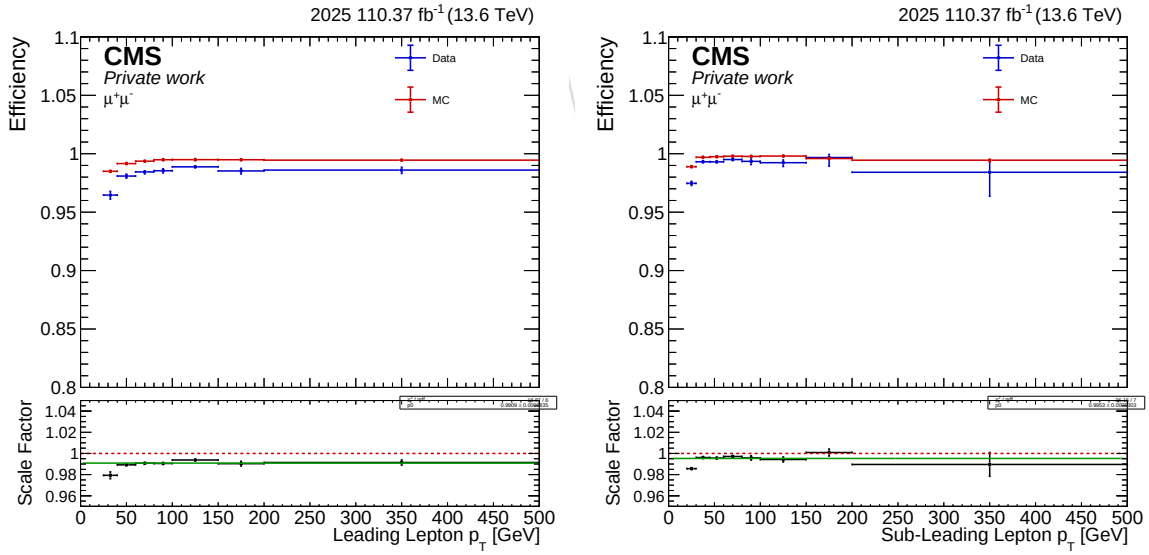


Figure 15: 2025: Trigger efficiencies and data-to-simulation scale factors in the $\mu\mu$ channel versus leading (left) and subleading (right) lepton transverse momentum. Upper panels show data and MC; lower panels show their ratio. The plotted errors and fit overlays are those of the supplied result set.

7.2.2 Two-dimensional correction maps

The central and accompanying uncertainty maps are presented for all three channels. White regions in same-flavour maps lie above the ordered-lepton diagonal and are not physical correction bins. The mixed-flavour map has no such ordering restriction. The associated uncertainty image is labelled here as a supplied uncertainty map: whether its entries are absolute or relative and which components it includes must be established from the producing code. The displayed values are generally close to unity, with more pronounced departures in some low-transverse-momentum bins. Closely spaced annotations at low momentum overlap, so the machine-readable bin content is required for quantitative use.

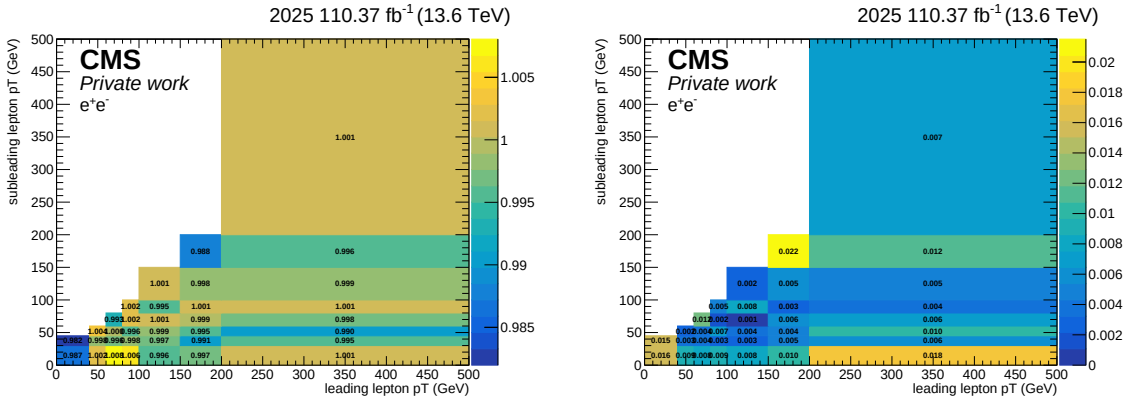


Figure 16: 2025: Two-dimensional central SF (left) and supplied uncertainty map (right) in the *ee* channel. The axes are leading and subleading lepton transverse momentum, respectively; only the ordered phase space is populated. Uncertainty content and lookup boundaries require code verification.

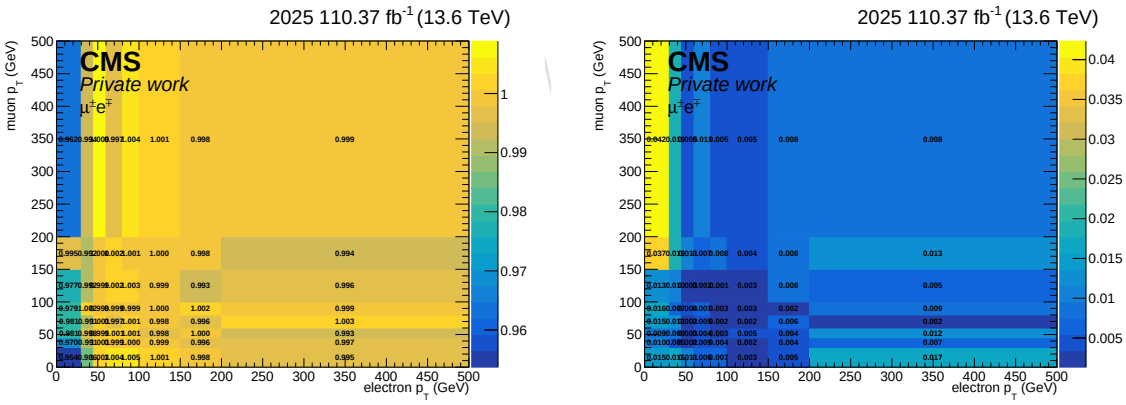


Figure 17: 2025: Two-dimensional central SF (left) and supplied uncertainty map (right) in the *eμ* channel. The horizontal and vertical axes are electron and muon transverse momentum, respectively. Uncertainty content and lookup boundaries require code verification.

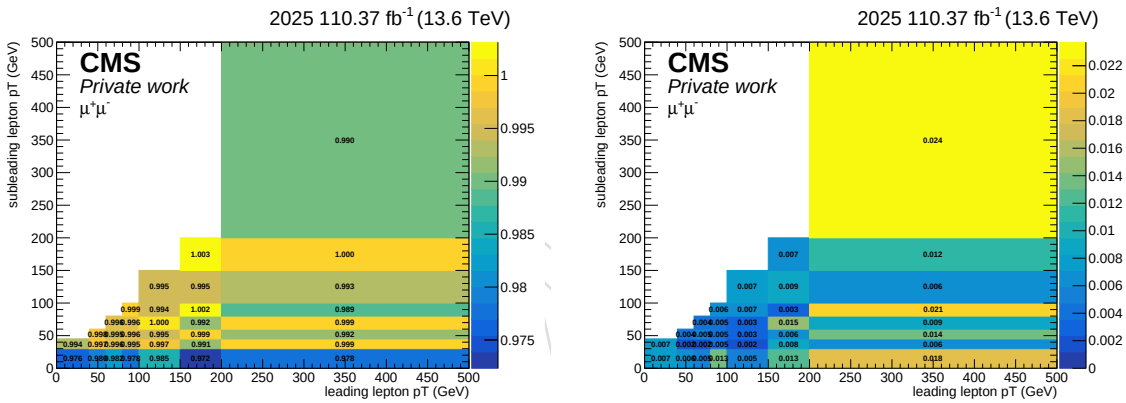


Figure 18: 2025: Two-dimensional central SF (left) and supplied uncertainty map (right) in the $\mu\mu$ channel. The axes are leading and subleading lepton transverse momentum, respectively; only the ordered phase space is populated. Uncertainty content and lookup boundaries require code verification.

7.3 Application conventions

For a simulated event passing the analysis trigger selection, the intended correction multiplies the non-trigger event weight by the year- and channel-specific SF evaluated at the selected lepton momenta. One must not additionally multiply the event by the measured trigger efficiency when using this data-to-MC ratio prescription. The final deployment must document the validity interval, object selection, trigger OR, bin edges, boundary policy, and nuisance-parameter correlations. No release recommendation is made before these checks are complete.

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8 Summary

This note documents the Run 3 dilepton trigger measurement. The current draft includes 2024 and 2025 results; other Run 3 years will be included in subsequent updates. The supplied 2024 and 2025 results provide dilepton trigger efficiency comparisons and two-dimensional data-to-simulation scale-factor maps for ee, $e\mu$, and $\mu\mu$. The one-dimensional figure labels quote 109.82 fb^{-1} for 2024 and 110.37 fb^{-1} for 2025 at 13.6 TeV. Same-flavour maps use ordered lepton momenta; the mixed-flavour map uses electron and muon momenta. Statistical, topology, pileup, era, and correlation studies accompany the measurement. Final release requires confirmation of the production selections and input samples, resolution of the correlation-error combination, validation of the ROOT lookup semantics, and an independent application cross-check.

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References

[1] S. Chenarani et al., "Measurement of the trigger efficiencies for a dilepton selection for a ttbar analysis with the full Run 2 dataset", CMS-AN-2019-140 (2022), internal analysis note.

[2] A. Arisi, "Trigger efficiencies and scale factors measurements with the Run 3 dataset for a dilepton selection", internal study slides (2026); metadata inherited from the supplied draft.

[3] CPV trigger working group, "Measurement of dilepton trigger efficiencies and scale factors with the Run 3 dataset", AN-26-XXX working draft, September 2026.

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A 2024: additional kinematic dependence

These distributions extend the main momentum comparisons. They use the same measurement sample and do not constitute independent closure tests.

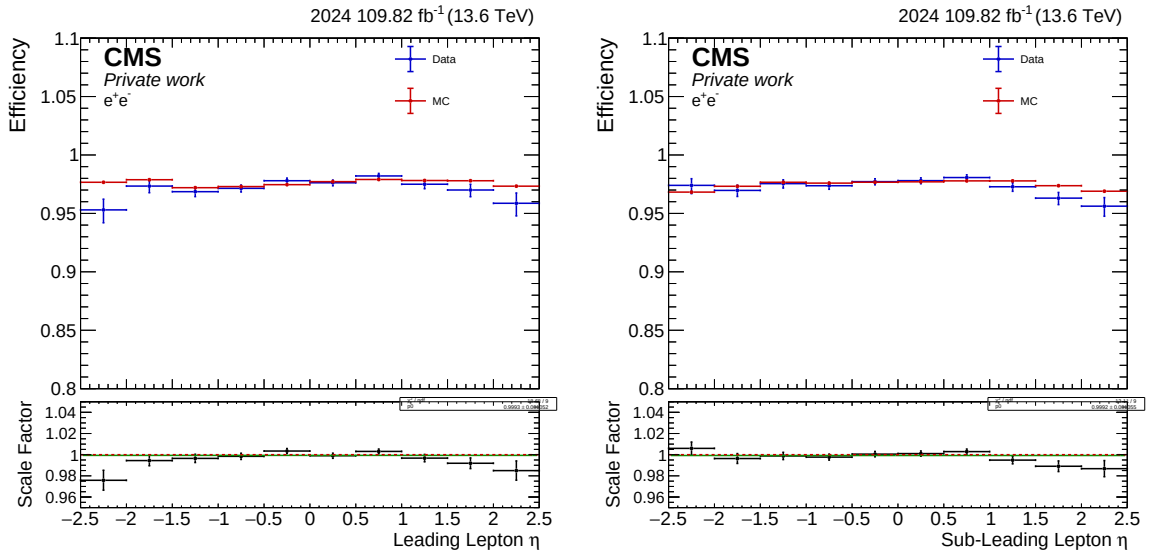


Figure 19: Additional one-dimensional efficiency and SF diagnostics in the *ee* channel. Variables and binning are shown on each axis.

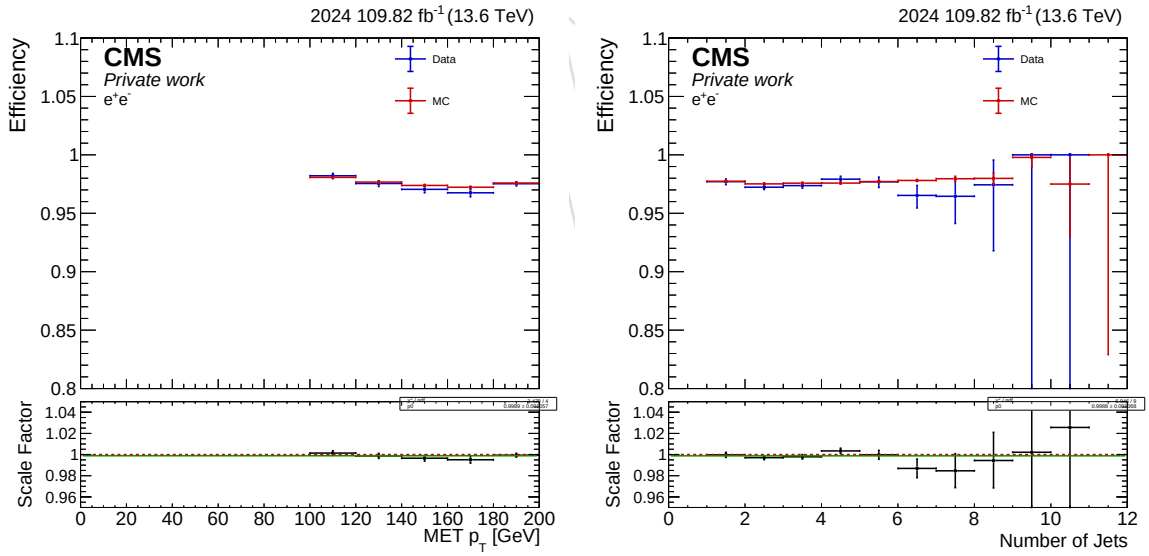


Figure 20: Additional one-dimensional efficiency and SF diagnostics in the *ee* channel. Variables and binning are shown on each axis.

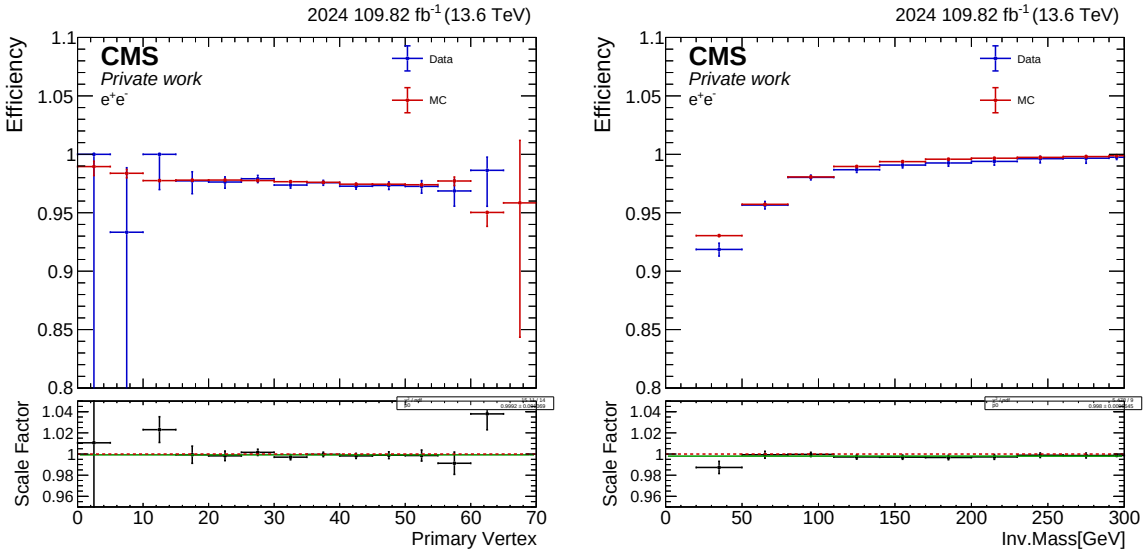


Figure 21: Additional one-dimensional efficiency and SF diagnostics in the *ee* channel. Variables and binning are shown on each axis.

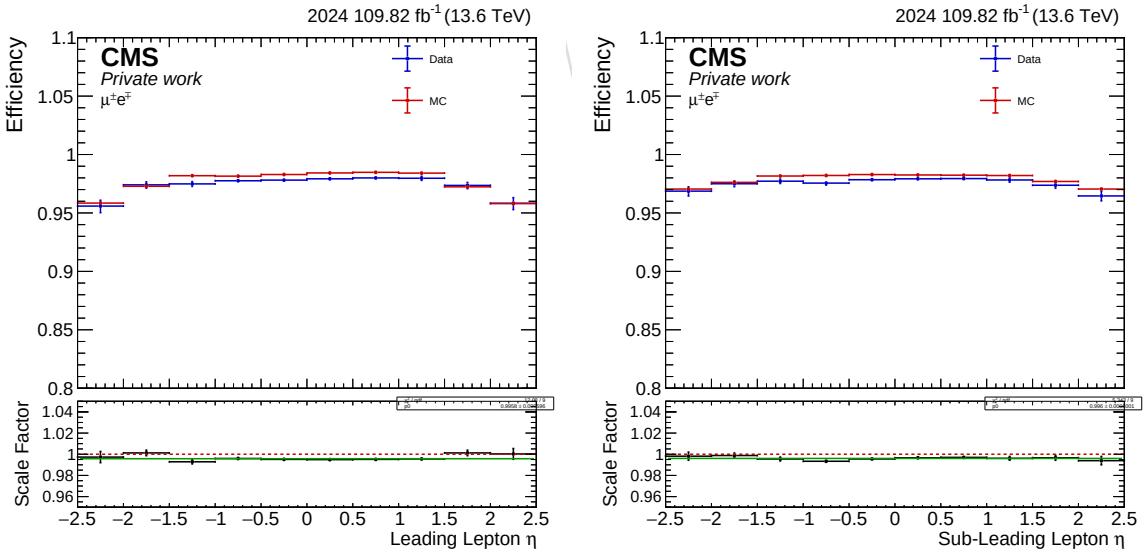


Figure 22: Additional one-dimensional efficiency and SF diagnostics in the *eμ* channel. Variables and binning are shown on each axis.

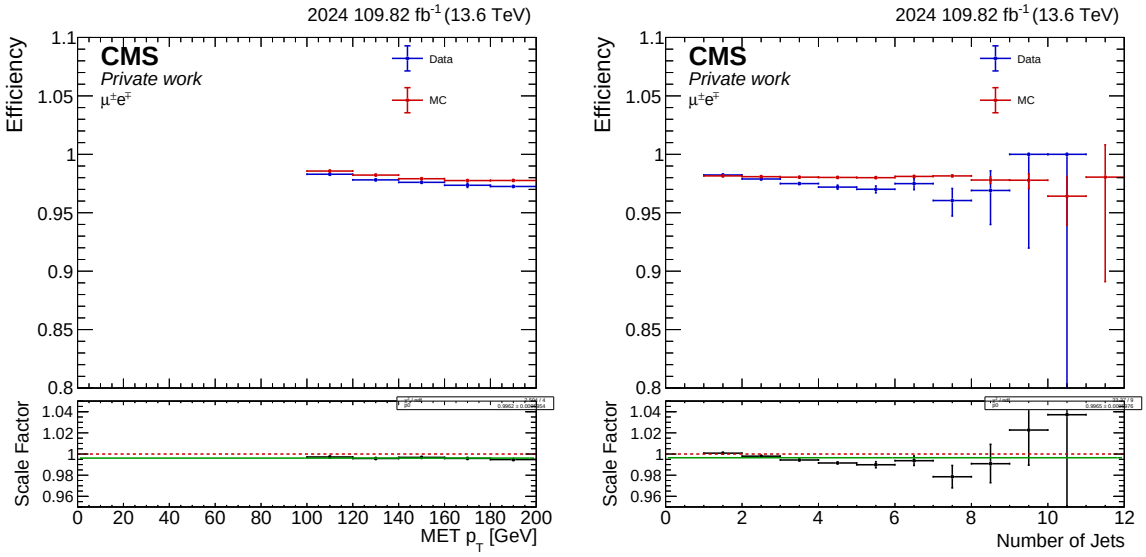


Figure 23: Additional one-dimensional efficiency and SF diagnostics in the $e\mu$ channel. Variables and binning are shown on each axis.

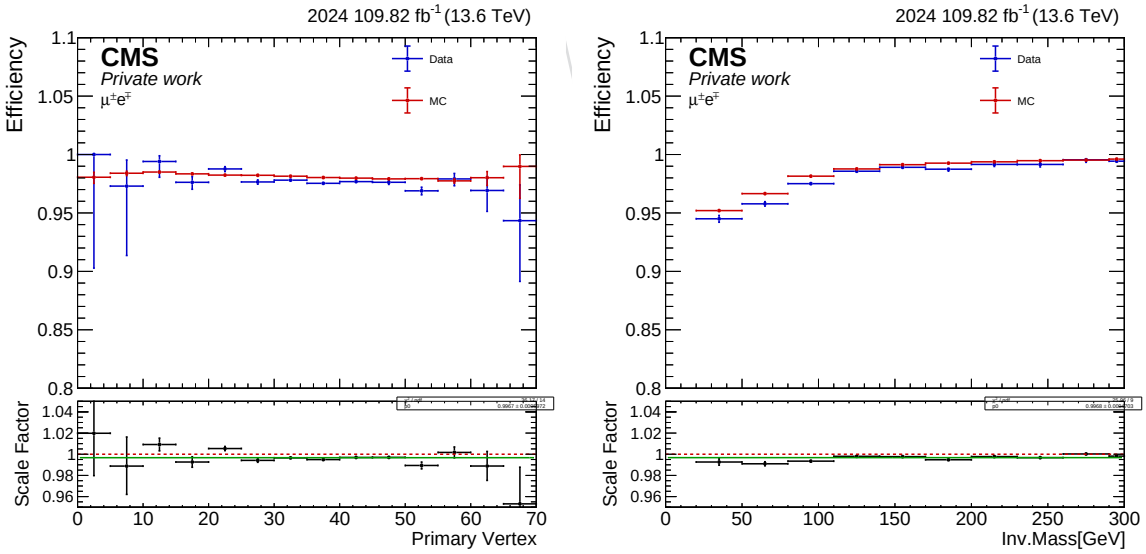


Figure 24: Additional one-dimensional efficiency and SF diagnostics in the $e\mu$ channel. Variables and binning are shown on each axis.

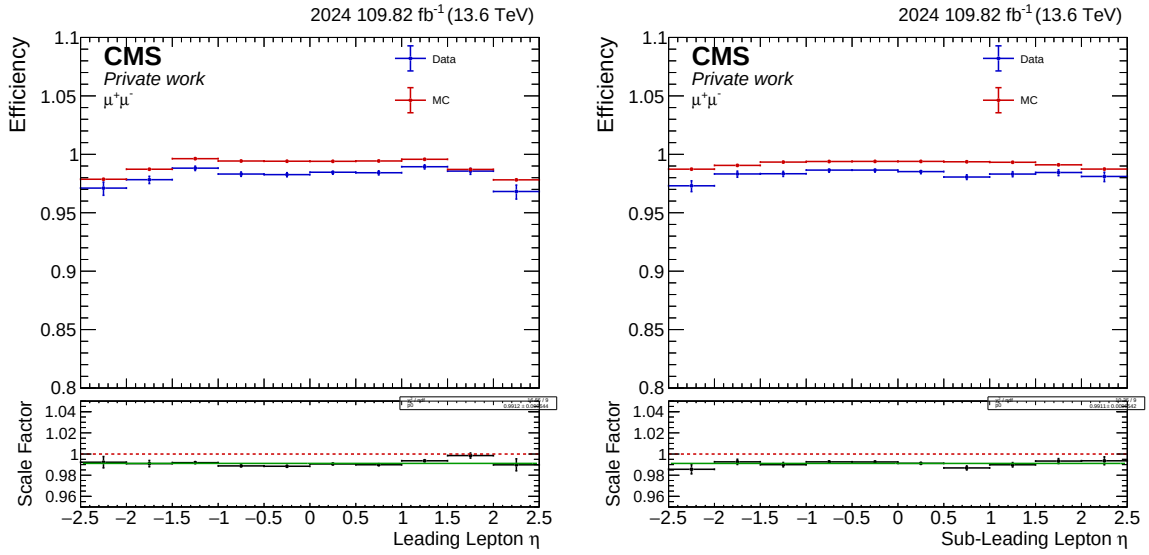


Figure 25: Additional one-dimensional efficiency and SF diagnostics in the $\mu\mu$ channel. Variables and binning are shown on each axis.

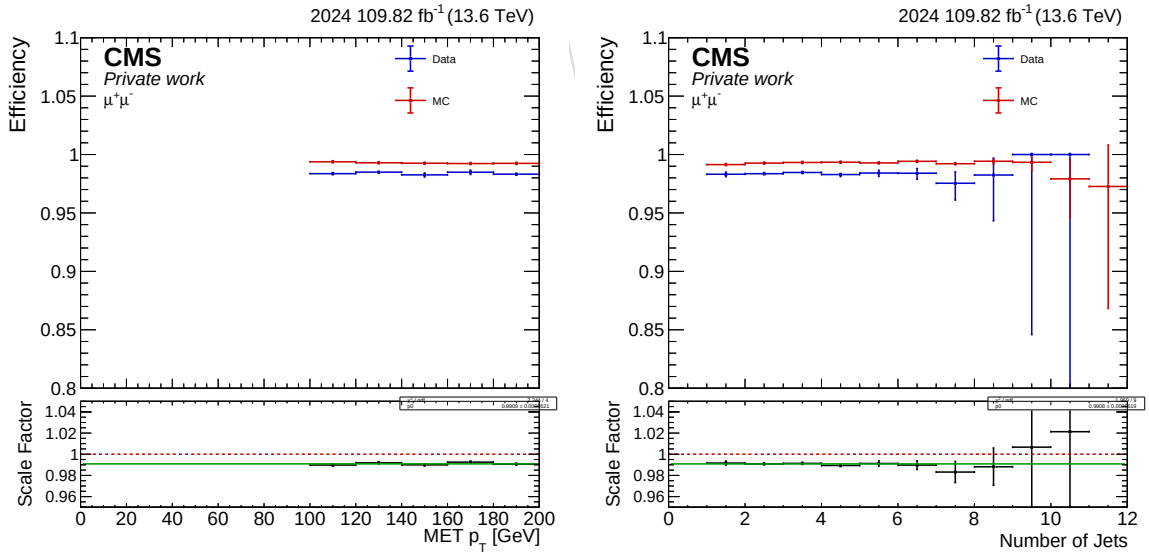


Figure 26: Additional one-dimensional efficiency and SF diagnostics in the $\mu\mu$ channel. Variables and binning are shown on each axis.

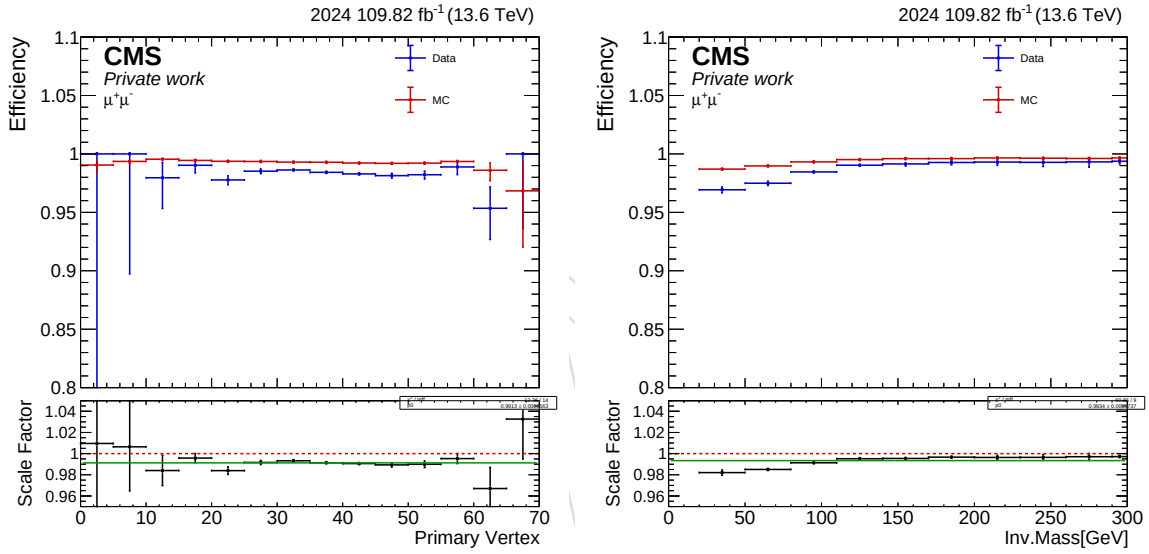


Figure 27: Additional one-dimensional efficiency and SF diagnostics in the $\mu\mu$ channel. Variables and binning are shown on each axis.

B 2024: detailed uncertainty studies

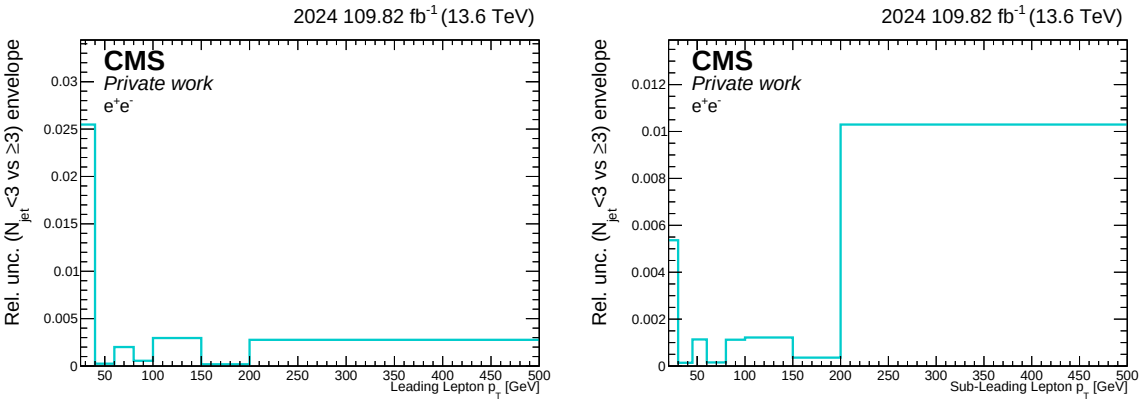


Figure 28: Jet Topology Uncertainty diagnostics in the *ee* channel versus leading and subleading lepton momentum. Curves are reproduced with the production definitions; their exact numerical prescription requires confirmation.

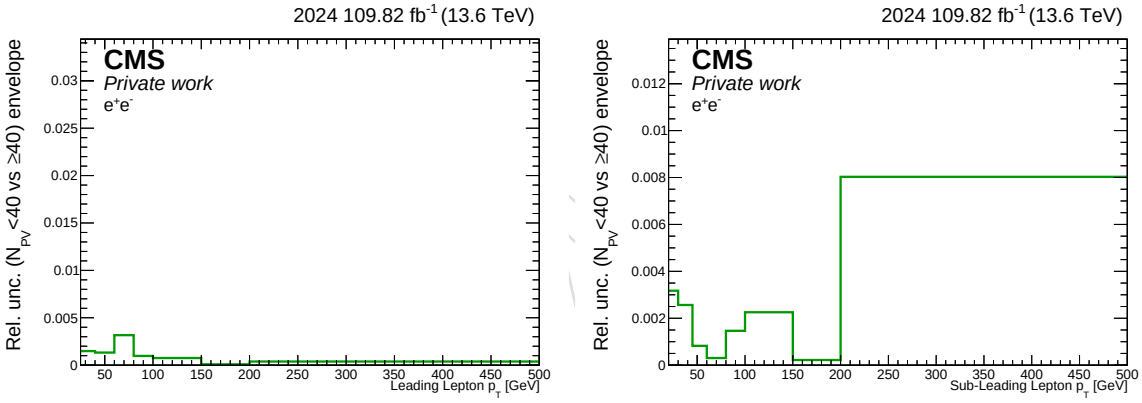


Figure 29: PV Topology Uncertainty diagnostics in the *ee* channel versus leading and subleading lepton momentum. Curves are reproduced with the production definitions; their exact numerical prescription requires confirmation.

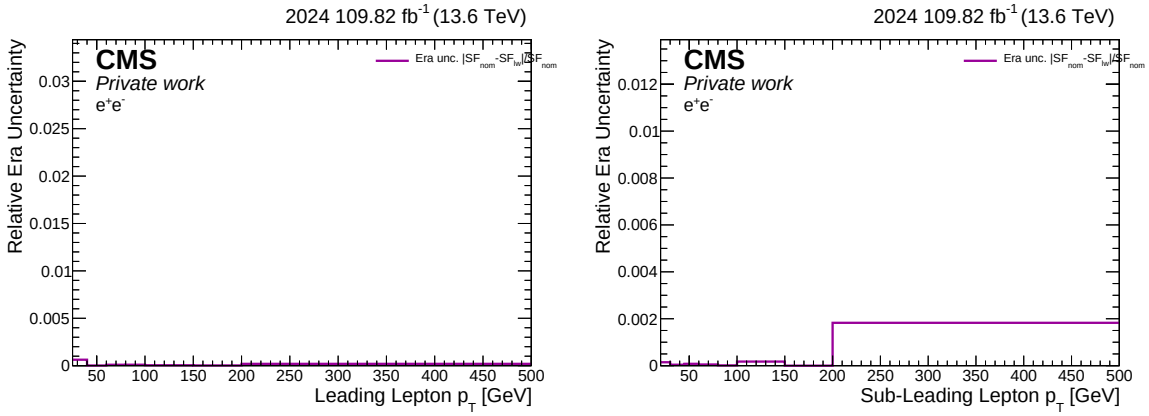


Figure 30: Era Uncertainty diagnostics in the *ee* channel versus leading and subleading lepton momentum. Curves are reproduced with the production definitions; their exact numerical prescription requires confirmation.

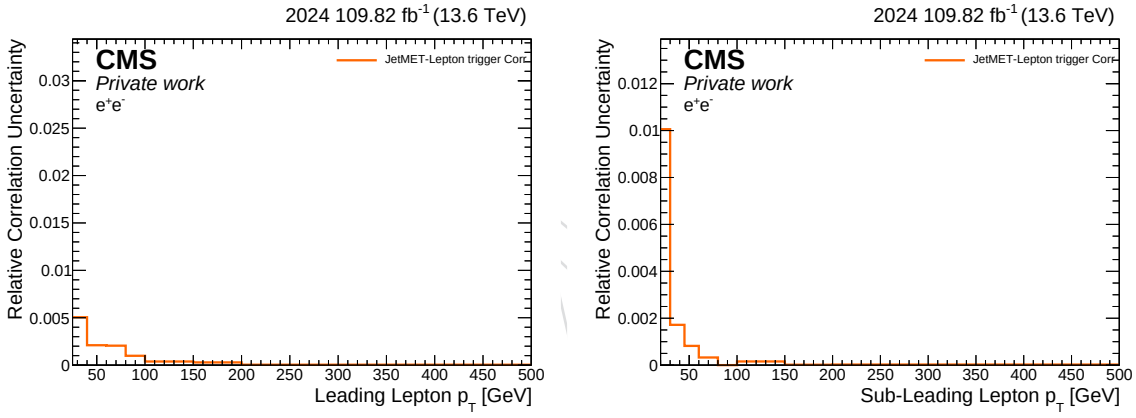


Figure 31: Correlation Uncertainty diagnostics in the *ee* channel versus leading and subleading lepton momentum. Curves are reproduced with the production definitions; their exact numerical prescription requires confirmation.

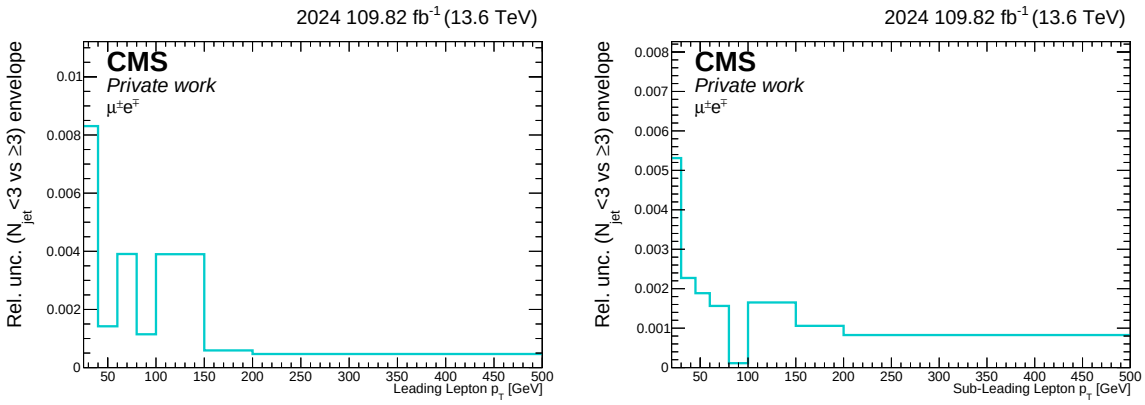


Figure 32: Jet Topology Uncertainty diagnostics in the $e\mu$ channel versus leading and subleading lepton momentum. Curves are reproduced with the production definitions; their exact numerical prescription requires confirmation.

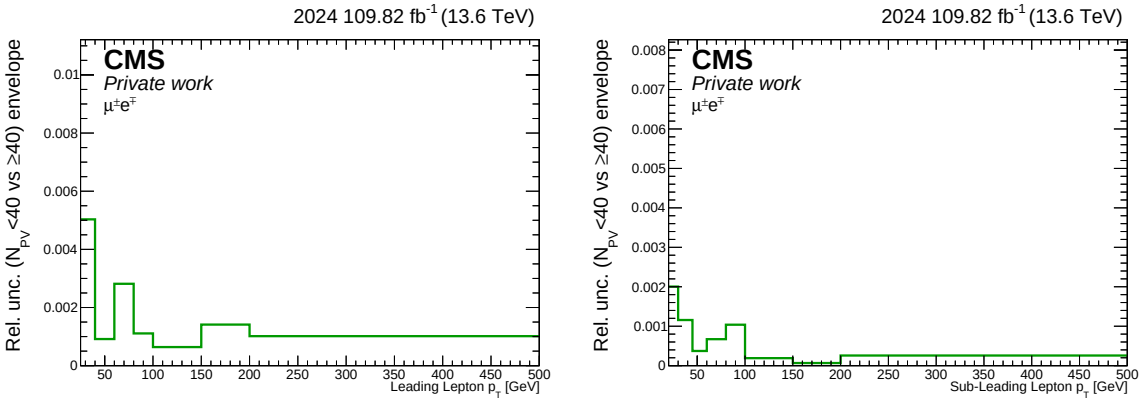


Figure 33: PV Topology Uncertainty diagnostics in the $e\mu$ channel versus leading and subleading lepton momentum. Curves are reproduced with the production definitions; their exact numerical prescription requires confirmation.

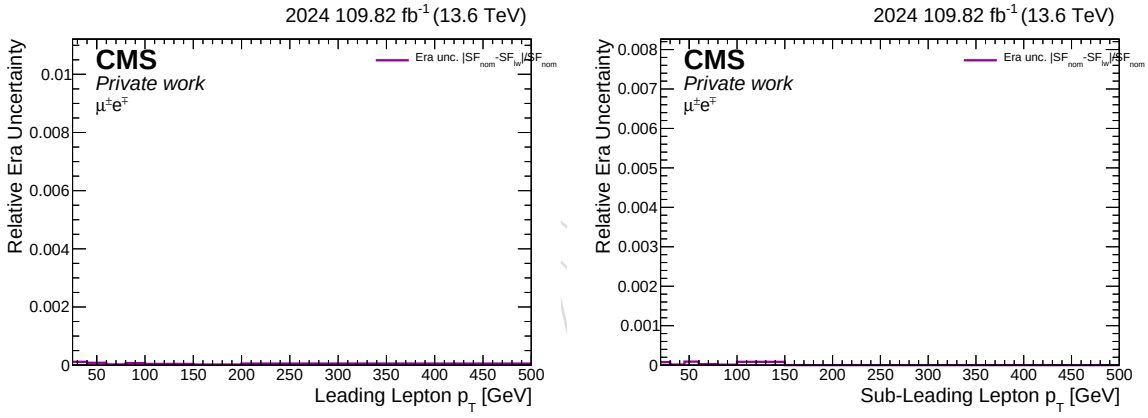


Figure 34: Era Uncertainty diagnostics in the $e\mu$ channel versus leading and subleading lepton momentum. Curves are reproduced with the production definitions; their exact numerical prescription requires confirmation.

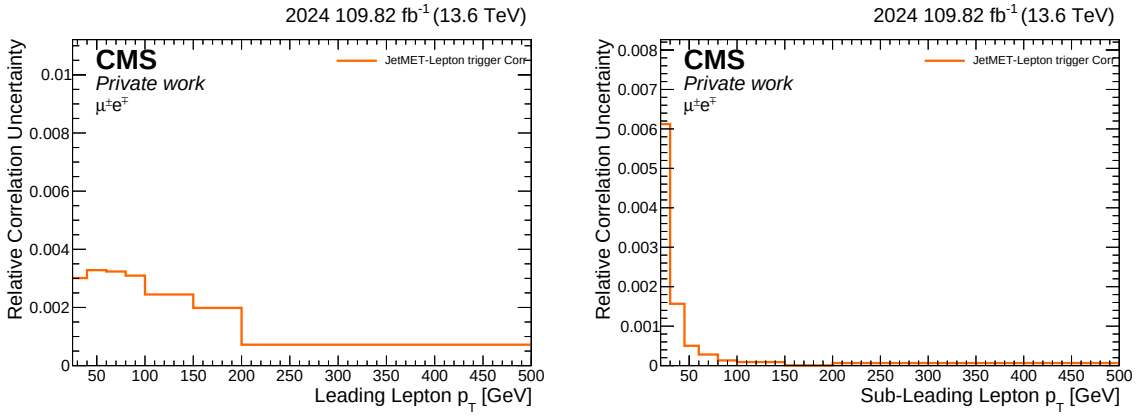


Figure 35: Correlation Uncertainty diagnostics in the $e\mu$ channel versus leading and subleading lepton momentum. Curves are reproduced with the production definitions; their exact numerical prescription requires confirmation.

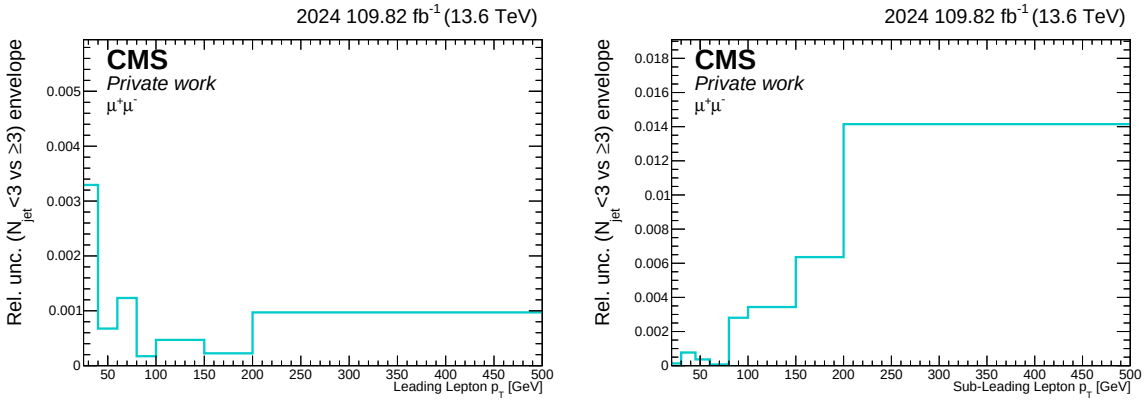


Figure 36: Jet Topology Uncertainty diagnostics in the $\mu\mu$ channel versus leading and subleading lepton momentum. Curves are reproduced with the production definitions; their exact numerical prescription requires confirmation.

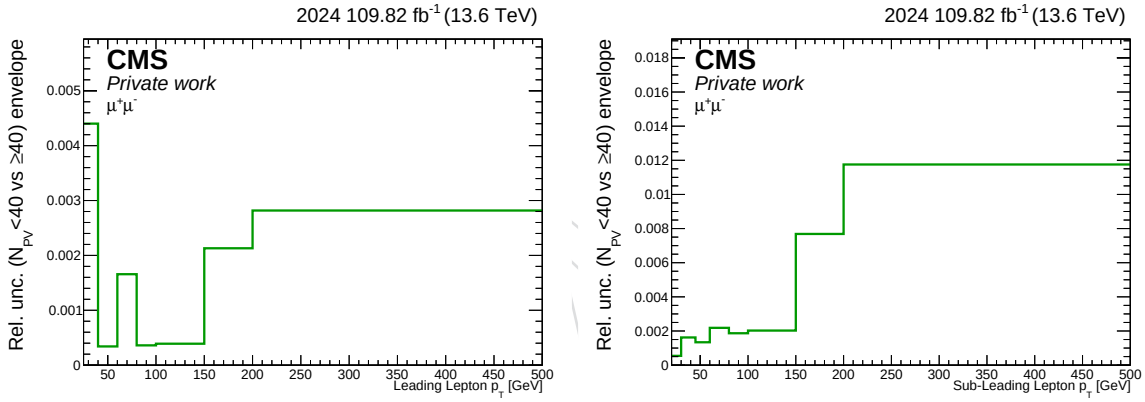


Figure 37: PV Topology Uncertainty diagnostics in the $\mu\mu$ channel versus leading and subleading lepton momentum. Curves are reproduced with the production definitions; their exact numerical prescription requires confirmation.

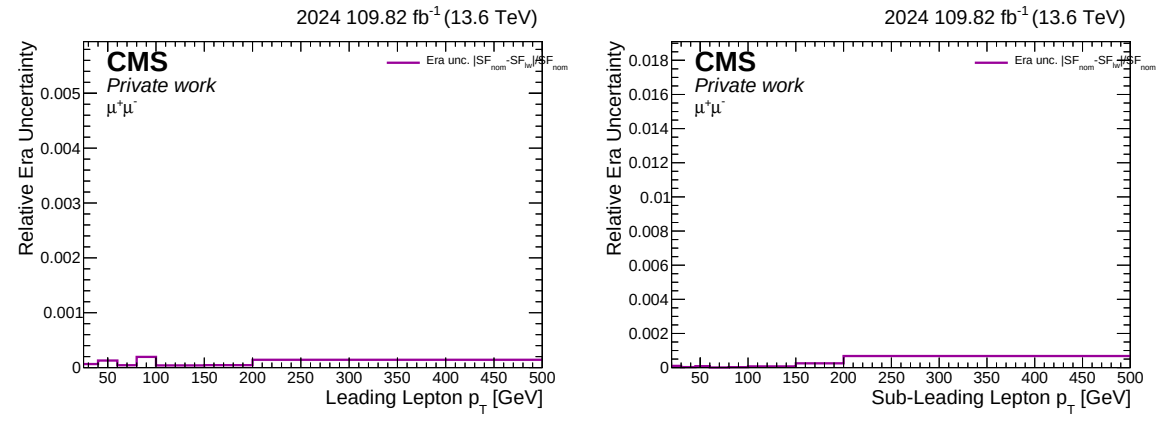


Figure 38: Era Uncertainty diagnostics in the $\mu\mu$ channel versus leading and subleading lepton momentum. Curves are reproduced with the production definitions; their exact numerical prescription requires confirmation.

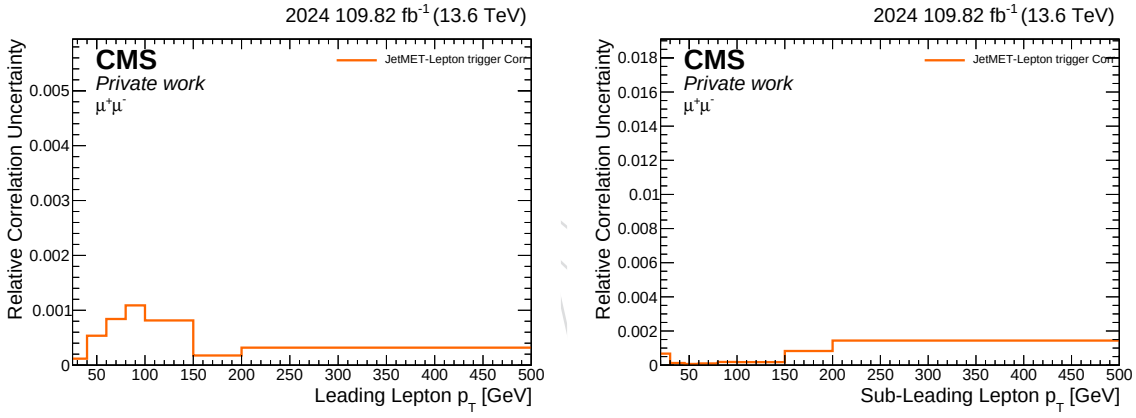


Figure 39: Correlation Uncertainty diagnostics in the $\mu\mu$ channel versus leading and subleading lepton momentum. Curves are reproduced with the production definitions; their exact numerical prescription requires confirmation.

C Selected 2024 supporting distributions

The following distributions document the efficiency inputs, statistical precision, and uncertainty composition for the 2024 measurement. Only inclusive distributions directly related to the correction or its principal checks are selected. The jet- and vertex-split variants and the full set of individual-jet and angular distributions are not reproduced. These plots use the measurement sample and are not independent closure tests.

C.1 *ee* channel

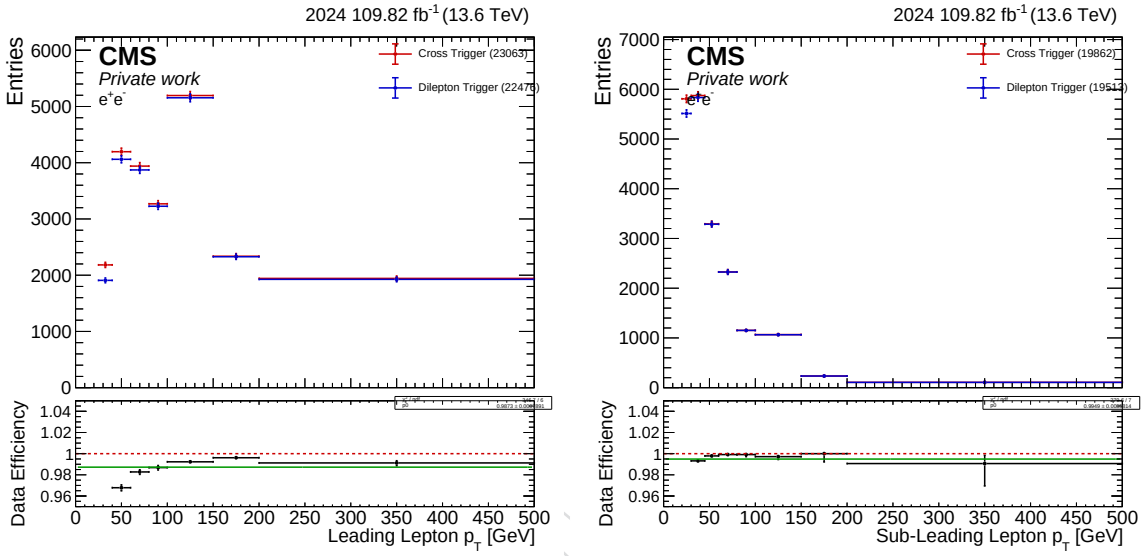


Figure 40: Data trigger numerator and reference-sample distributions (upper panels), with their efficiency ratios (lower panels) in the *ee* channel. Left and right: leading and subleading lepton p_{T}.

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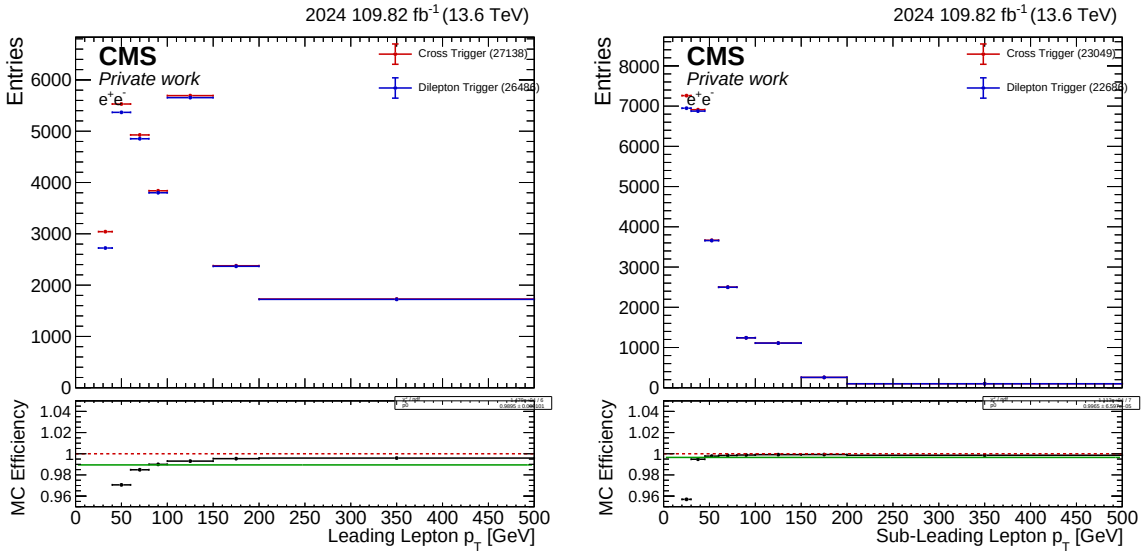


Figure 41: Simulated trigger numerator and reference-sample distributions (upper panels), with their efficiency ratios (lower panels) in the *ee* channel. Left and right: leading and subleading lepton p_{T}.

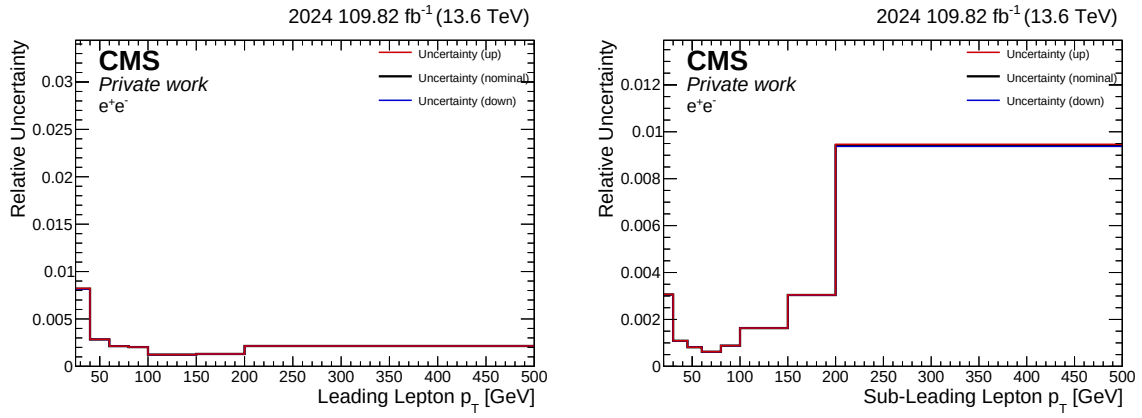


Figure 42: Relative statistical uncertainty diagnostics, retaining the nominal, upward, and downward curves of the supplied production in the *ee* channel. Left and right: leading and subleading lepton p_{T}.

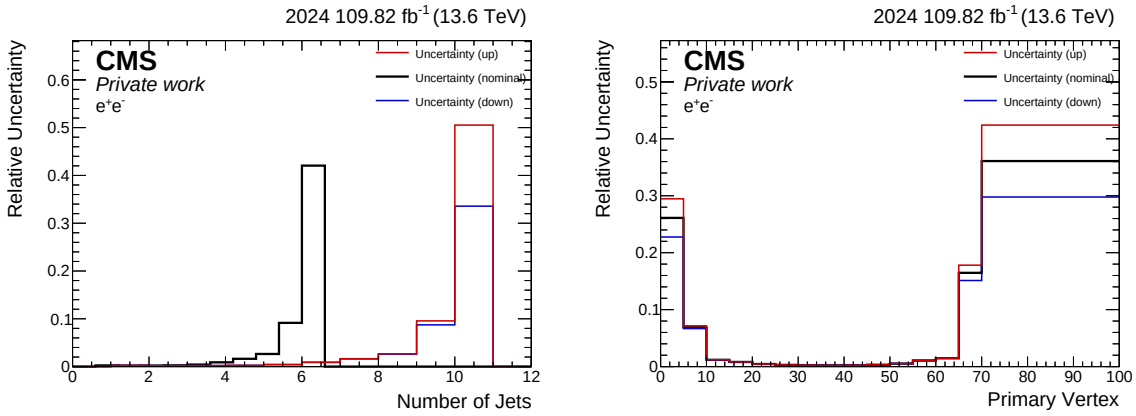


Figure 43: Relative statistical uncertainty diagnostics, retaining the nominal, upward, and downward curves of the supplied production in the *ee* channel. Left and right: jet and primary-vertex multiplicities.

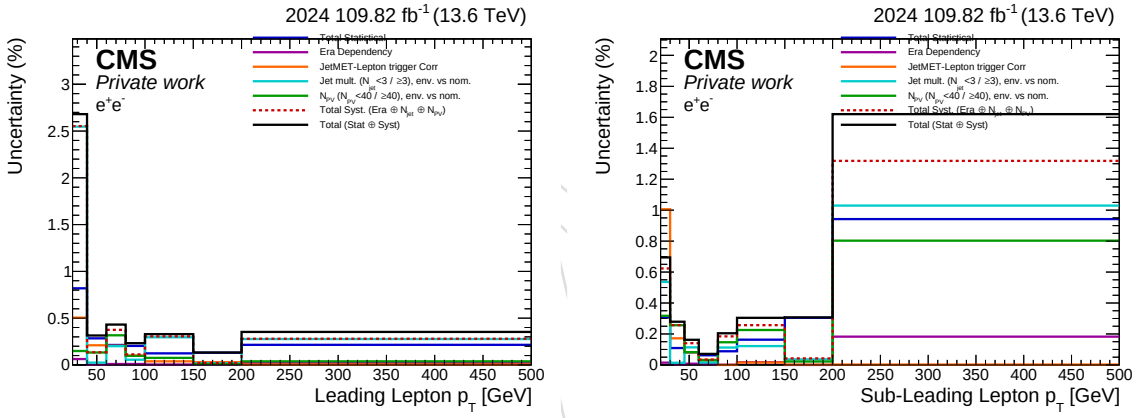


Figure 44: Absolute uncertainty component breakdown in the *ee* channel. Left and right: leading and subleading lepton p_T.

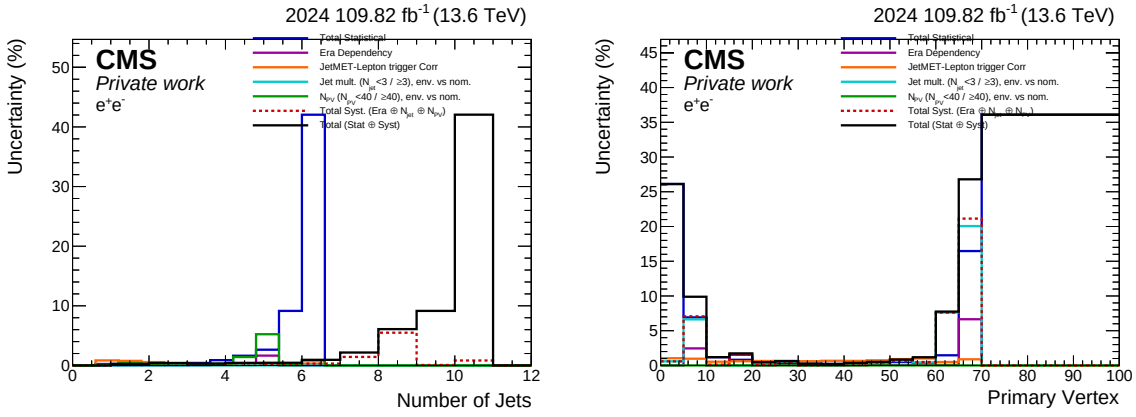


Figure 45: Absolute uncertainty component breakdown in the *ee* channel. Left and right: jet and primary-vertex multiplicities.

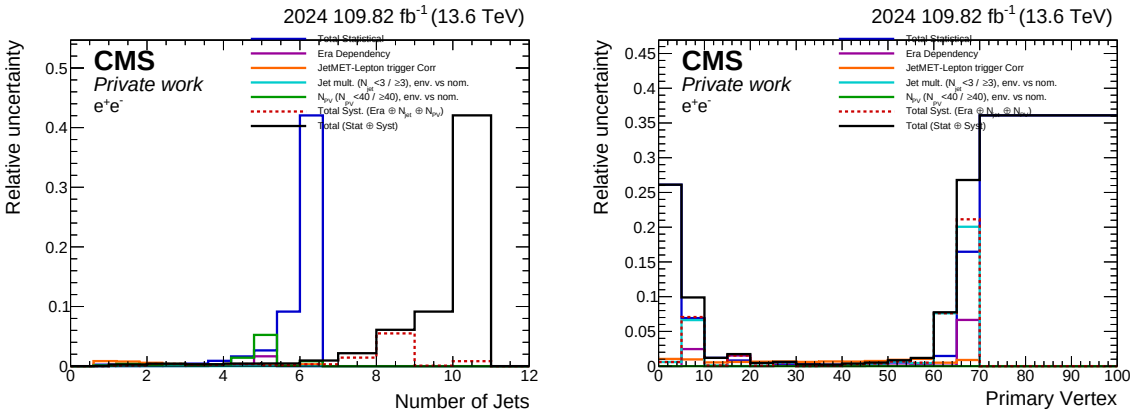


Figure 46: Relative uncertainty component breakdown versus event activity in the *ee* channel. Left and right: jet and primary-vertex multiplicities.

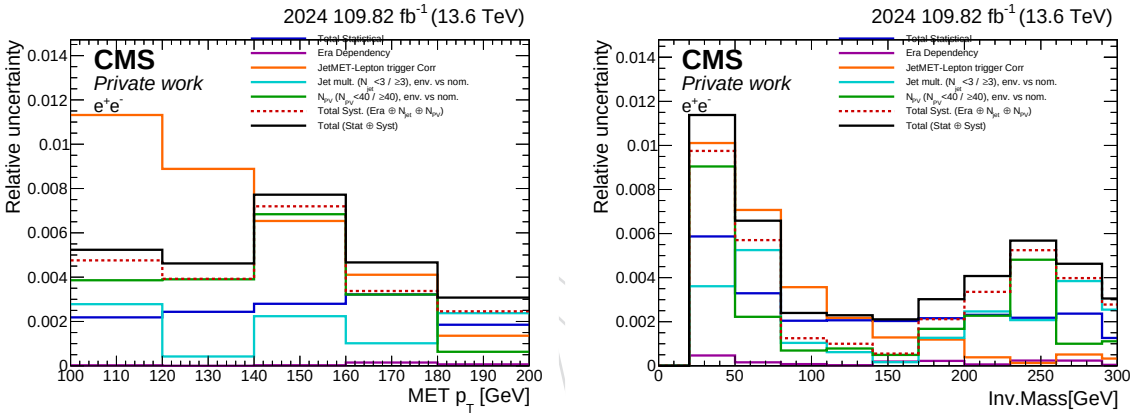


Figure 47: Relative uncertainty component breakdown versus event kinematics in the *ee* channel. Left and right: missing transverse momentum and dilepton invariant mass.

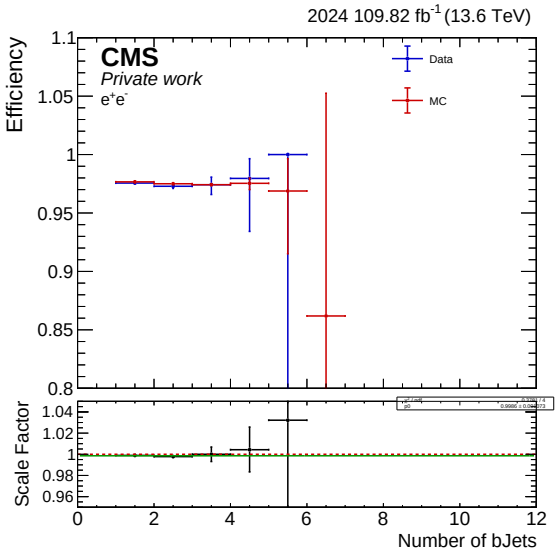


Figure 48: Efficiency comparison and scale factor versus heavy-flavour activity in the *ee* channel. *b*-tagged jet multiplicity.

C.2 $e\mu$ channel

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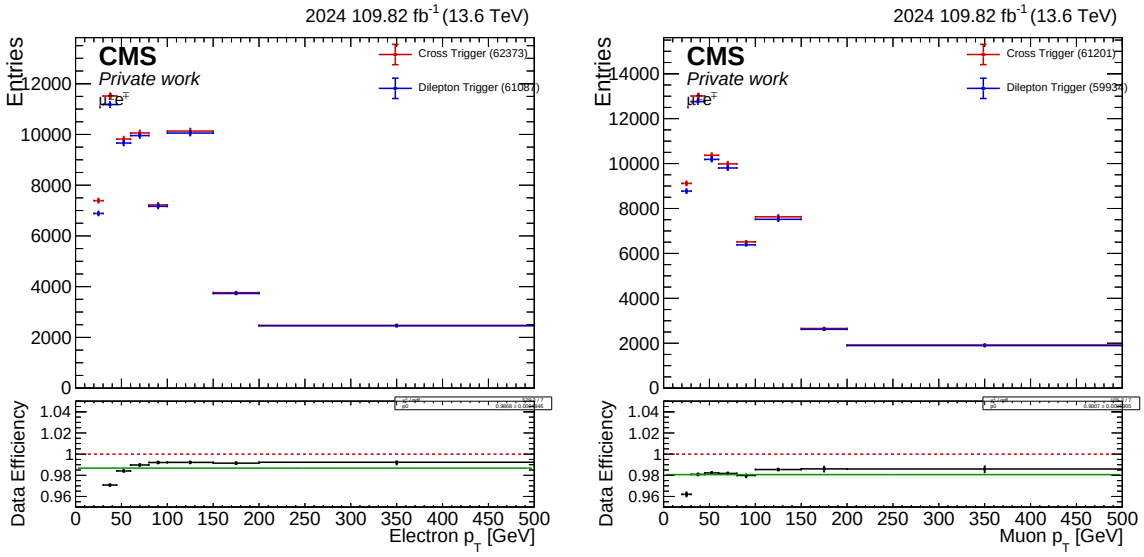


Figure 49: Data trigger numerator and reference-sample distributions (upper panels), with their efficiency ratios (lower panels) in the $e\mu$ channel. Left and right: electron and muon p_T.

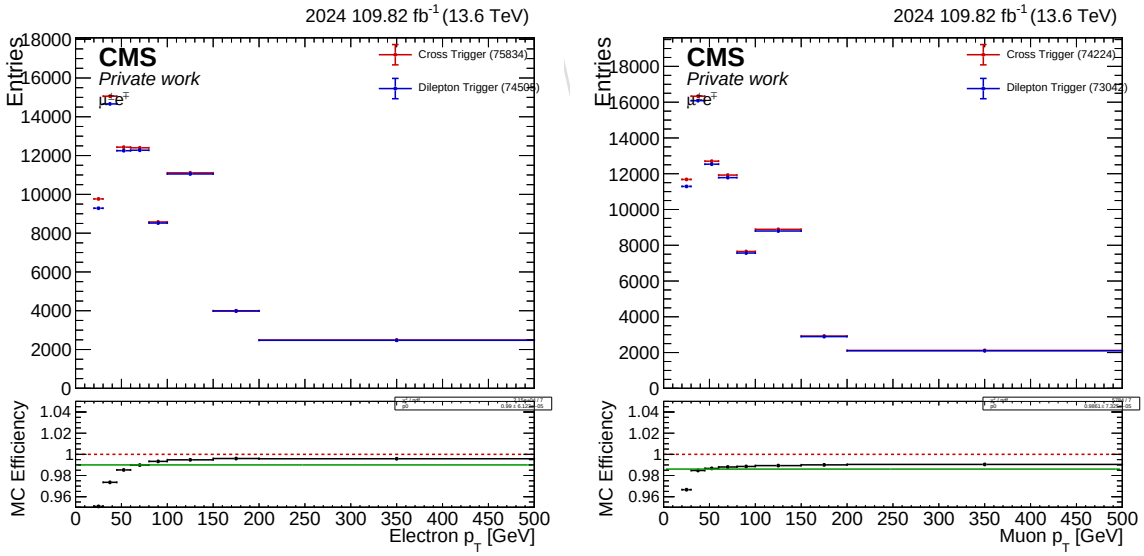


Figure 50: Simulated trigger numerator and reference-sample distributions (upper panels), with their efficiency ratios (lower panels) in the $e\mu$ channel. Left and right: electron and muon p_T.

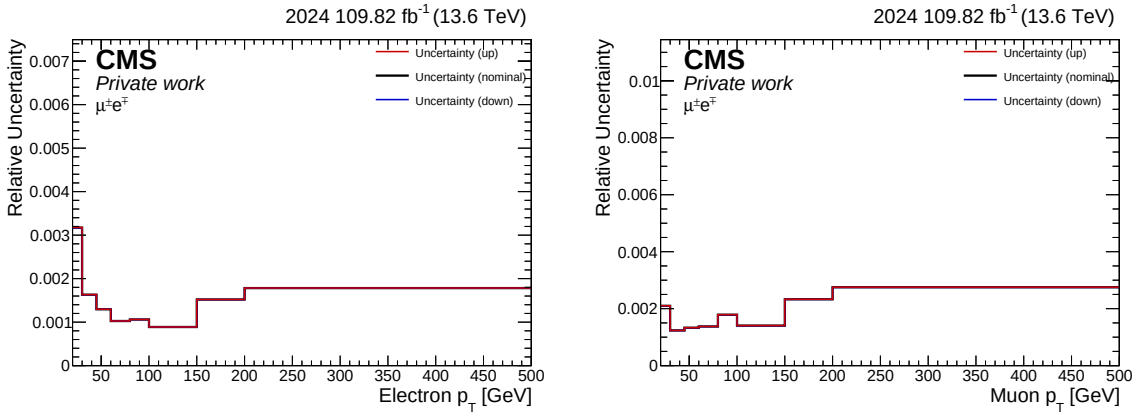


Figure 51: Relative statistical uncertainty diagnostics, retaining the nominal, upward, and downward curves of the supplied production in the $e\mu$ channel. Left and right: electron and muon p_{T}.

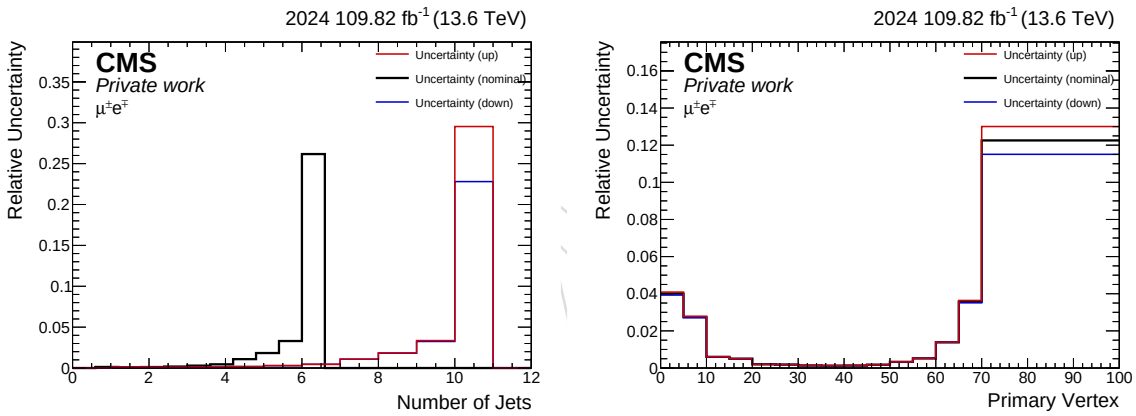


Figure 52: Relative statistical uncertainty diagnostics, retaining the nominal, upward, and downward curves of the supplied production in the $e\mu$ channel. Left and right: jet and primary-vertex multiplicities.

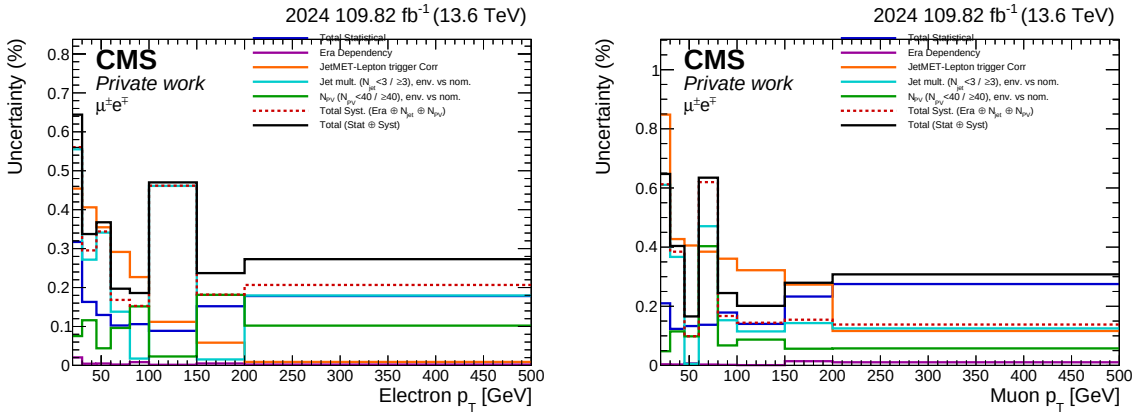


Figure 53: Absolute uncertainty component breakdown in the $e\mu$ channel. Left and right: electron and muon p_{T}.

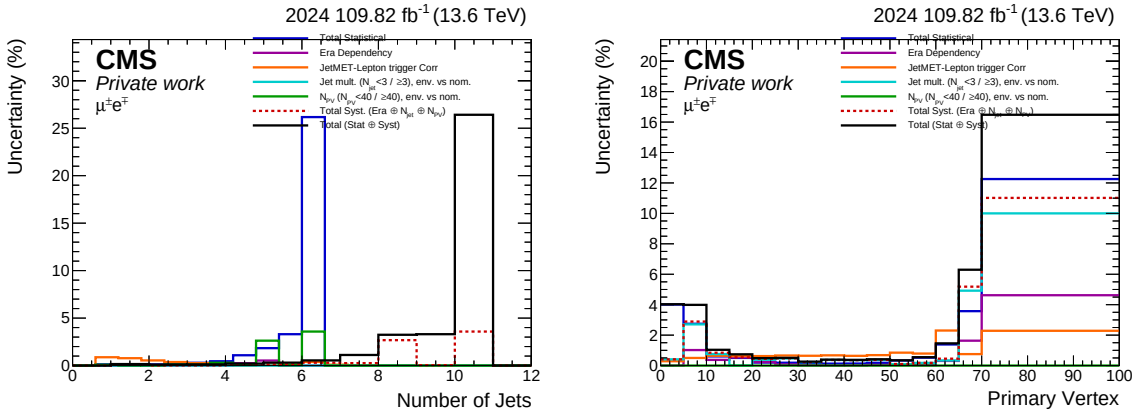


Figure 54: Absolute uncertainty component breakdown in the $e\mu$ channel. Left and right: jet and primary-vertex multiplicities.

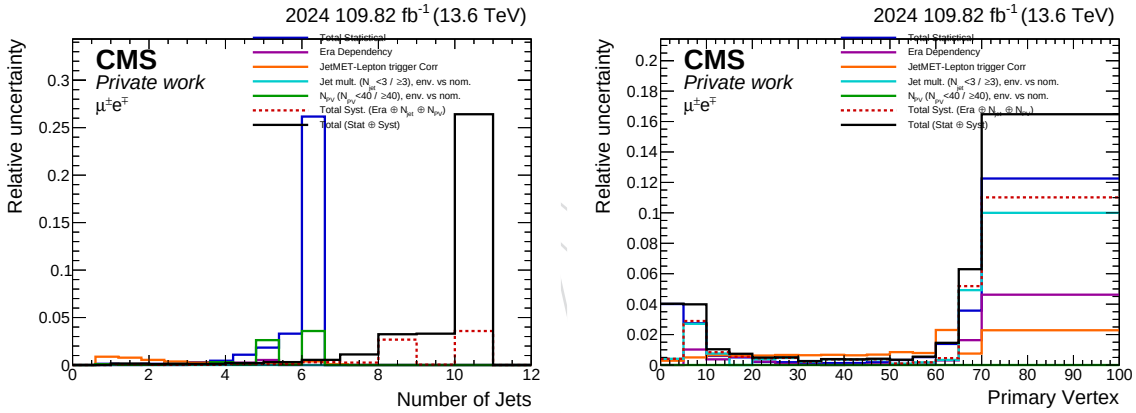


Figure 55: Relative uncertainty component breakdown versus event activity in the $e\mu$ channel. Left and right: jet and primary-vertex multiplicities.

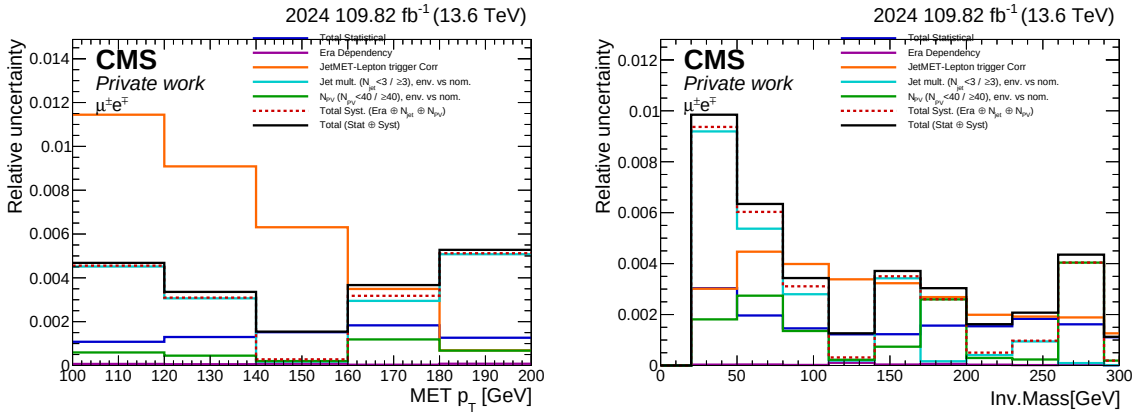


Figure 56: Relative uncertainty component breakdown versus event kinematics in the $e\mu$ channel. Left and right: missing transverse momentum and dilepton invariant mass.

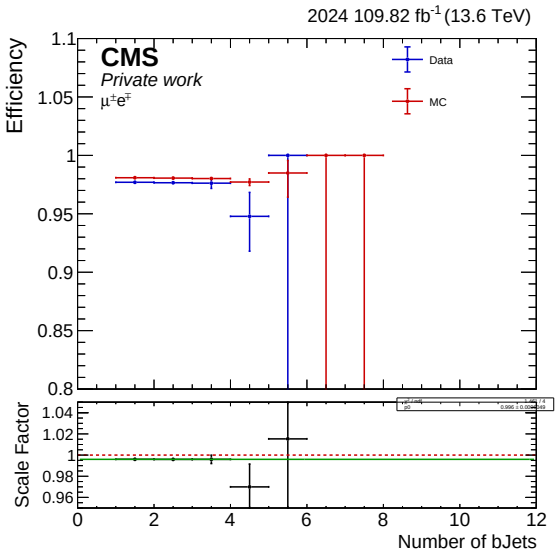


Figure 57: Efficiency comparison and scale factor versus heavy-flavour activity in the $e\mu$ channel. b-tagged jet multiplicity.

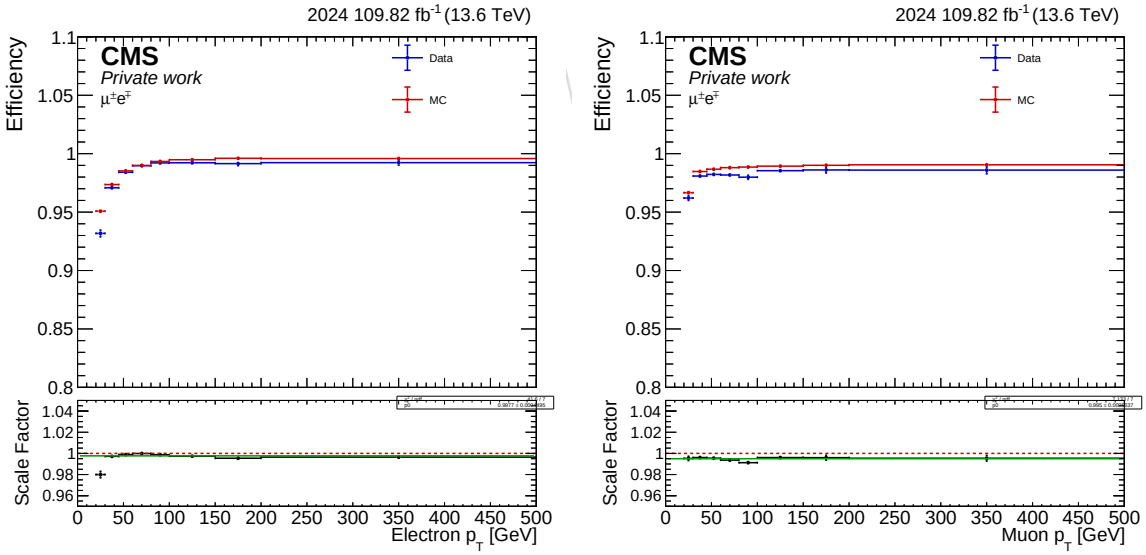


Figure 58: Efficiency comparison and scale factor in the flavour-specific momentum variables used by the two-dimensional correction in the $e\mu$ channel. Left and right: electron and muon p_{T}.

C.3 $\mu\mu$ channel

D 2025: additional kinematic dependence

These distributions extend the main momentum comparisons. They use the same measurement sample and do not constitute independent closure tests.

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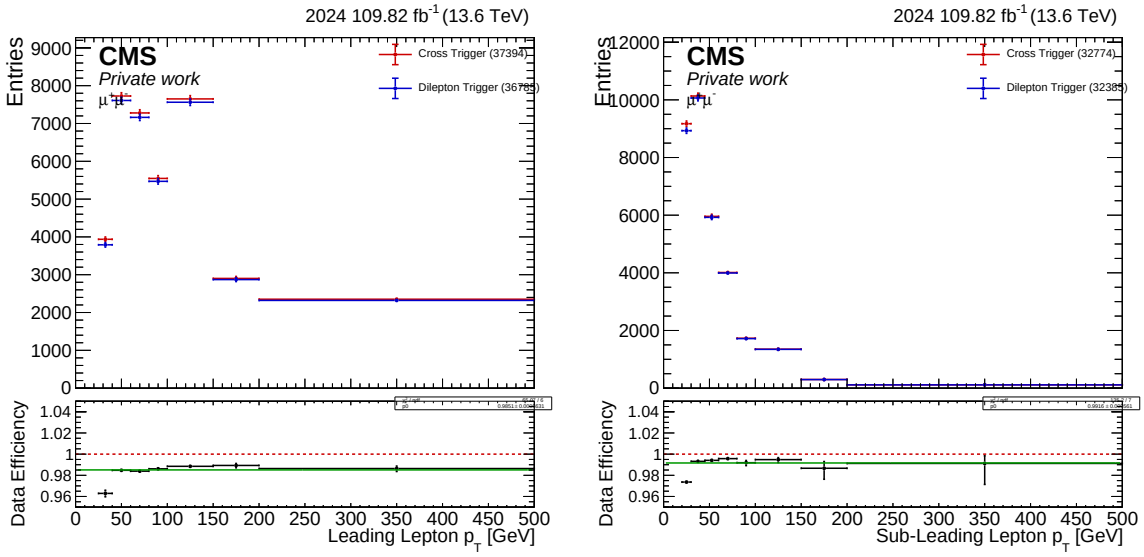


Figure 59: Data trigger numerator and reference-sample distributions (upper panels), with their efficiency ratios (lower panels) in the $\mu\mu$ channel. Left and right: leading and subleading lepton p_{T}.

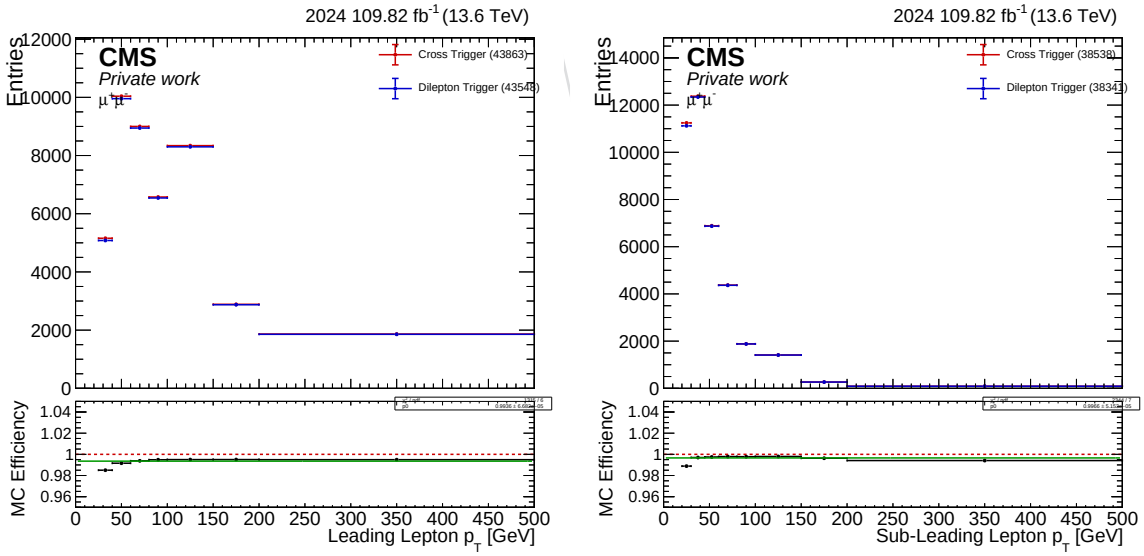


Figure 60: Simulated trigger numerator and reference-sample distributions (upper panels), with their efficiency ratios (lower panels) in the $\mu\mu$ channel. Left and right: leading and subleading lepton p_{T}.

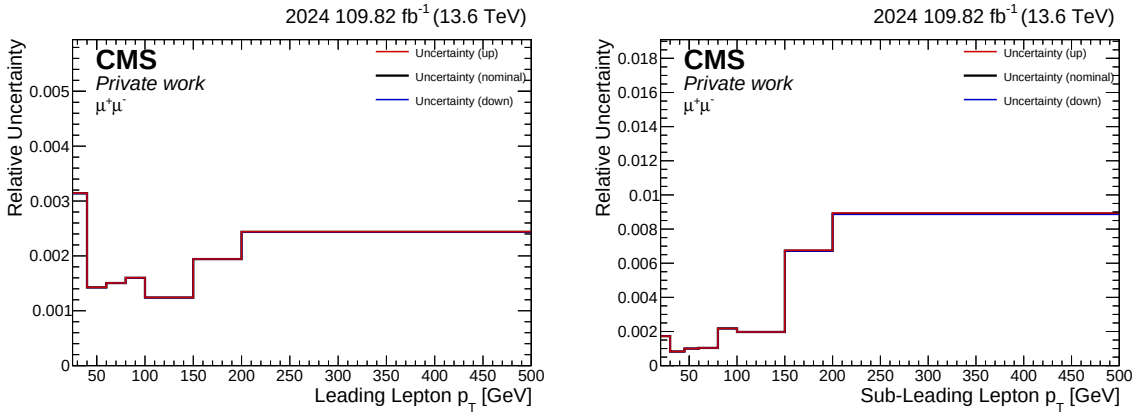


Figure 61: Relative statistical uncertainty diagnostics, retaining the nominal, upward, and downward curves of the supplied production in the $\mu\mu$ channel. Left and right: leading and subleading lepton p_{T}.

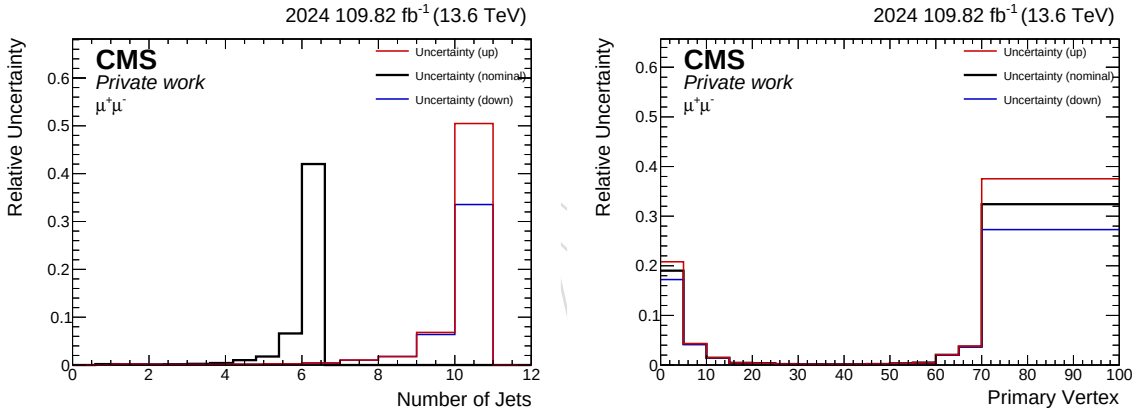


Figure 62: Relative statistical uncertainty diagnostics, retaining the nominal, upward, and downward curves of the supplied production in the $\mu\mu$ channel. Left and right: jet and primary-vertex multiplicities.

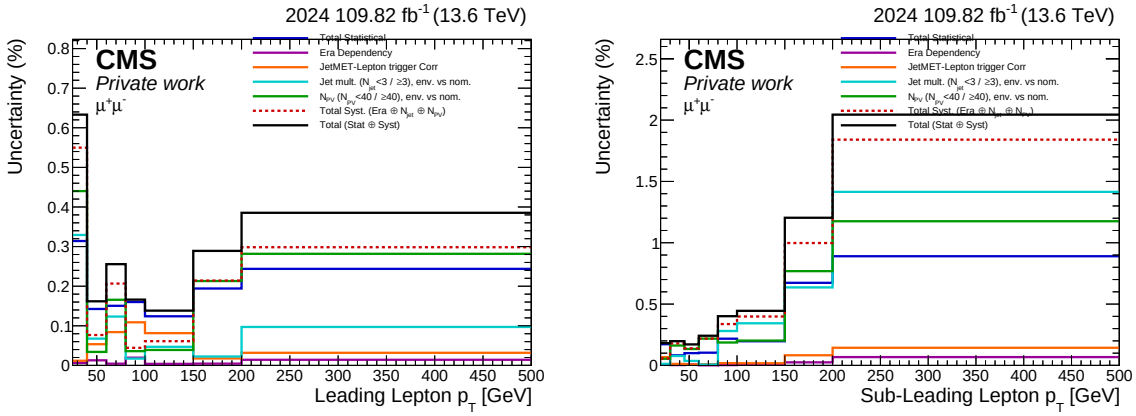


Figure 63: Absolute uncertainty component breakdown in the $\mu\mu$ channel. Left and right: leading and subleading lepton p_{T}.

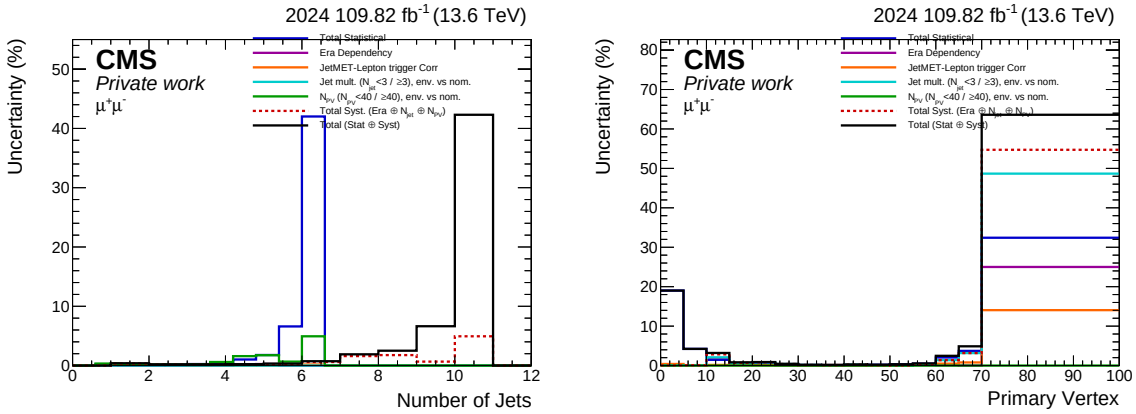


Figure 64: Absolute uncertainty component breakdown in the $\mu\mu$ channel. Left and right: jet and primary-vertex multiplicities.

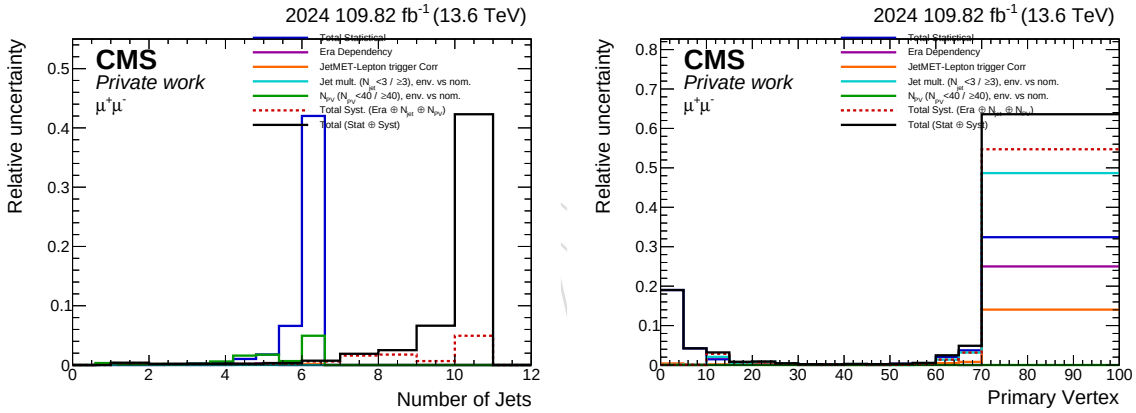


Figure 65: Relative uncertainty component breakdown versus event activity in the $\mu\mu$ channel. Left and right: jet and primary-vertex multiplicities.

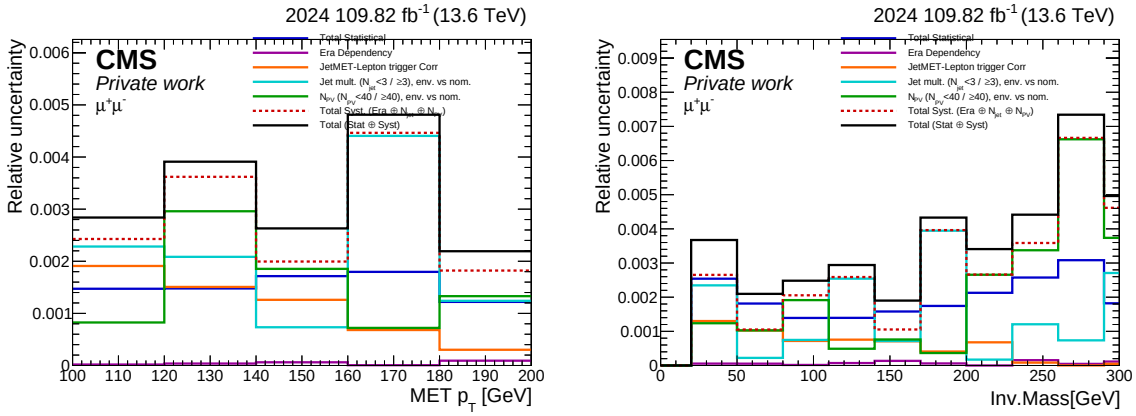


Figure 66: Relative uncertainty component breakdown versus event kinematics in the $\mu\mu$ channel. Left and right: missing transverse momentum and dilepton invariant mass.

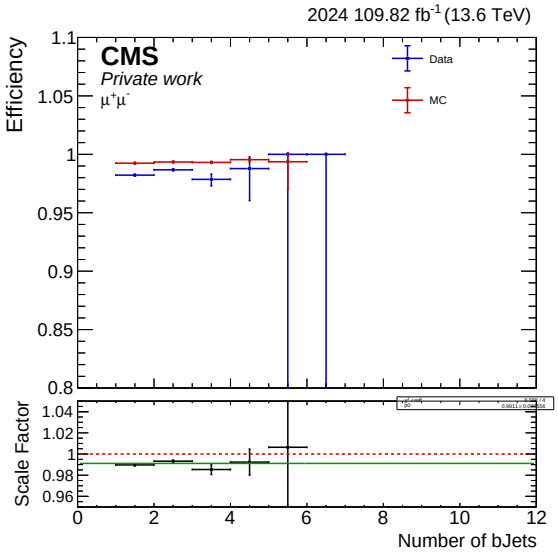


Figure 67: Efficiency comparison and scale factor versus heavy-flavour activity in the $\mu\mu$ channel. b-tagged jet multiplicity.

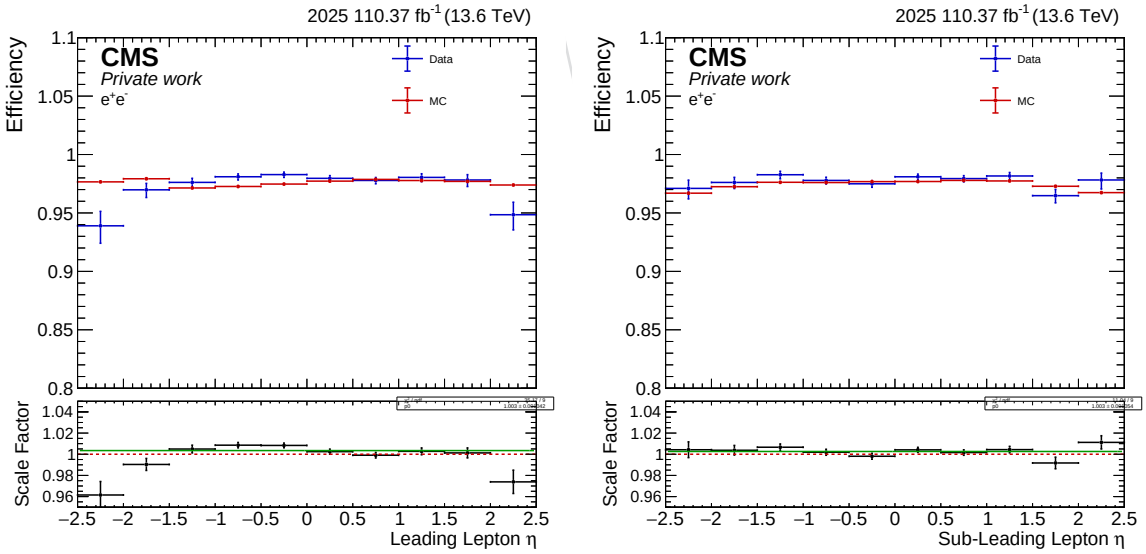


Figure 68: Additional one-dimensional efficiency and SF diagnostics in the ee channel. Variables and binning are shown on each axis.

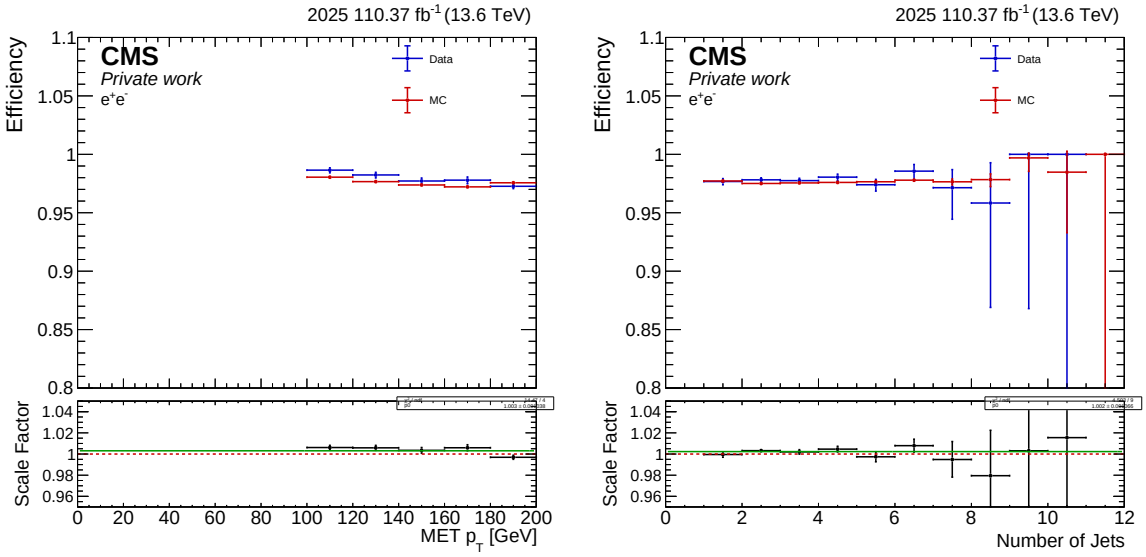


Figure 69: Additional one-dimensional efficiency and SF diagnostics in the *ee* channel. Variables and binning are shown on each axis.

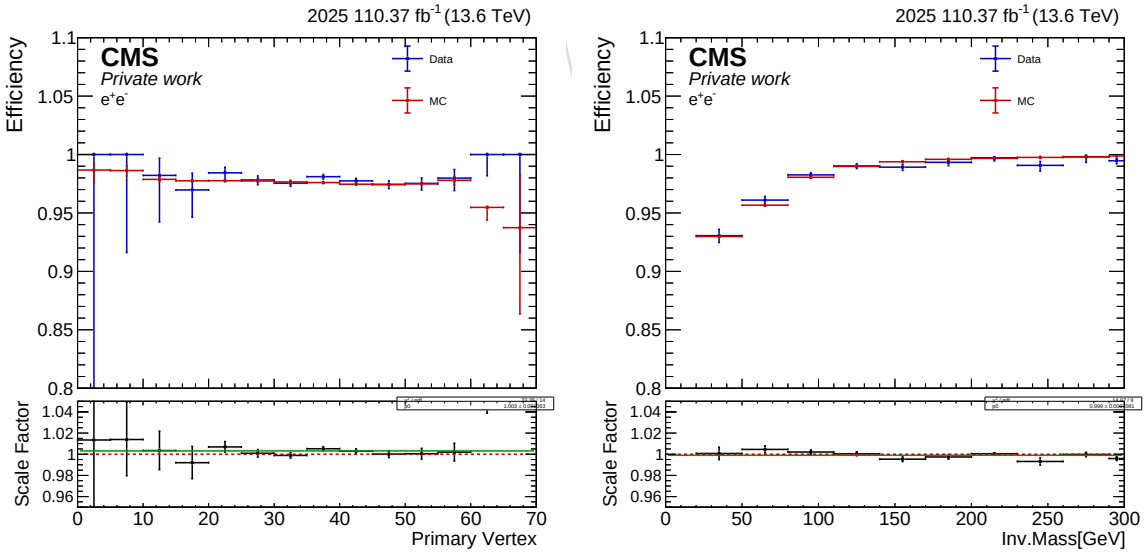


Figure 70: Additional one-dimensional efficiency and SF diagnostics in the *ee* channel. Variables and binning are shown on each axis.

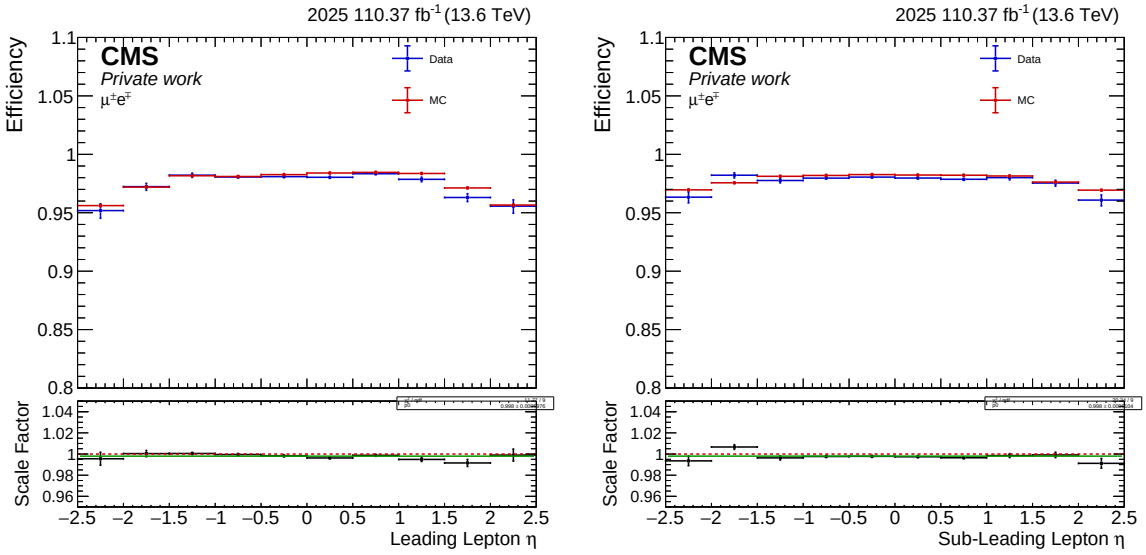


Figure 71: Additional one-dimensional efficiency and SF diagnostics in the $e\mu$ channel. Variables and binning are shown on each axis.

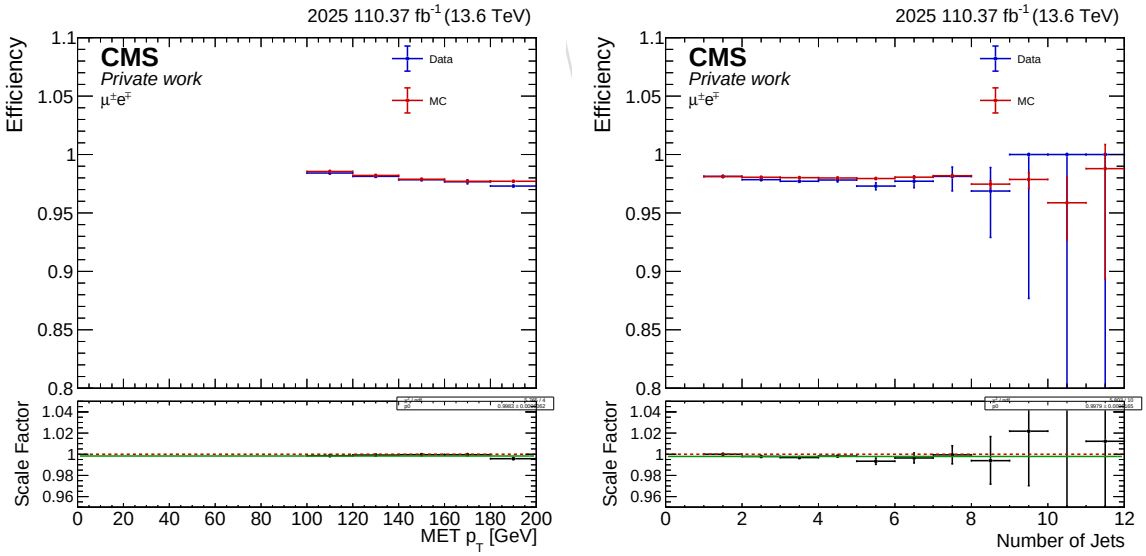


Figure 72: Additional one-dimensional efficiency and SF diagnostics in the $e\mu$ channel. Variables and binning are shown on each axis.

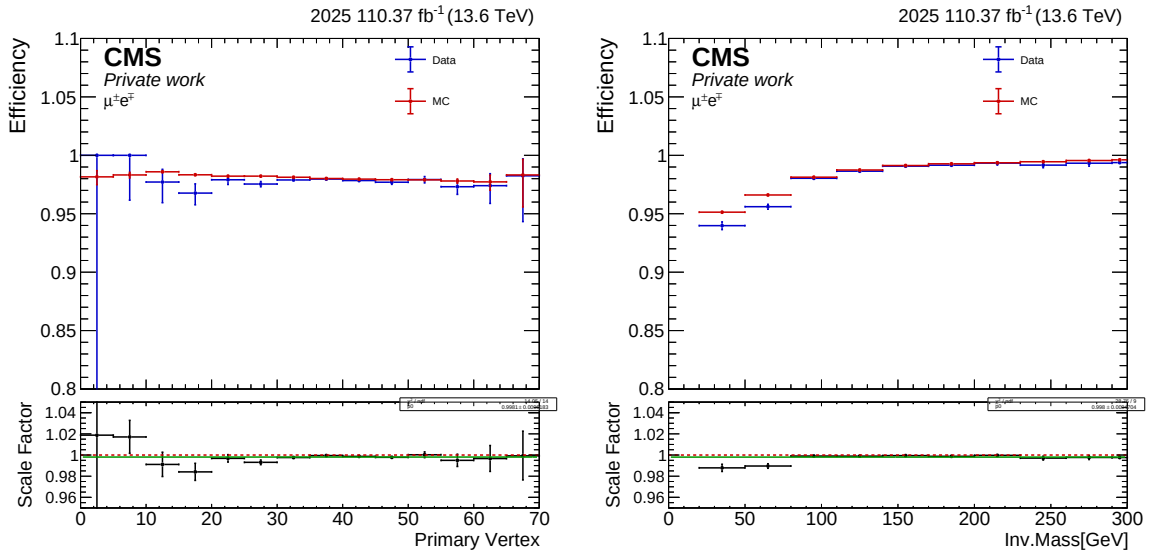


Figure 73: Additional one-dimensional efficiency and SF diagnostics in the $e\mu$ channel. Variables and binning are shown on each axis.

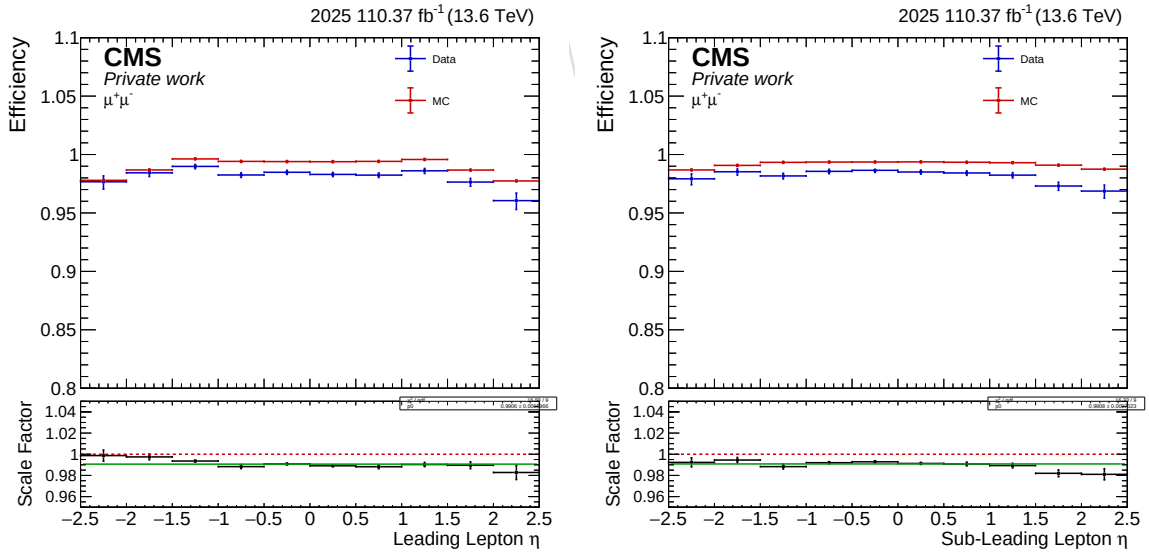


Figure 74: Additional one-dimensional efficiency and SF diagnostics in the $\mu\mu$ channel. Variables and binning are shown on each axis.

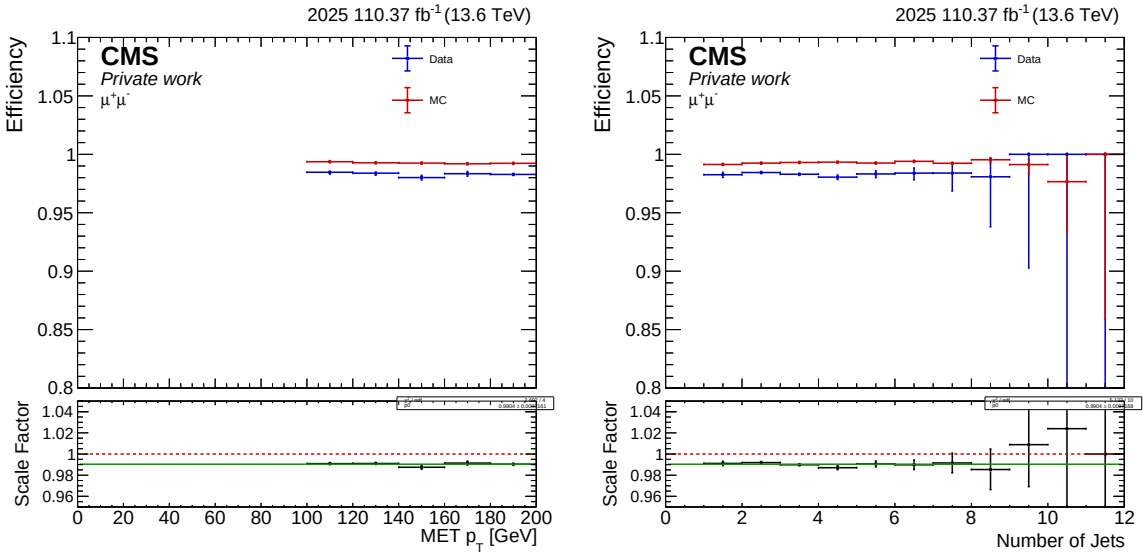


Figure 75: Additional one-dimensional efficiency and SF diagnostics in the $\mu\mu$ channel. Variables and binning are shown on each axis.

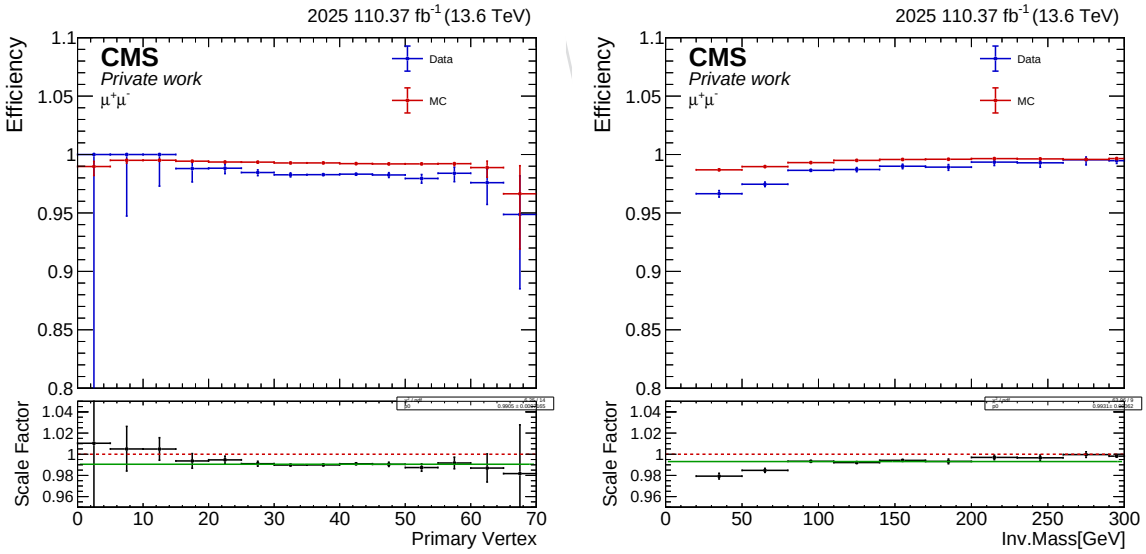


Figure 76: Additional one-dimensional efficiency and SF diagnostics in the $\mu\mu$ channel. Variables and binning are shown on each axis.

E 2025: detailed uncertainty studies

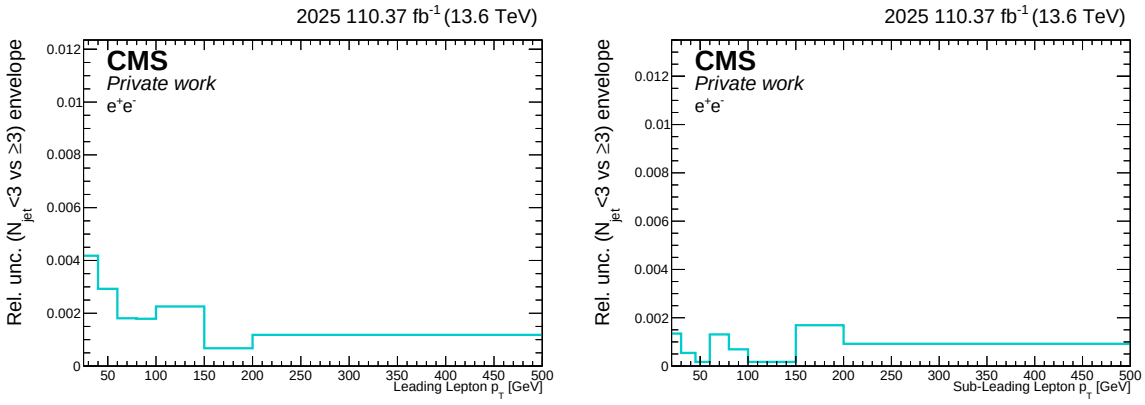


Figure 77: Jet Topology Uncertainty diagnostics in the *ee* channel versus leading and subleading lepton momentum. Curves are reproduced with the production definitions; their exact numerical prescription requires confirmation.

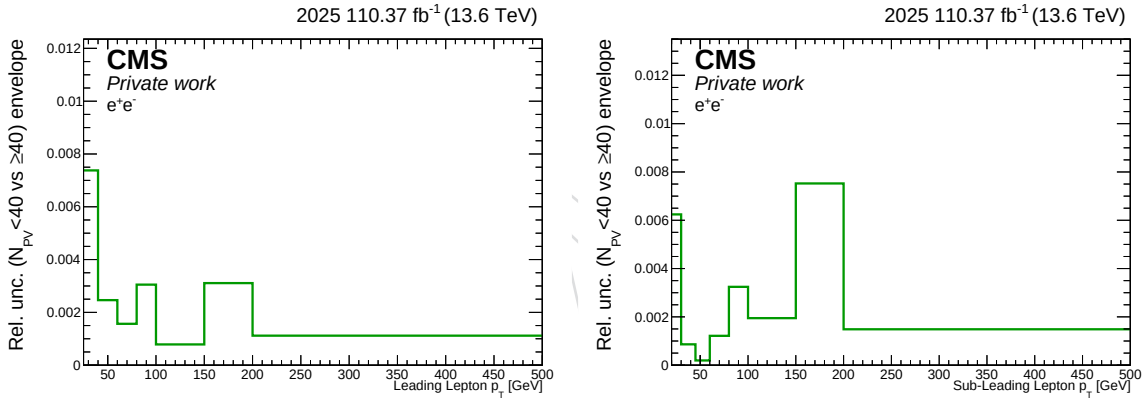


Figure 78: PV Topology Uncertainty diagnostics in the *ee* channel versus leading and subleading lepton momentum. Curves are reproduced with the production definitions; their exact numerical prescription requires confirmation.

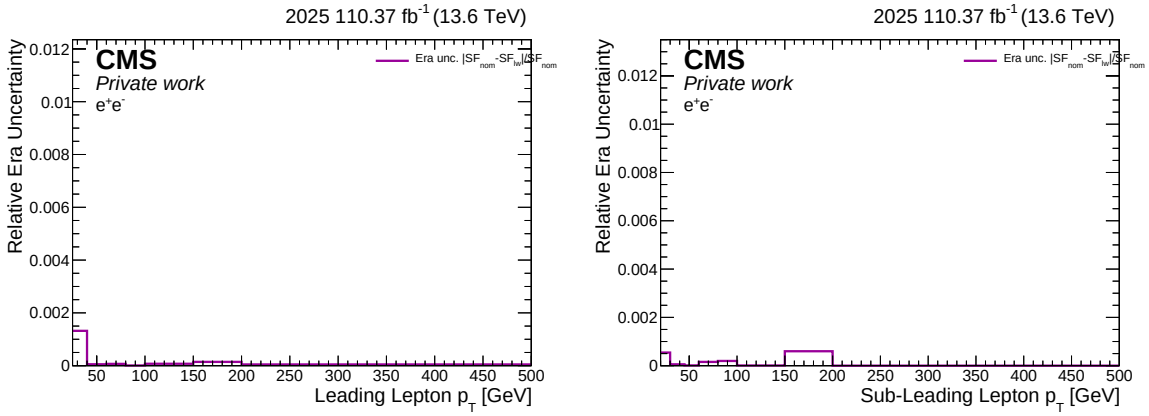


Figure 79: Era Uncertainty diagnostics in the *ee* channel versus leading and subleading lepton momentum. Curves are reproduced with the production definitions; their exact numerical prescription requires confirmation.

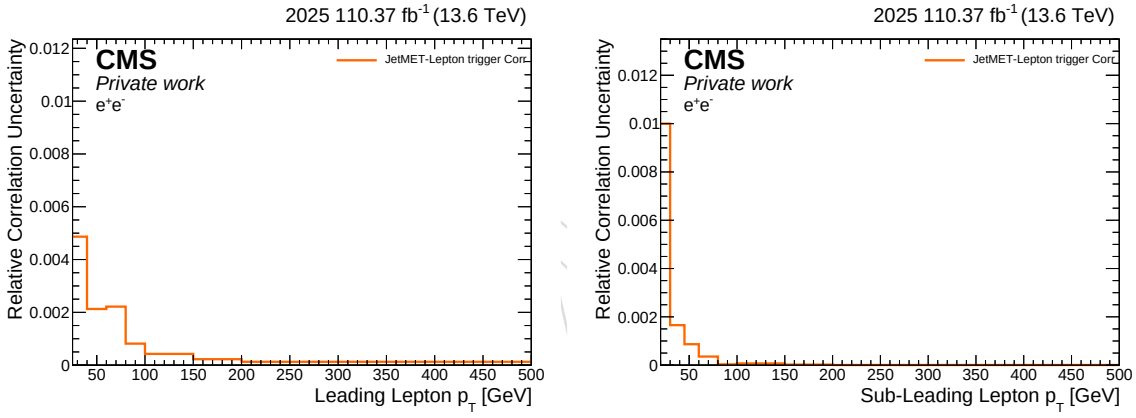


Figure 80: Correlation Uncertainty diagnostics in the *ee* channel versus leading and subleading lepton momentum. Curves are reproduced with the production definitions; their exact numerical prescription requires confirmation.

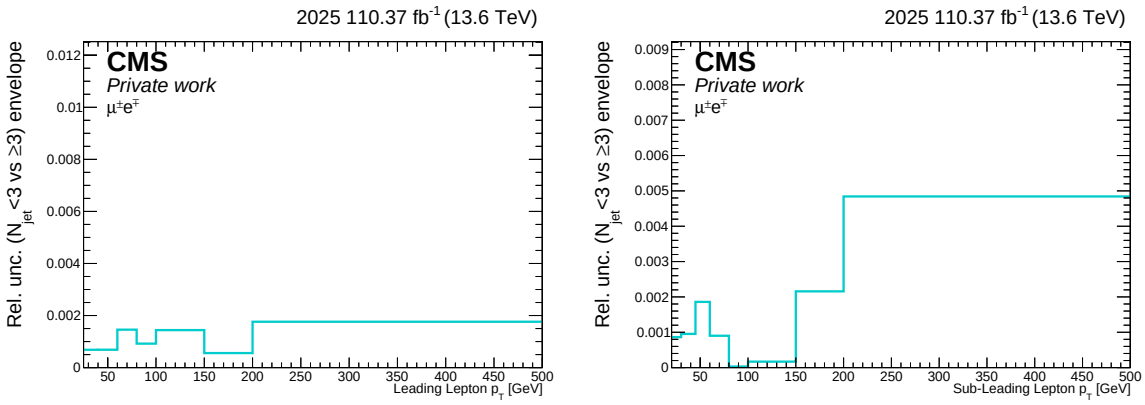


Figure 81: Jet Topology Uncertainty diagnostics in the $e\mu$ channel versus leading and subleading lepton momentum. Curves are reproduced with the production definitions; their exact numerical prescription requires confirmation.

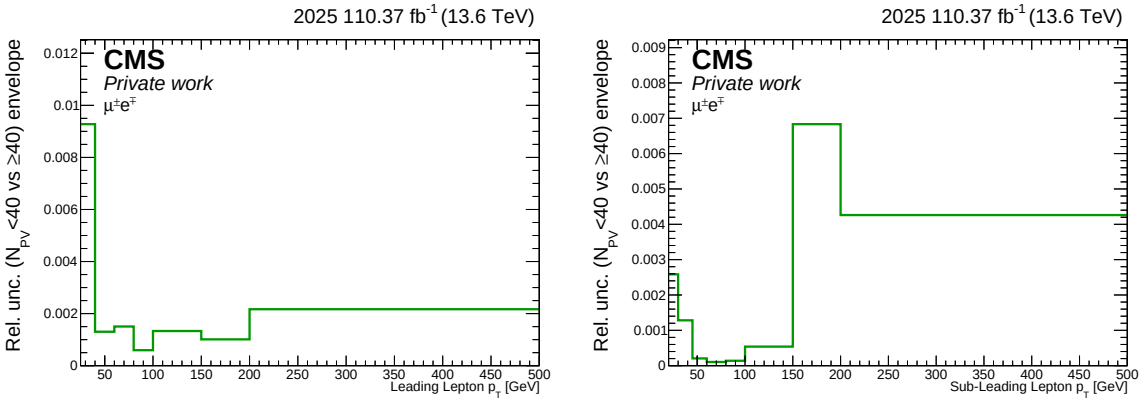


Figure 82: PV Topology Uncertainty diagnostics in the $e\mu$ channel versus leading and subleading lepton momentum. Curves are reproduced with the production definitions; their exact numerical prescription requires confirmation.

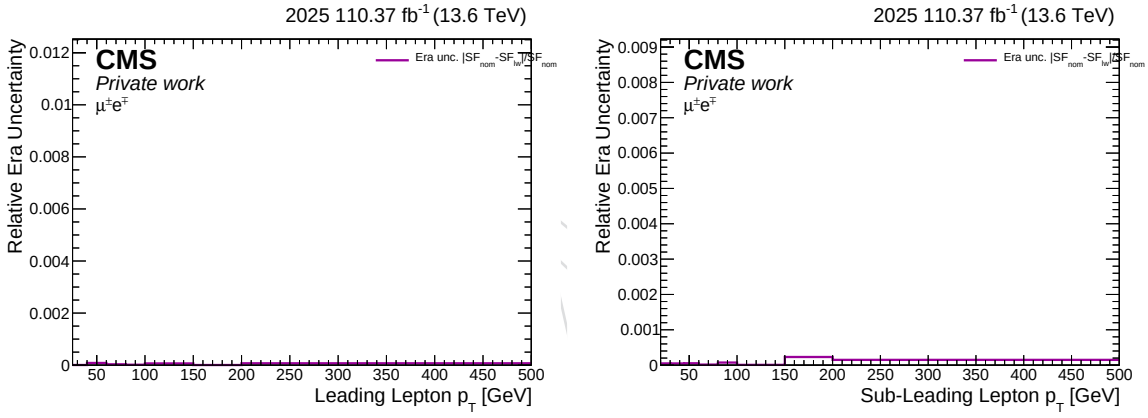


Figure 83: Era Uncertainty diagnostics in the $e\mu$ channel versus leading and subleading lepton momentum. Curves are reproduced with the production definitions; their exact numerical prescription requires confirmation.

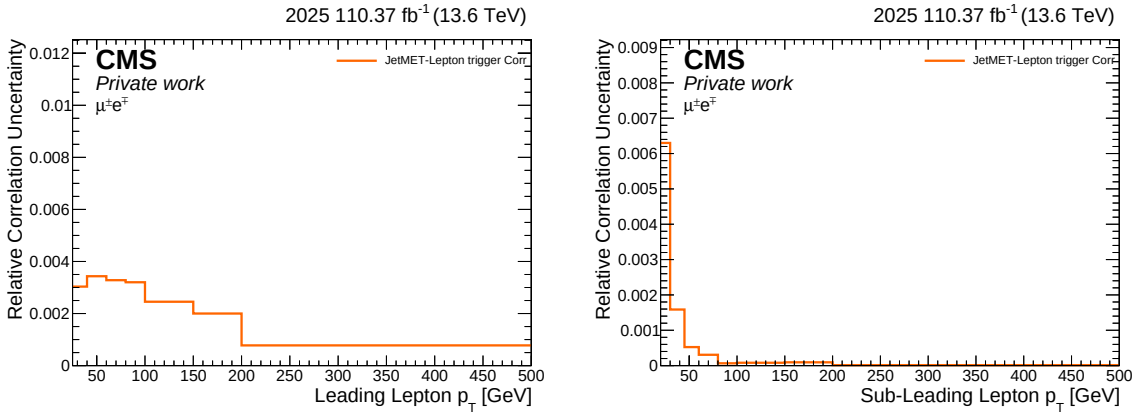


Figure 84: Correlation Uncertainty diagnostics in the $e\mu$ channel versus leading and subleading lepton momentum. Curves are reproduced with the production definitions; their exact numerical prescription requires confirmation.

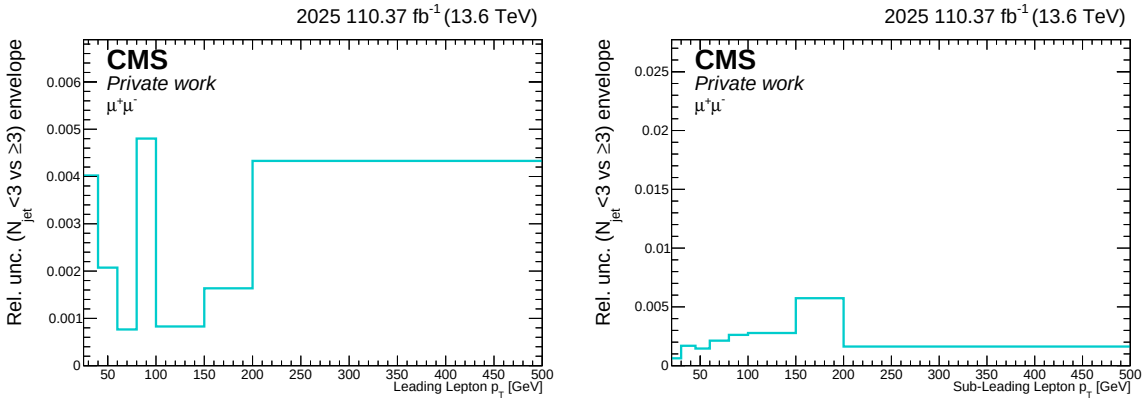


Figure 85: Jet Topology Uncertainty diagnostics in the $\mu\mu$ channel versus leading and subleading lepton momentum. Curves are reproduced with the production definitions; their exact numerical prescription requires confirmation.

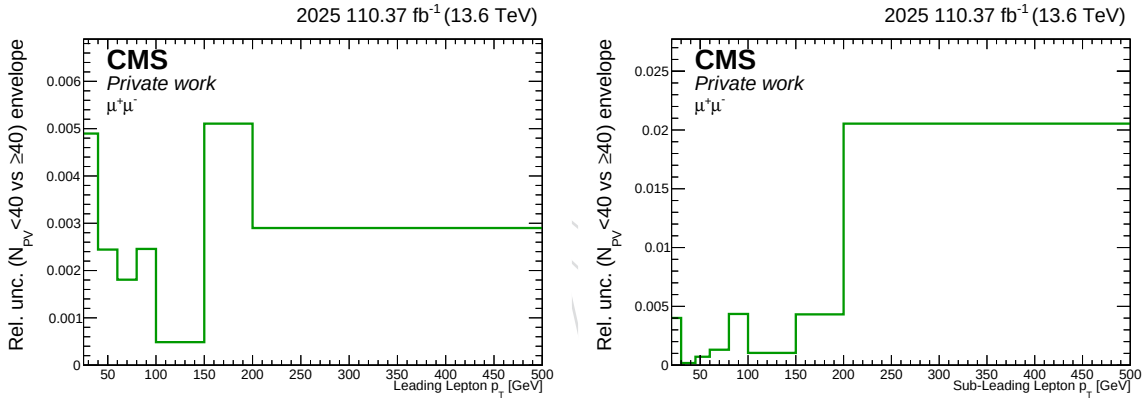


Figure 86: PV Topology Uncertainty diagnostics in the $\mu\mu$ channel versus leading and subleading lepton momentum. Curves are reproduced with the production definitions; their exact numerical prescription requires confirmation.

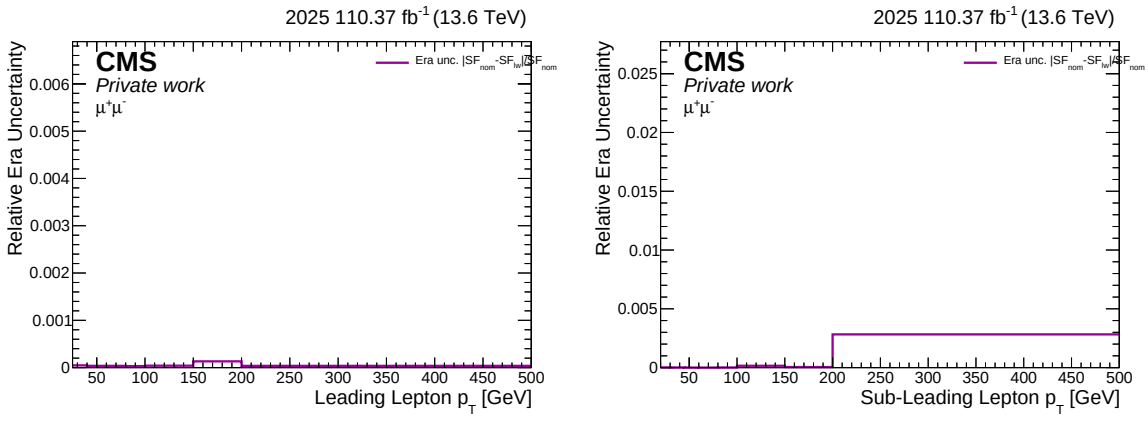


Figure 87: Era Uncertainty diagnostics in the $\mu\mu$ channel versus leading and subleading lepton momentum. Curves are reproduced with the production definitions; their exact numerical prescription requires confirmation.

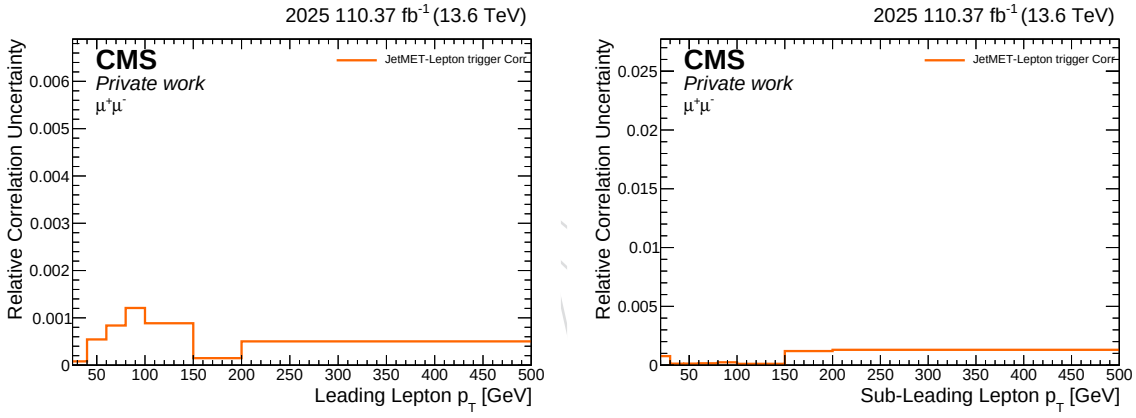


Figure 88: Correlation Uncertainty diagnostics in the $\mu\mu$ channel versus leading and subleading lepton momentum. Curves are reproduced with the production definitions; their exact numerical prescription requires confirmation.

F Selected 2025 supporting distributions

The following distributions document the efficiency inputs, statistical precision, and uncertainty composition for the 2025 measurement. Only inclusive distributions directly related to the correction or its principal checks are selected. The jet- and vertex-split variants and the full set of individual-jet and angular distributions are not reproduced. These plots use the measurement sample and are not independent closure tests.

F.1 *ee* channel

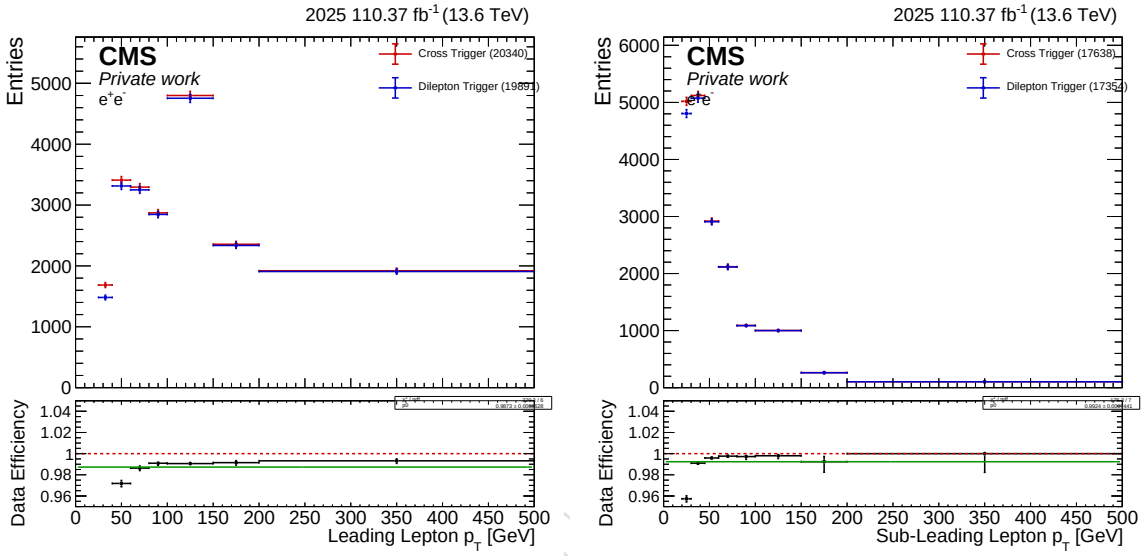


Figure 89: Data trigger numerator and reference-sample distributions (upper panels), with their efficiency ratios (lower panels) in the *ee* channel. Left and right: leading and subleading lepton p_{T}.

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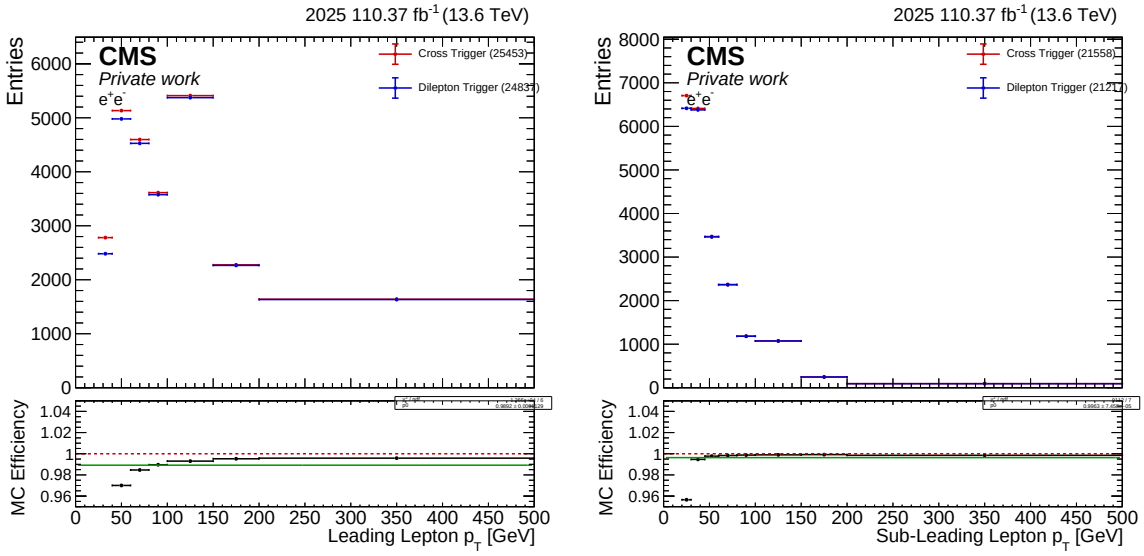


Figure 90: Simulated trigger numerator and reference-sample distributions (upper panels), with their efficiency ratios (lower panels) in the *ee* channel. Left and right: leading and subleading lepton p_{T}.

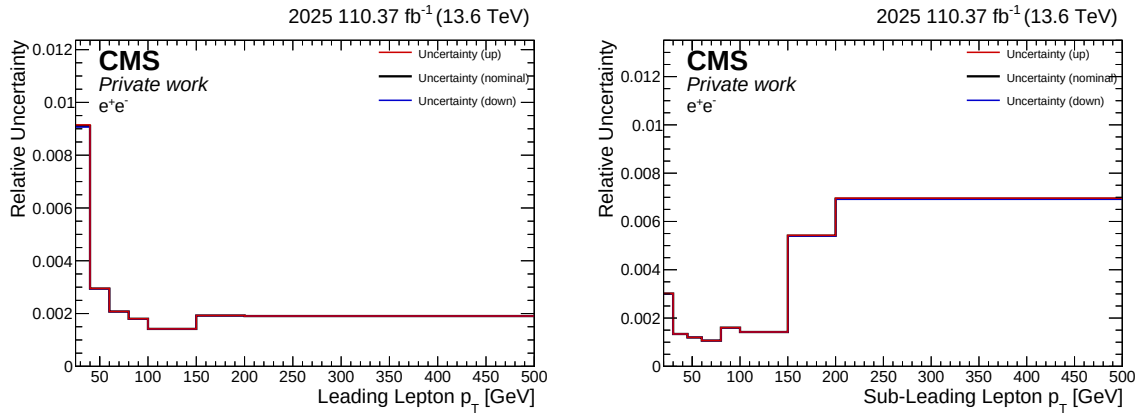


Figure 91: Relative statistical uncertainty diagnostics, retaining the nominal, upward, and downward curves of the supplied production in the *ee* channel. Left and right: leading and subleading lepton p_{T}.

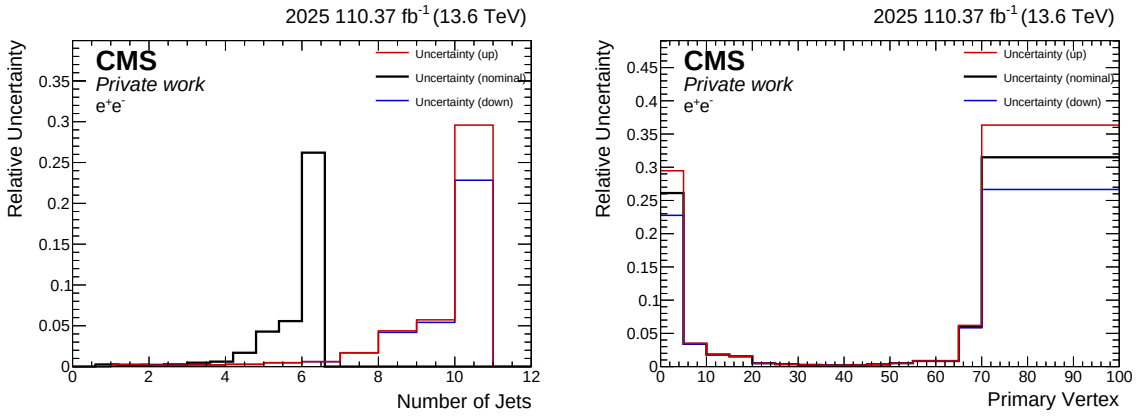


Figure 92: Relative statistical uncertainty diagnostics, retaining the nominal, upward, and downward curves of the supplied production in the *ee* channel. Left and right: jet and primary-vertex multiplicities.

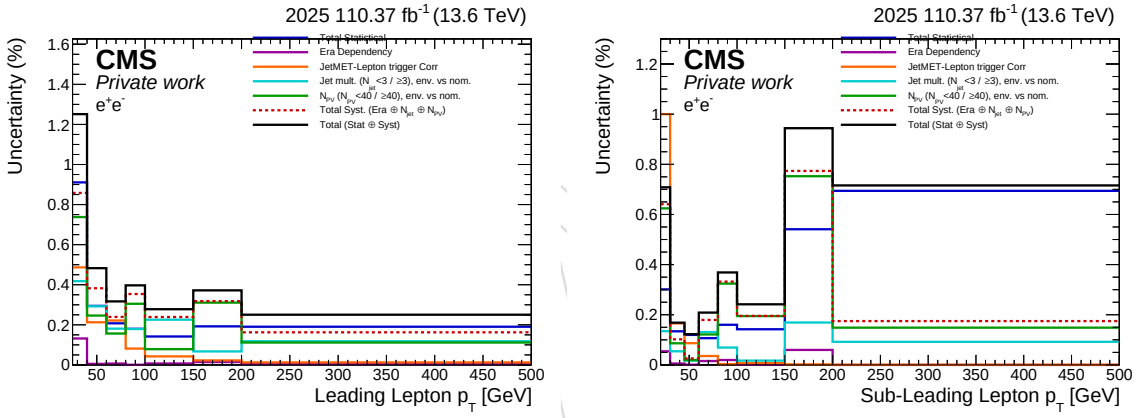


Figure 93: Absolute uncertainty component breakdown in the *ee* channel. Left and right: leading and subleading lepton p_T.

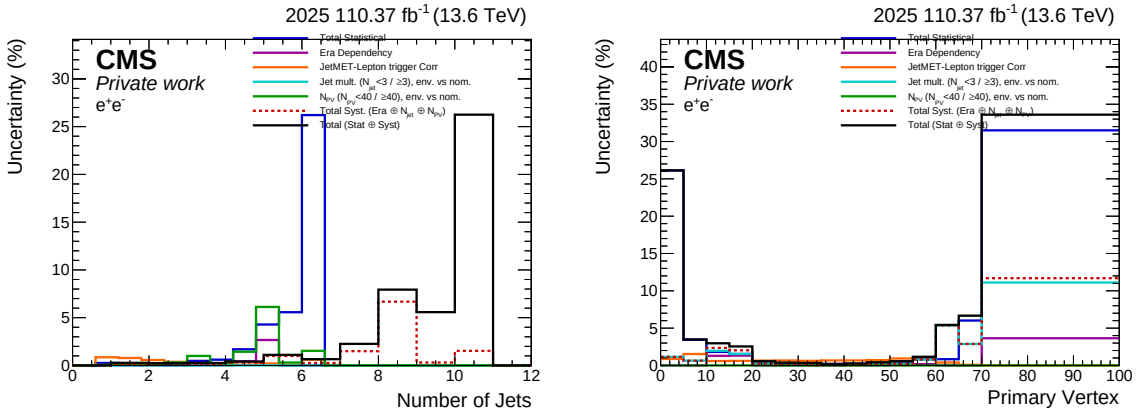


Figure 94: Absolute uncertainty component breakdown in the *ee* channel. Left and right: jet and primary-vertex multiplicities.

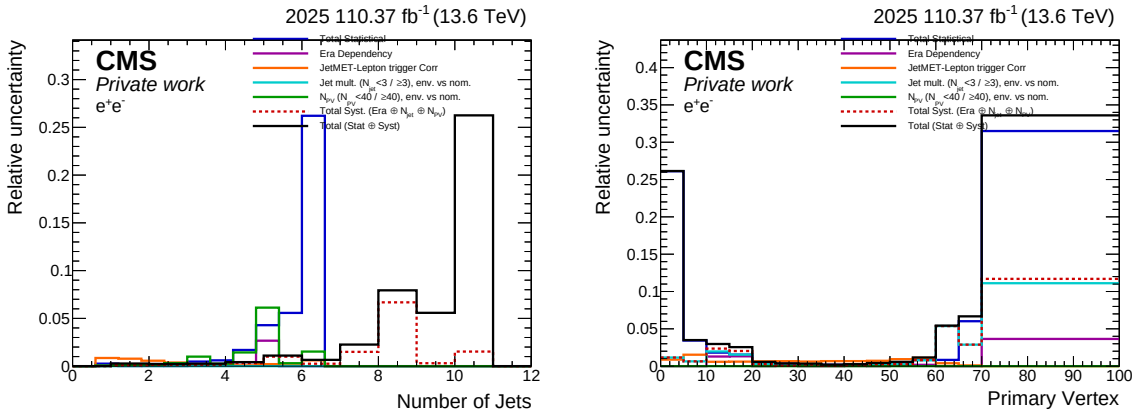


Figure 95: Relative uncertainty component breakdown versus event activity in the *ee* channel. Left and right: jet and primary-vertex multiplicities.

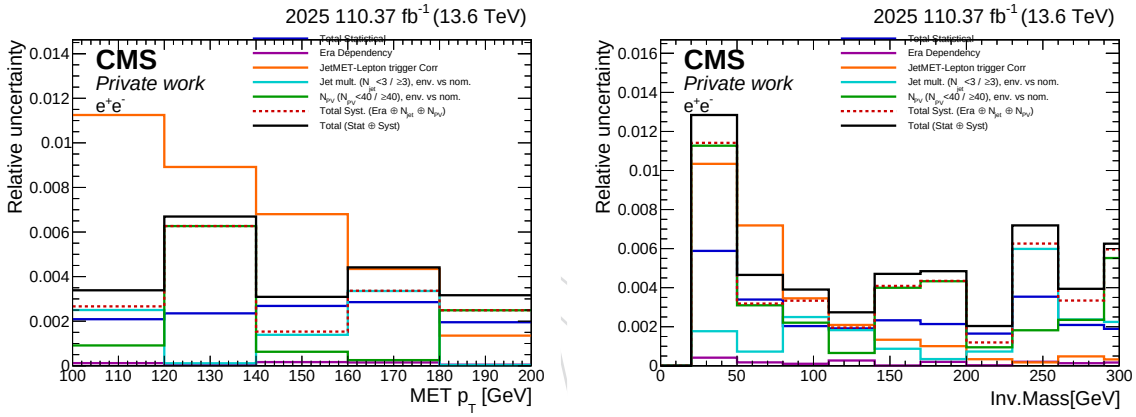


Figure 96: Relative uncertainty component breakdown versus event kinematics in the *ee* channel. Left and right: missing transverse momentum and dilepton invariant mass.

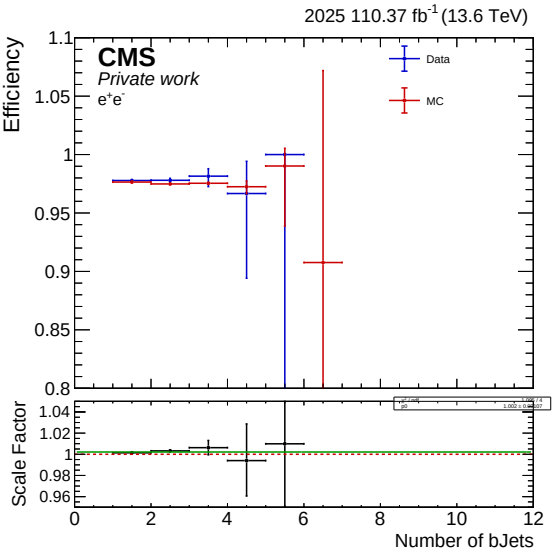


Figure 97: Efficiency comparison and scale factor versus heavy-flavour activity in the *ee* channel. *b*-tagged jet multiplicity.

F.2 $e\mu$ channel

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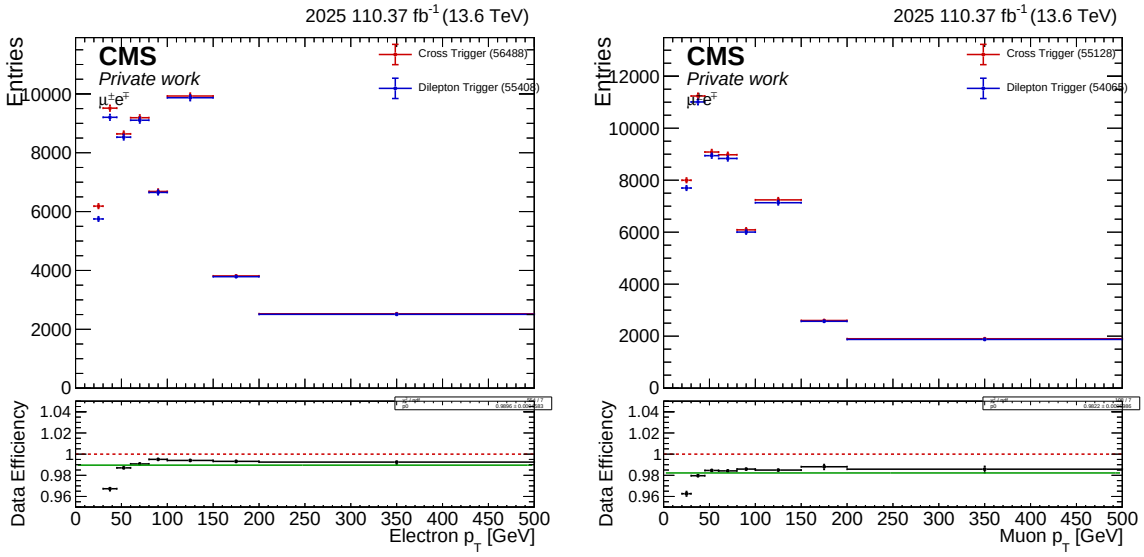


Figure 98: Data trigger numerator and reference-sample distributions (upper panels), with their efficiency ratios (lower panels) in the $e\mu$ channel. Left and right: electron and muon p_{T}.

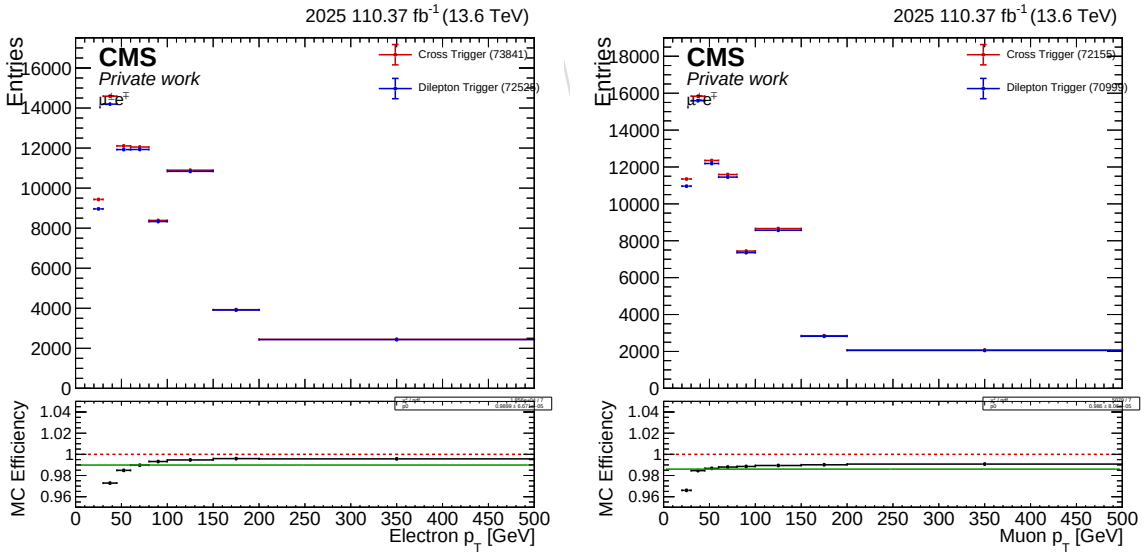


Figure 99: Simulated trigger numerator and reference-sample distributions (upper panels), with their efficiency ratios (lower panels) in the $e\mu$ channel. Left and right: electron and muon p_{T}.

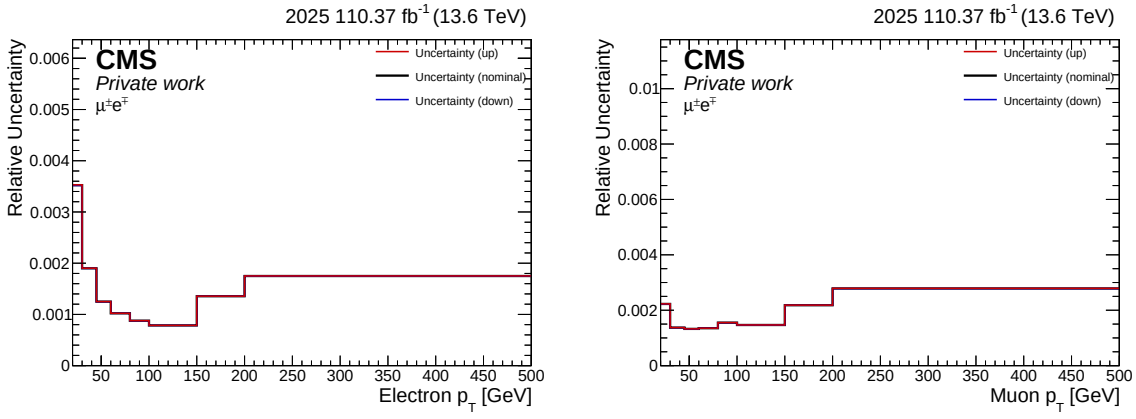


Figure 100: Relative statistical uncertainty diagnostics, retaining the nominal, upward, and downward curves of the supplied production in the $e\mu$ channel. Left and right: electron and muon p_{T}.

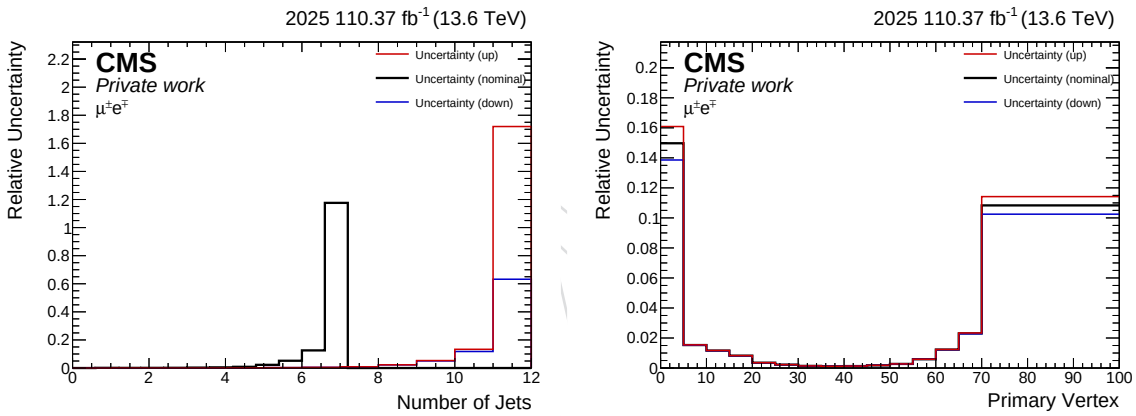


Figure 101: Relative statistical uncertainty diagnostics, retaining the nominal, upward, and downward curves of the supplied production in the $e\mu$ channel. Left and right: jet and primary-vertex multiplicities.

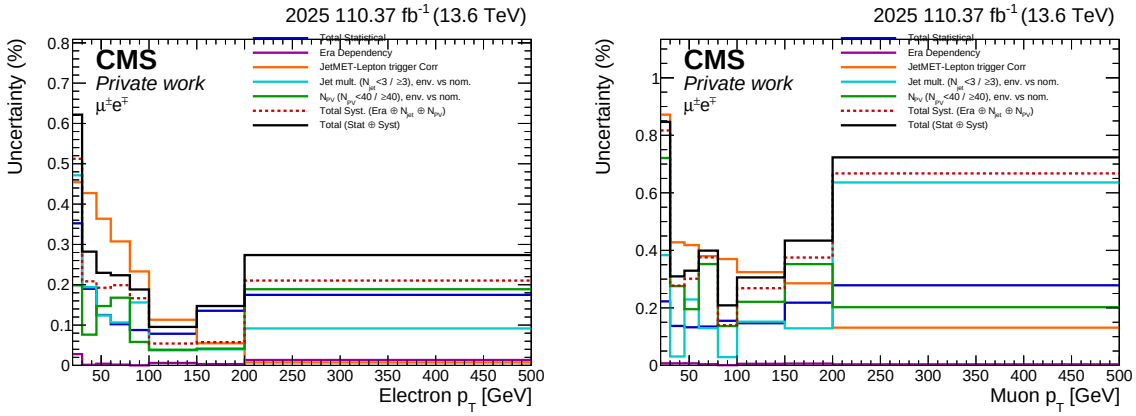


Figure 102: Absolute uncertainty component breakdown in the $e\mu$ channel. Left and right: electron and muon p_{T}.

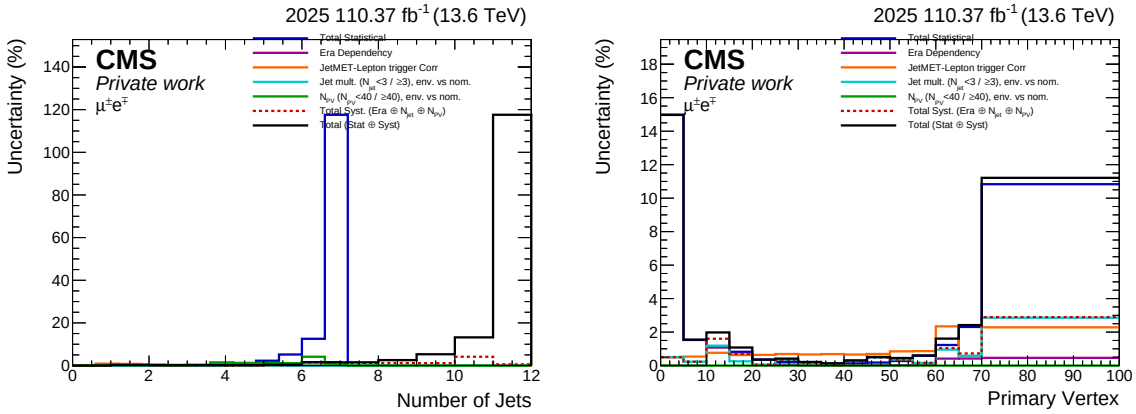


Figure 103: Absolute uncertainty component breakdown in the $e\mu$ channel. Left and right: jet and primary-vertex multiplicities.

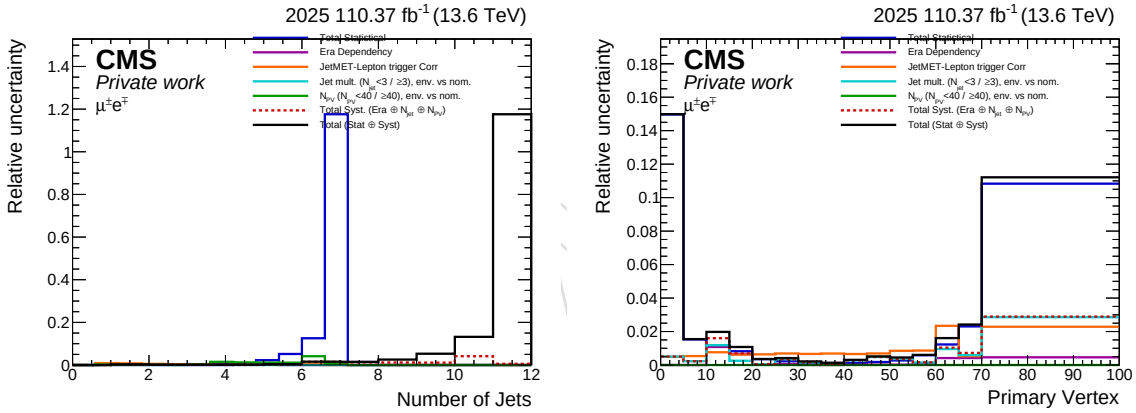


Figure 104: Relative uncertainty component breakdown versus event activity in the $e\mu$ channel. Left and right: jet and primary-vertex multiplicities.

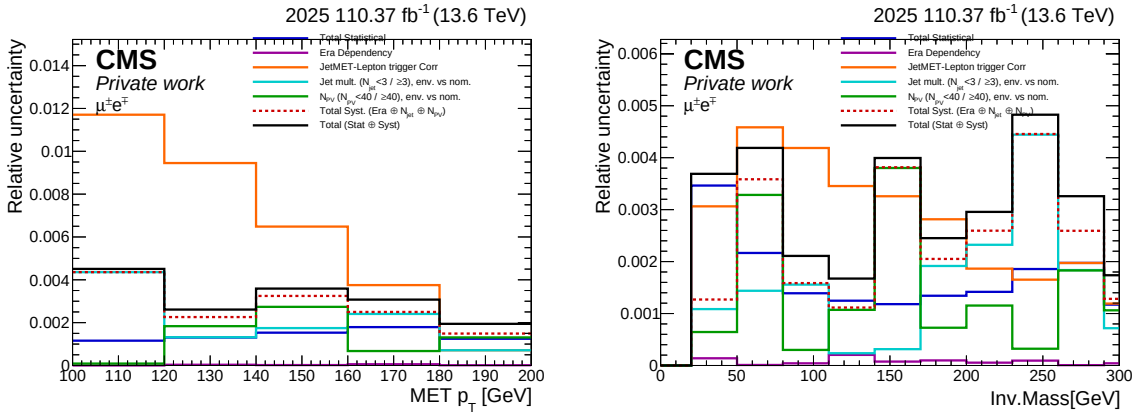


Figure 105: Relative uncertainty component breakdown versus event kinematics in the $e\mu$ channel. Left and right: missing transverse momentum and dilepton invariant mass.

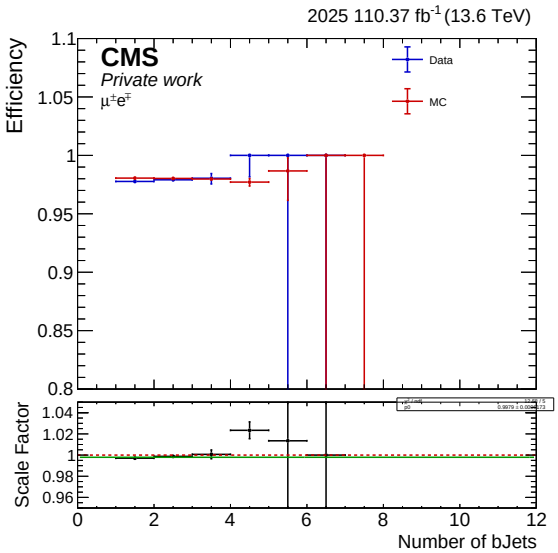


Figure 106: Efficiency comparison and scale factor versus heavy-flavour activity in the $e\mu$ channel. b-tagged jet multiplicity.

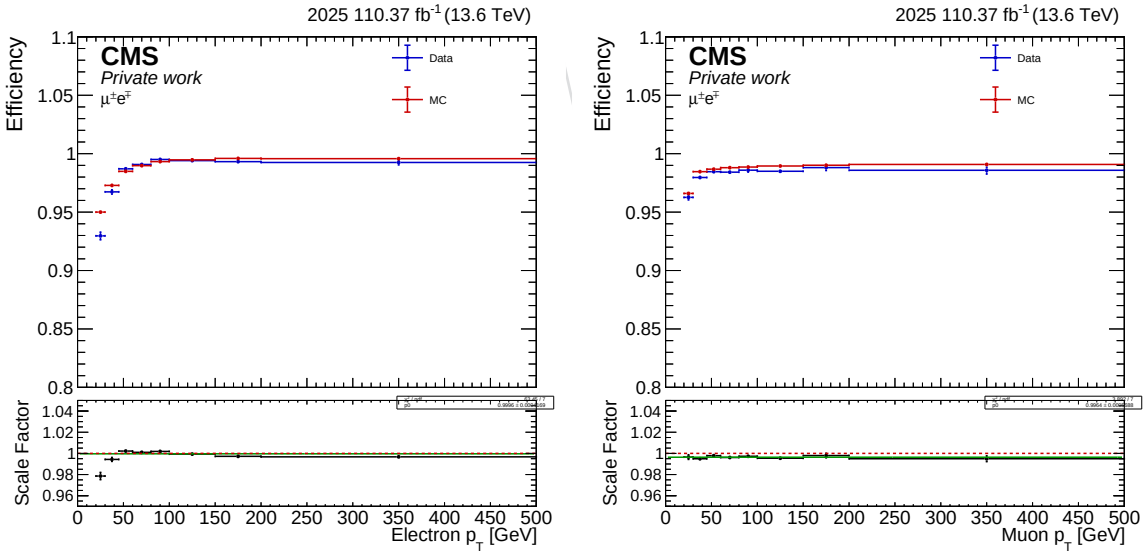


Figure 107: Efficiency comparison and scale factor in the flavour-specific momentum variables used by the two-dimensional correction in the $e\mu$ channel. Left and right: electron and muon p_{T}.

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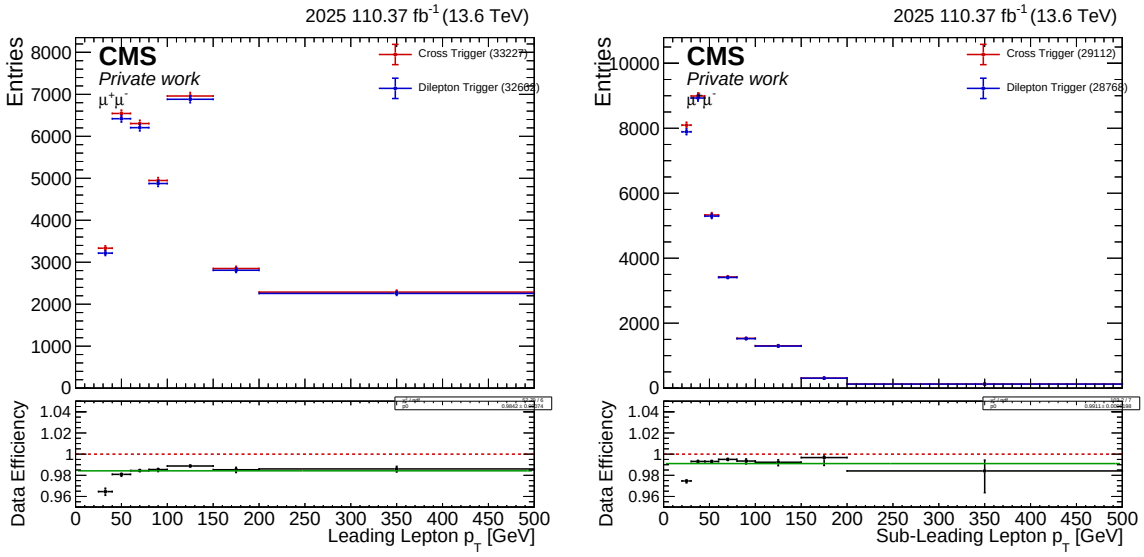


Figure 108: Data trigger numerator and reference-sample distributions (upper panels), with their efficiency ratios (lower panels) in the $\mu\mu$ channel. Left and right: leading and subleading lepton p_{T}.

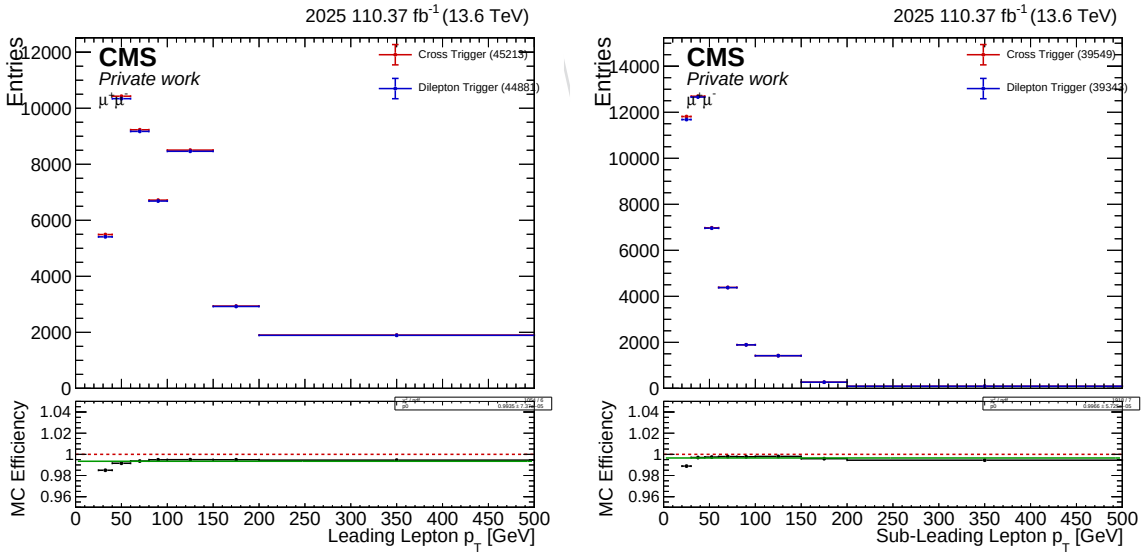


Figure 109: Simulated trigger numerator and reference-sample distributions (upper panels), with their efficiency ratios (lower panels) in the $\mu\mu$ channel. Left and right: leading and subleading lepton p_{T}.

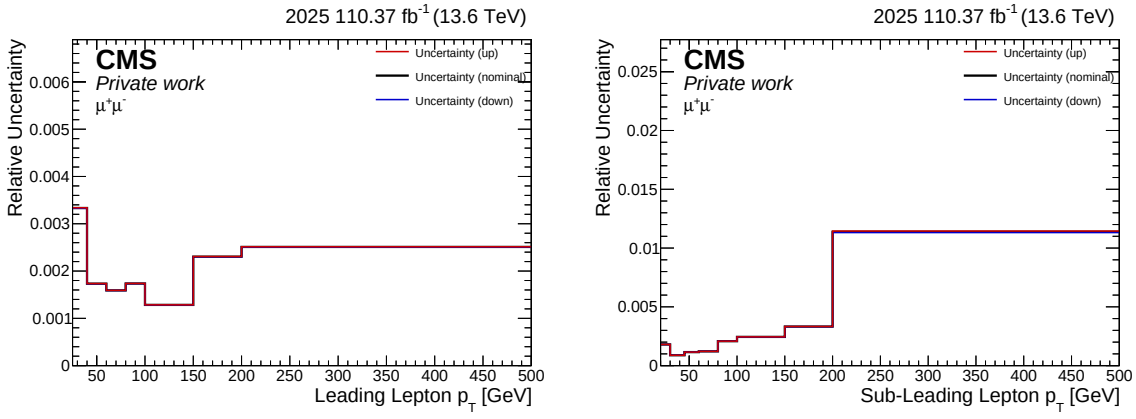


Figure 110: Relative statistical uncertainty diagnostics, retaining the nominal, upward, and downward curves of the supplied production in the $\mu\mu$ channel. Left and right: leading and subleading lepton p_{T}.

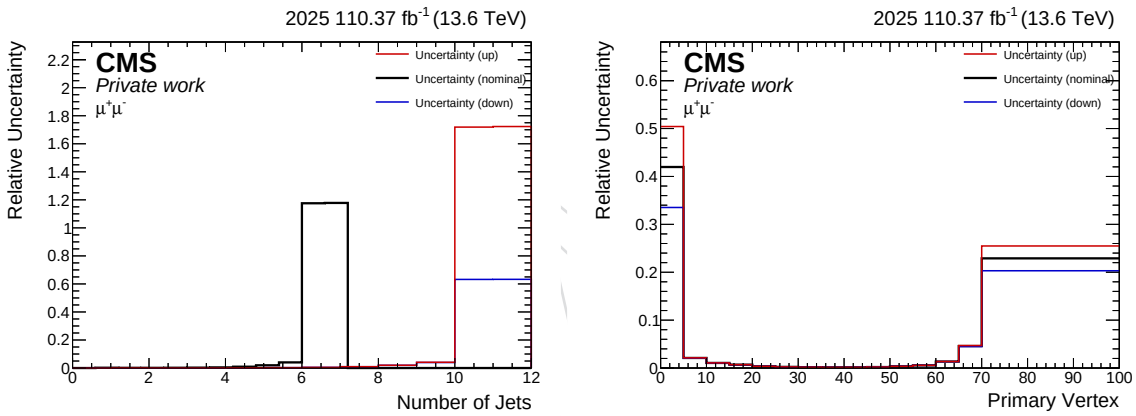


Figure 111: Relative statistical uncertainty diagnostics, retaining the nominal, upward, and downward curves of the supplied production in the $\mu\mu$ channel. Left and right: jet and primary-vertex multiplicities.

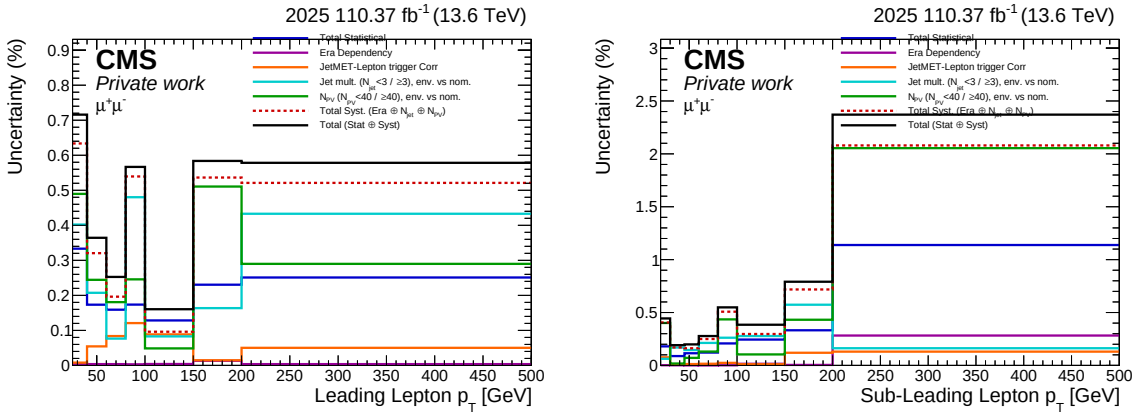


Figure 112: Absolute uncertainty component breakdown in the $\mu\mu$ channel. Left and right: leading and subleading lepton p_{T}.

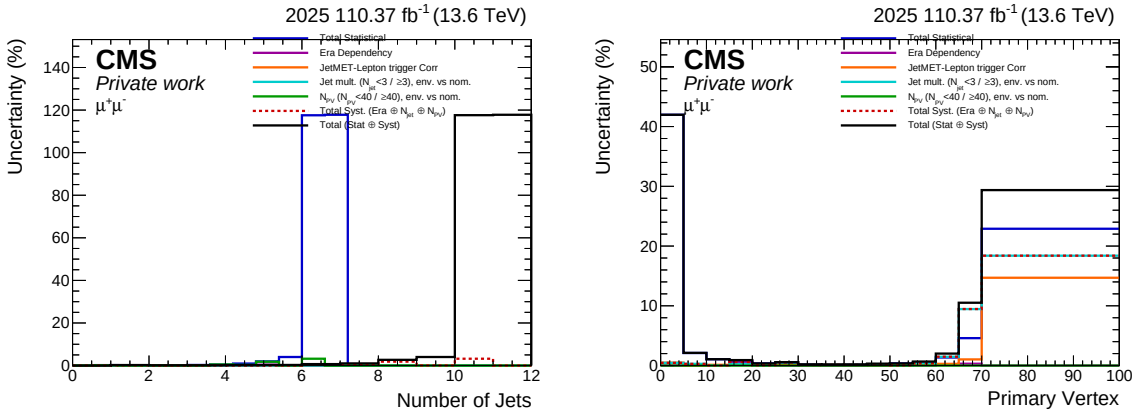


Figure 113: Absolute uncertainty component breakdown in the $\mu\mu$ channel. Left and right: jet and primary-vertex multiplicities.

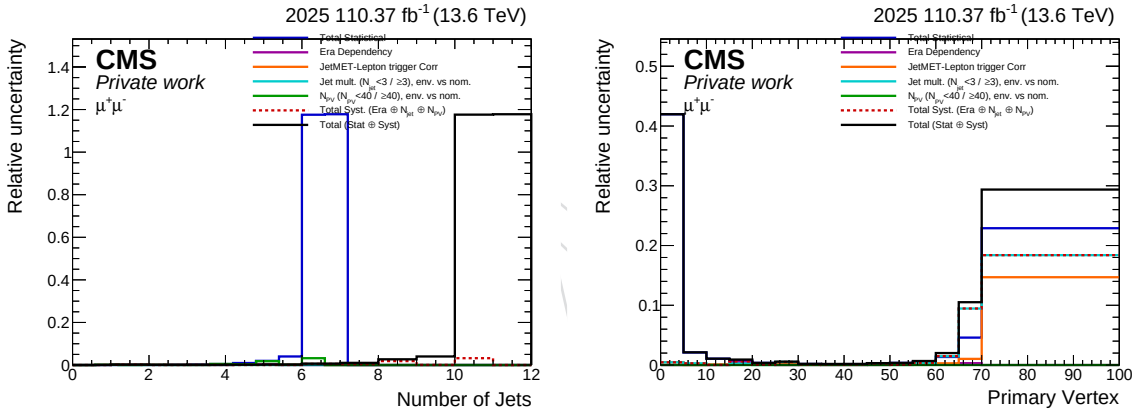


Figure 114: Relative uncertainty component breakdown versus event activity in the $\mu\mu$ channel. Left and right: jet and primary-vertex multiplicities.

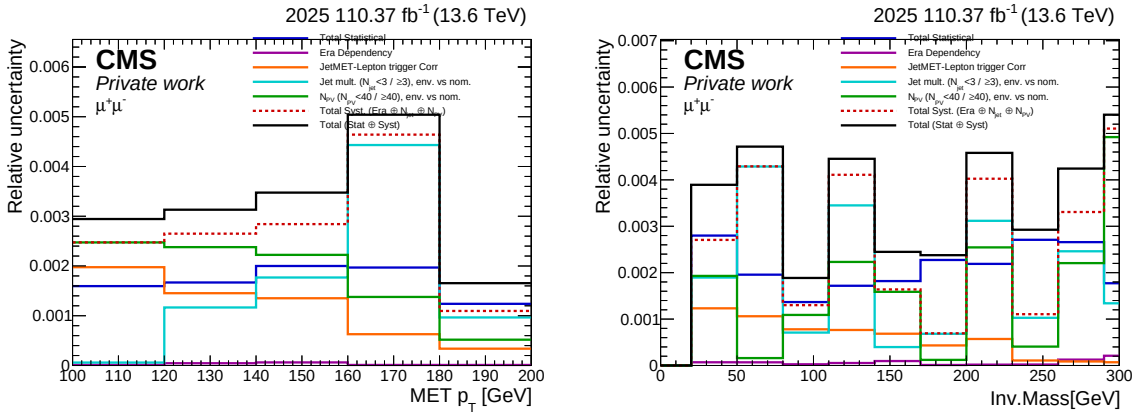


Figure 115: Relative uncertainty component breakdown versus event kinematics in the $\mu\mu$ channel. Left and right: missing transverse momentum and dilepton invariant mass.

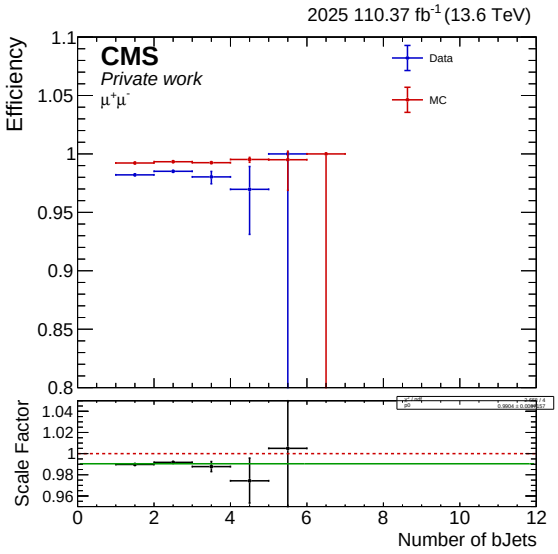


Figure 116: Efficiency comparison and scale factor versus heavy-flavour activity in the $\mu\mu$ channel. b-tagged jet multiplicity.